

#1 · Primary Care Medicine

What are the ACC/AHA blood pressure categories and their cutoffs?

Answer

Normal is below 120/80 mmHg. Elevated is systolic 120-129 with diastolic below 80. Stage 1 hypertension is 130-139/80-89 mmHg. Stage 2 hypertension is at or above 140/90 mmHg.

#2 · Primary Care Medicine

When does elevated blood pressure (120-129/<80) require medication?

Answer

Elevated BP does not require medication. First-line treatment is lifestyle modification: weight loss, DASH diet, sodium restriction below 2,300 mg/day, aerobic exercise 90-150 min/week, and alcohol limitation. Reassess in 3-6 months.

#3 · Primary Care Medicine

What is the blood pressure treatment threshold and target for Stage 1 hypertension in a patient with 10-year ASCVD risk above 10%?

Answer

Initiate pharmacotherapy when BP is 130-139/80-89 mmHg if 10-year ASCVD risk is at or above 10%. The treatment target is below 130/80 mmHg. Thiazide diuretics, ACE inhibitors, ARBs, or calcium channel blockers are all acceptable first-line agents.

#4 · Primary Care Medicine

What fasting glucose level and A1c value are diagnostic for type 2 diabetes mellitus?

Answer

A fasting plasma glucose of 126 mg/dL or above on two separate occasions confirms diabetes. An A1c of 6.5% or above also confirms the diagnosis. A single abnormal result in a symptomatic patient is sufficient.

#5 · Primary Care Medicine

What glucose and A1c values define prediabetes?

Answer

Prediabetes is defined as a fasting glucose of 100-125 mg/dL (impaired fasting glucose) or an A1c of 5.7-6.4%. A 2-hour glucose of 140-199 mg/dL on a 75 g OGTT also meets the criteria. Lifestyle changes reduce progression to DM by about 58%.

#6 · Primary Care Medicine

What is the A1c target for most non-pregnant adults with type 2 diabetes?

Answer

The ADA target A1c is below 7% for most adults with type 2 diabetes. A less stringent target of below 8% is acceptable for elderly patients, those with limited life expectancy, or those with hypoglycemia unawareness. Targets are individualized based on patient factors.

#7 · Primary Care Medicine

What is the first-line oral agent for type 2 diabetes in a patient without contraindications?

Answer

Metformin is first-line for type 2 diabetes. It is contraindicated when eGFR falls below 30 mL/min/1.73m² and should be used with caution when eGFR is 30-45. GLP-1 agonists or SGLT-2 inhibitors are added when A1c is above goal or in patients with ASCVD, CKD, or heart failure.

#8 · Primary Care Medicine

What ejection fraction defines heart failure with reduced ejection fraction (HFrEF) and what drugs reduce mortality?

Answer

HFrEF is defined as an ejection fraction below 40%. Mortality-reducing medications include ACE inhibitors or ARBs (or ARNI sacubitril-valsartan), beta-blockers (carvedilol, metoprolol succinate, or bisoprolol), mineralocorticoid receptor antagonists (spironolactone), and SGLT-2 inhibitors (dapagliflozin, empagliflozin).

#9 · Primary Care Medicine

What is sacubitril-valsartan (Entresto) and when is it used in heart failure?

Answer

Sacubitril-valsartan is an ARNI (angiotensin receptor-neprilysin inhibitor) used in HFrEF (EF below 40%) to reduce hospitalizations and mortality. It replaces an ACE inhibitor or ARB. A 36-hour washout is required before switching from an ACE inhibitor to prevent angioedema.

#10 · Primary Care Medicine

What ejection fraction defines HFpEF and what is the primary treatment approach?

Answer

HFpEF is defined as an ejection fraction at or above 50%. Treatment focuses on diuretics (furosemide) to relieve congestion and control of underlying comorbidities including hypertension, atrial fibrillation, and obesity. SGLT-2 inhibitors (empagliflozin, dapagliflozin) reduce the composite risk of cardiovascular death and heart failure hospitalization in HFpEF, driven mainly by fewer hospitalizations, and are now guideline-recommended for heart failure regardless of ejection fraction.

#11 · Primary Care Medicine

What FEV1/FVC ratio confirms obstructive lung disease on spirometry?

Answer

A post-bronchodilator FEV1/FVC ratio below 0.70 confirms airflow obstruction consistent with COPD. This is the diagnostic criterion per GOLD guidelines. Spirometry must be performed after bronchodilator administration to confirm fixed obstruction.

#12 · Primary Care Medicine

What are the four GOLD spirometric stages of COPD based on FEV1 percent predicted?

Answer

GOLD 1 (mild): FEV1 at or above 80% predicted. GOLD 2 (moderate): FEV1 50-79%. GOLD 3 (severe): FEV1 30-49%. GOLD 4 (very severe): FEV1 below 30%. All stages require FEV1/FVC below 0.70 to qualify as COPD.

#13 · Primary Care Medicine

What is first-line pharmacotherapy for a newly diagnosed COPD patient with low symptoms and low exacerbation risk?

Answer

A short-acting bronchodilator (SABA such as albuterol or SAMA such as ipratropium) is used as needed for immediate relief. For persistent symptoms, a long-acting bronchodilator (LAMA such as tiotropium or LABA such as salmeterol) is added. ICS is not first-line unless eosinophils are elevated or exacerbations occur.

#14 · Primary Care Medicine

What are the four asthma severity classifications and the corresponding step therapy for each?

Answer

Intermittent asthma: SABA as needed only.
Mild persistent: low-dose ICS daily.
Moderate persistent: medium-dose ICS plus LABA.
Severe persistent: high-dose ICS plus LABA, with consideration of oral corticosteroids or biologic agents such as omalizumab or dupilumab.

#15 · Primary Care Medicine

What lab pattern confirms primary hypothyroidism?

Answer

Primary hypothyroidism shows a high TSH with a low free T4. Subclinical hypothyroidism shows a high TSH with a normal free T4. Treatment with levothyroxine is indicated for TSH above 10 mIU/L or symptomatic patients with lower TSH elevation.

#16 · Primary Care Medicine

What lab pattern confirms hyperthyroidism?

Answer

Hyperthyroidism shows a suppressed (low) TSH with an elevated free T4 or elevated free T3. Subclinical hyperthyroidism shows suppressed TSH with normal free T4 and T3. Common causes include Graves disease, toxic multinodular goiter, and toxic adenoma.

#17 · Primary Care Medicine

What is the first-line treatment for newly diagnosed hypothyroidism?

Answer

Levothyroxine (synthetic T4) is the standard treatment for hypothyroidism. Starting dose is typically 1.6 mcg/kg/day in healthy adults. TSH should be rechecked 6-8 weeks after initiating or adjusting therapy. Goal TSH is 0.5-4.5 mIU/L for most patients.

#18 · Primary Care Medicine

What are the six CKD stages based on GFR?

Answer

G1: GFR at or above 90 mL/min/1.73m² (with kidney damage markers). G2: 60-89. G3a: 45-59. G3b: 30-44. G4: 15-29. G5: below 15 (kidney failure). Staging requires a marker of kidney damage or GFR below 60 persisting more than 3 months.

#19 · Primary Care Medicine

What are the most common causes of CKD and which drug class is first-line to slow progression?

Answer

Diabetes mellitus and hypertension account for over 70% of CKD cases. ACE inhibitors or ARBs are first-line to slow CKD progression, especially in diabetic nephropathy with proteinuria. SGLT-2 inhibitors (canagliflozin, dapagliflozin) also reduce CKD progression and are now guideline-recommended.

#20 · Primary Care Medicine

At what GFR should metformin be avoided in CKD patients?

Answer

Metformin should be stopped when eGFR falls below 30 mL/min/1.73m² due to risk of lactic acidosis. It should be used with caution and dose-reduced when eGFR is 30-45 mL/min/1.73m². Renal function should be rechecked every 3-6 months in CKD patients on metformin.

#21 · Primary Care Medicine

What LDL-lowering drugs are classified as high-intensity statins and what percent reduction do they achieve?

Answer

High-intensity statins include atorvastatin 40-80 mg and rosuvastatin 20-40 mg daily. They reduce LDL by at or above 50%. These are indicated for patients with clinical ASCVD, LDL at or above 190 mg/dL, diabetes aged 40-75, or 10-year ASCVD risk at or above 7.5-10%.

#22 · Primary Care Medicine

At what 10-year ASCVD risk threshold should statin therapy be initiated?

Answer

ACC/AHA guidelines recommend initiating a statin when 10-year ASCVD risk is at or above 7.5% (intermediate risk, 7.5% to less than 20%) in adults aged 40-75 without prior ASCVD. A risk discussion is appropriate in the 5% to 7.5% borderline-risk range, and CAC scoring can help decide in uncertain cases. Pooled Cohort Equations are used to calculate 10-year risk.

#23 · Primary Care Medicine

What is the CHA2DS2-VASc score and when should anticoagulation be started for atrial fibrillation?

Answer

CHA2DS2-VASc assigns points for CHF, Hypertension, Age 65-74 (1 pt) or 75+ (2 pts), Diabetes, Stroke/TIA (2 pts), Vascular disease, and female Sex. Anticoagulate when score is at or above 2 in men or at or above 3 in women. Direct oral anticoagulants (DOACs) are preferred over warfarin.

#24 · Primary Care Medicine

What is first-line treatment for uncomplicated GERD and how long should it be continued?

Answer

Proton pump inhibitors (PPIs) such as omeprazole 20 mg daily are first-line for GERD. Treat for 4-8 weeks for uncomplicated erosive esophagitis. Long-term PPI use requires justification due to risks including *C. difficile*, hypomagnesemia, and fractures. H2 blockers are an alternative for mild symptoms.

#25 · Primary Care Medicine

What organism causes most uncomplicated urinary tract infections and what are two first-line oral treatments?

Answer

Escherichia coli causes 80-85% of uncomplicated UTIs. First-line treatment options are nitrofurantoin 100 mg ER twice daily for 5 days or trimethoprim-sulfamethoxazole (TMP-SMX) DS twice daily for 3 days. Fluoroquinolones are not recommended first-line due to resistance concerns and side effects.

#26 · Primary Care Medicine

When is nitrofurantoin contraindicated for UTI treatment?

Answer

Nitrofurantoin should be avoided when GFR is below 30 mL/min/1.73m² due to inadequate urinary drug concentration and risk of peripheral neuropathy. It should also be avoided in the third trimester of pregnancy. Use with caution in GFR 30-45 mL/min/1.73m².

#27 · Primary Care Medicine

What is the CURB-65 score and what score allows outpatient treatment of community-acquired pneumonia?

Answer

CURB-65 scores one point each for Confusion, BUN above 19 mg/dL, Respiratory rate at or above 30/min, BP below 90 systolic or at or below 60 diastolic, and Age at or above 65. A score of 0-1 supports outpatient treatment. A score of 2 suggests inpatient care; 3 or above warrants ICU consideration.

#28 · Primary Care Medicine

What is the empiric antibiotic regimen for outpatient community-acquired pneumonia in a healthy adult without comorbidities?

Answer

Amoxicillin 1 g three times daily for 5 days is first-line for outpatient CAP in healthy adults. Doxycycline 100 mg twice daily for 5 days is an alternative. Azithromycin monotherapy is no longer preferred due to increasing pneumococcal resistance. Duration is typically 5 days if clinical improvement occurs.

#29 · Primary Care Medicine

What ankle-brachial index (ABI) value is considered abnormal and suggests peripheral artery disease?

Answer

An ABI below 0.9 is abnormal and indicates peripheral artery disease (PAD). An ABI of 0.41-0.69 indicates moderate PAD, and below 0.4 indicates severe or critical limb ischemia. An ABI above 1.3 suggests non-compressible vessels (as seen in severe diabetes or calcification) and is also abnormal.

#30 · Primary Care Medicine

What are the classic symptoms of peripheral artery disease and what is first-line medical treatment?

Answer

Intermittent claudication (leg pain with walking that resolves with rest) is the hallmark symptom of PAD. Medical treatment includes antiplatelet therapy (aspirin 75-100 mg or clopidogrel 75 mg daily), a statin, ACE inhibitor or ARB, and supervised exercise therapy. Smoking cessation is critical.

#31 · Primary Care Medicine

What blood pressure target is recommended for patients with confirmed hypertension and CKD?

Answer

The target BP for patients with CKD is below 130/80 mmHg per ACC/AHA and KDIGO guidelines. ACE inhibitors or ARBs are preferred in CKD with proteinuria because they reduce intraglomerular pressure and slow progression. Avoid combining ACEi and ARB due to hyperkalemia and acute kidney injury risk.

#32 · Primary Care Medicine

What are the first-line antihypertensive drug classes per ACC/AHA guidelines?

Answer

ACC/AHA recommends thiazide-type diuretics (chlorthalidone preferred over hydrochlorothiazide), ACE inhibitors, ARBs, and dihydropyridine calcium channel blockers (amlodipine) as first-line agents. ACEi and ARB should not be combined. In Black patients without CKD, thiazides or CCBs are preferred.

#33 · Primary Care Medicine

What blood pressure medication class is most effective as initial monotherapy in Black patients without CKD or heart failure?

Answer

Thiazide-type diuretics (chlorthalidone) and calcium channel blockers (amlodipine) are more effective as initial therapy in Black patients due to lower renin activity. ACE inhibitors and ARBs are less effective as monotherapy but are indicated when CKD with proteinuria is present.

#34 · Primary Care Medicine

What is the fasting lipid panel value at which statins are started regardless of ASCVD risk calculation?

Answer

An LDL at or above 190 mg/dL indicates familial hypercholesterolemia and warrants high-intensity statin therapy without needing to calculate 10-year ASCVD risk. The goal is at least a 50% LDL reduction. If LDL remains above 100 mg/dL, ezetimibe or PCSK9 inhibitors may be added.

#35 · Primary Care Medicine

What is the mechanism of ezetimibe and in what clinical situations is it added to statin therapy?

Answer

Ezetimibe blocks cholesterol absorption at the intestinal brush border, reducing LDL by approximately 15-20%. It is added to high-intensity statin therapy when LDL remains above goal (below 70 mg/dL for very high risk, below 100 mg/dL for high risk). It is also used in patients who are statin-intolerant.

#36 · Primary Care Medicine

What are the classic symptoms of hypothyroidism and what drug interaction can reduce levothyroxine absorption?

Answer

Classic hypothyroid symptoms include fatigue, weight gain, constipation, cold intolerance, dry skin, bradycardia, and depression. Calcium, iron supplements, and PPIs reduce levothyroxine absorption. Patients should take levothyroxine on an empty stomach 30-60 minutes before eating or other medications.

#37 · Primary Care Medicine

What type of diuretic is furosemide and what electrolyte imbalances does it commonly cause?

Answer

Furosemide is a loop diuretic that acts on the thick ascending limb of the loop of Henle. Common electrolyte imbalances include hypokalemia, hyponatremia, hypomagnesemia, and metabolic alkalosis. Monitor potassium closely, especially in patients also taking digoxin, as hypokalemia increases digoxin toxicity.

#38 · Primary Care Medicine

What are the major side effects of ACE inhibitors and what is the mechanism of ACE inhibitor cough?

Answer

ACE inhibitors commonly cause a dry, persistent cough due to accumulation of bradykinin, which is normally degraded by ACE. Other side effects include hyperkalemia, acute kidney injury (especially with bilateral renal artery stenosis), and angioedema. ARBs do not cause cough and are used as alternatives.

#39 · Primary Care Medicine

What is the recommended screening interval for colorectal cancer starting at age 45?

Answer

USPSTF recommends colorectal cancer screening starting at age 45 for average-risk adults. Options include colonoscopy every 10 years, annual high-sensitivity fecal occult blood test (FIT or gFOBT), CT colonography every 5 years, or stool DNA test every 1-3 years. Screening continues through age 75.

#40 · Primary Care Medicine

What are the USPSTF breast cancer screening recommendations for average-risk women?

Answer

USPSTF (2024) recommends biennial mammography for women aged 40-74. Women aged 75 and older have insufficient evidence to guide recommendations. High-risk women (BRCA mutation, first-degree relative with BRCA, prior chest radiation) should start earlier with annual MRI plus mammography.

#41 · Primary Care Medicine

What is the criteria for diagnosing metabolic syndrome?

Answer

Metabolic syndrome requires 3 of 5 criteria: waist circumference above 102 cm (men) or 88 cm (women); triglycerides at or above 150 mg/dL; HDL below 40 mg/dL (men) or below 50 mg/dL (women); BP at or above 130/85 mmHg; and fasting glucose at or above 100 mg/dL. It significantly increases cardiovascular and DM risk.

#42 · Primary Care Medicine

What is the significance of microalbuminuria in a diabetic patient and what is the first-line treatment?

Answer

Microalbuminuria (urine albumin-to-creatinine ratio 30-300 mg/g) is the earliest sign of diabetic nephropathy. ACE inhibitors or ARBs reduce proteinuria and slow progression of CKD in diabetic patients regardless of blood pressure. SGLT-2 inhibitors (canagliflozin, dapagliflozin) also reduce CKD progression.

#43 · Primary Care Medicine

What are the major complications of uncontrolled type 2 diabetes?

Answer

Microvascular complications include diabetic retinopathy (leading cause of adult blindness), nephropathy (leading cause of ESRD), and peripheral neuropathy. Macrovascular complications include CAD, stroke, and PAD. Annual dilated eye exams, urine microalbumin, and foot exams are recommended.

#44 · Primary Care Medicine

What defines diabetic ketoacidosis (DKA) and what is the initial treatment?

Answer

DKA is characterized by glucose above 250 mg/dL, pH below 7.3 or bicarbonate below 18 mEq/L, and positive ketones (ketonemia or ketonuria). It primarily occurs in type 1 DM. Initial treatment is aggressive IV normal saline hydration, insulin infusion (0.1 units/kg/hr), and potassium replacement when K⁺ is below 5.3 mEq/L.

#45 · Primary Care Medicine

What are the criteria for diagnosing obstructive sleep apnea (OSA) on polysomnography?

Answer

OSA is diagnosed by apnea-hypopnea index (AHI): mild is 5-14 events/hour, moderate is 15-29, severe is at or above 30. Symptoms include snoring, witnessed apneas, and daytime sleepiness (Epworth Sleepiness Scale above 10). CPAP is first-line treatment for moderate-severe OSA.

#46 · Primary Care Medicine

What is the STOP-BANG questionnaire used for and what score indicates high risk?

Answer

STOP-BANG is a screening tool for obstructive sleep apnea assessing Snoring, Tiredness, Observed apnea, Pressure (high BP), BMI above 35, Age above 50, Neck circumference above 40 cm, and male Gender. A score of 3 or above indicates high risk and warrants polysomnography.

#47 · Primary Care Medicine

What are the classic ECG findings of ST-elevation myocardial infarction (STEMI)?

Answer

STEMI shows ST elevation at or above 1 mm in two contiguous limb leads or at or above 2 mm in two contiguous precordial leads. New left bundle branch block may also represent STEMI. Immediate percutaneous coronary intervention (PCI) within 90 minutes is the gold standard reperfusion strategy.

#48 · Primary Care Medicine

What troponin level confirms acute myocardial infarction and when does it peak?

Answer

Elevated high-sensitivity troponin I or T above the 99th percentile upper reference limit confirms myocardial injury. Troponin rises 3-4 hours after injury onset, peaks at 12-24 hours, and remains elevated for 7-14 days. Serial troponins at 0 and 3 hours are used for rapid rule-out protocols.

#49 · Primary Care Medicine

What is *Helicobacter pylori* and what test is preferred to confirm eradication after treatment?

Answer

H. pylori is a gram-negative bacterium causing peptic ulcer disease and is the most common cause of duodenal ulcers. First-line treatment is clarithromycin-based triple therapy (PPI + clarithromycin + amoxicillin) for 14 days. Urea breath test or fecal antigen test (not serology) confirms eradication at least 4 weeks after completing treatment.

#50 · Primary Care Medicine

What are the diagnostic criteria for irritable bowel syndrome (IBS) per the Rome IV criteria?

Answer

Rome IV requires recurrent abdominal pain at least 1 day per week over the past 3 months (with symptom onset at least 6 months prior) associated with 2 or more of: related to defecation, change in stool frequency, or change in stool form. IBS-C (constipation predominant), IBS-D (diarrhea), and IBS-M (mixed) are subtypes.

#51 · Primary Care Medicine

What are red flag symptoms in a patient with IBS that require further evaluation?

Answer

Red flag symptoms requiring colonoscopy or further workup include rectal bleeding, unintentional weight loss above 10 lbs, nocturnal symptoms, age above 45 at first presentation, family history of colon cancer or IBD, iron deficiency anemia, and elevated inflammatory markers (ESR, CRP). These suggest organic pathology.

#52 · Primary Care Medicine

What is the most common cause of acute low back pain and how long does it typically resolve?

Answer

Muscle strain or ligamentous sprain is the most common cause of acute low back pain. Most cases resolve within 4-6 weeks with conservative treatment. First-line management includes NSAIDs, heat application, and continued activity. Imaging is not recommended in the first 6 weeks without red flag symptoms.

#53 · Primary Care Medicine

What are red flag symptoms in back pain that require urgent imaging?

Answer

Red flags warranting urgent MRI or CT include new bowel or bladder dysfunction (cauda equina syndrome), saddle anesthesia, fever with back pain (epidural abscess or osteomyelitis), history of cancer, osteoporosis with minor trauma, and progressive neurological deficits. Cauda equina syndrome is a surgical emergency.

#54 · Primary Care Medicine

What is the Wells score used for and what score warrants imaging for DVT?

Answer

The Wells DVT score estimates pretest probability of deep vein thrombosis. High probability (score 3 or above) warrants compression ultrasound. A score of 0-1 with negative D-dimer effectively rules out DVT. Components include active cancer, paralysis, bedridden more than 3 days, localized tenderness, pitting edema, calf swelling, and prior DVT.

#55 · Primary Care Medicine

What is first-line anticoagulation for a patient with unprovoked DVT and no contraindications?

Answer

Direct oral anticoagulants (DOACs) are first-line for DVT treatment: rivaroxaban 15 mg twice daily for 21 days then 20 mg daily, or apixaban 10 mg twice daily for 7 days then 5 mg twice daily. Treatment duration is typically 3 months for provoked DVT and at least 3-6 months (often indefinitely) for unprovoked DVT.

#56 · Primary Care Medicine

What is the diagnostic workup for a patient presenting with hemoptysis?

Answer

Initial workup includes chest X-ray and CBC. CT chest with contrast is the preferred imaging for identifying the source. Bronchoscopy is indicated for massive hemoptysis (above 300 mL/24 hours) or when CT is non-diagnostic. Common causes include bronchitis, bronchiectasis, TB, lung cancer, and PE.

#57 · Primary Care Medicine

What is the recommended lung cancer screening protocol per USPSTF?

Answer

Annual low-dose CT (LDCT) of the chest is recommended for adults aged 50-80 with a 20-pack-year smoking history who currently smoke or quit within the past 15 years. Screening should be discontinued if the patient has not smoked for 15 years or develops a condition limiting treatment or life expectancy.

#58 · Primary Care Medicine

What are the USPSTF recommendations for osteoporosis screening?

Answer

USPSTF recommends bone mineral density (DEXA scan) screening for women aged 65 and older. Younger postmenopausal women should be screened if their 10-year fracture risk (by FRAX) is equal to or greater than that of a 65-year-old White woman (approximately 9.3%). No recommendation for men (insufficient evidence).

#59 · Primary Care Medicine

What is the mechanism of beta-blockers used in HFrEF and which three are mortality-proven?

Answer

Beta-blockers in HFrEF reduce heart rate and block the harmful effects of chronic sympathetic nervous system activation on the myocardium. Only three beta-blockers have proven mortality benefit: carvedilol (non-selective), metoprolol succinate (beta-1 selective, extended-release only), and bisoprolol (beta-1 selective).

#60 · Primary Care Medicine

What is the recommended blood pressure target for a patient with established cardiovascular disease?

Answer

ACC/AHA guidelines recommend a blood pressure target below 130/80 mmHg for patients with established cardiovascular disease (prior MI, stroke, or high ASCVD risk). Achieving this target is associated with reduced recurrent cardiovascular events. Beta-blockers and ACE inhibitors or ARBs are preferred agents post-MI.

#61 · Primary Care Medicine

What is the diagnosis and treatment of hypertensive urgency vs. hypertensive emergency?

Answer

Hypertensive urgency is BP above 180/120 mmHg without acute target organ damage; reduce BP over 24-48 hours with oral agents (amlodipine, lisinopril). Hypertensive emergency is BP above 180/120 mmHg WITH end-organ damage (encephalopathy, AKI, flash pulmonary edema, aortic dissection); requires IV agents (nicardipine, labetalol) and ICU admission with BP reduction by 25% in first hour.

#62 · Primary Care Medicine

What lab values define anemia and what are the three most common types by MCV?

Answer

Anemia is hemoglobin below 13 g/dL in men or below 12 g/dL in women. Microcytic (MCV below 80 fL): iron deficiency anemia (most common worldwide), thalassemia, anemia of chronic disease (sometimes). Normocytic (MCV 80-100): anemia of chronic disease. Macrocytic (MCV above 100): B12 or folate deficiency, hypothyroidism, alcohol use.

#63 · Primary Care Medicine

What lab pattern differentiates iron deficiency anemia from anemia of chronic disease?

Answer

Iron deficiency anemia: low serum iron, low ferritin (below 30 ng/mL), high TIBC. Anemia of chronic disease: low serum iron, normal or high ferritin (above 100 ng/mL), low or normal TIBC. Ferritin is an acute phase reactant and may be falsely elevated in inflammatory states even with true iron deficiency.

#64 · Primary Care Medicine

What is the most common presenting symptom of aortic stenosis and what is the expected survival without intervention?

Answer

Aortic stenosis classically presents with the triad of syncope, angina, and exertional dyspnea (heart failure). Once symptomatic, survival averages 2-3 years without intervention. Severe AS is defined as aortic valve area below 1.0 cm², mean gradient above 40 mmHg, or peak velocity above 4 m/s.

#65 · Primary Care Medicine

What is the criteria for chronic stable angina treatment and what is first-line pharmacotherapy?

Answer

Chronic stable angina symptoms occur with exertion and resolve with rest or nitroglycerin within 5 minutes. First-line pharmacotherapy includes beta-blockers (metoprolol) to reduce myocardial oxygen demand, sublingual nitroglycerin PRN for acute episodes, aspirin 75-100 mg daily, and high-intensity statin therapy.

#66 · Primary Care Medicine

What is the diagnostic test and treatment for H. pylori-positive peptic ulcer disease?

Answer

H. pylori can be detected by urea breath test, stool antigen, or endoscopy with biopsy (CLO test). First-line treatment is clarithromycin-based triple therapy: PPI + clarithromycin 500 mg + amoxicillin 1 g, all twice daily for 14 days. In areas with high clarithromycin resistance, bismuth quadruple therapy is preferred.

#67 · Primary Care Medicine

What is the diagnostic approach to a thyroid nodule found incidentally on imaging?

Answer

Thyroid ultrasound characterizes nodule features (size, echogenicity, calcifications, vascularity). Per ATA guidelines, fine needle aspiration (FNA) biopsy is recommended for nodules at or above 1 cm with suspicious features or at or above 1.5 cm with intermediate-risk features. TSH should also be checked to evaluate function.

#68 · Primary Care Medicine

What are the USPSTF recommendations for abdominal aortic aneurysm (AAA) screening?

Answer

USPSTF recommends one-time screening ultrasound for AAA in men aged 65-75 who have ever smoked (at least 100 cigarettes lifetime). No recommendation for women who never smoked. AAA is defined as aortic diameter at or above 3 cm; surgical repair is considered when diameter reaches 5.5 cm in men or 5.0 cm in women.

#69 · Primary Care Medicine

What is the definition of atrial fibrillation and what are the two main treatment strategies?

Answer

Atrial fibrillation is an irregularly irregular rhythm with absent P waves and variable rate, caused by chaotic atrial electrical activity. The two main strategies are rate control (beta-blockers or calcium channel blockers to keep resting heart rate below 80 bpm) and rhythm control (cardioversion or antiarrhythmics such as amiodarone or flecainide). Anticoagulation is addressed separately by CHA2DS2-VASc score.

#70 · Primary Care Medicine

What is the role of aspirin in primary prevention of cardiovascular disease?

Answer

USPSTF (2022) recommends against initiating low-dose aspirin for primary prevention of CVD in adults aged 60 and older. For adults aged 40-59 with 10-year CVD risk at or above 10%, the decision is individualized. Aspirin increases bleeding risk and this outweighs benefits in primary prevention for most patients.

#71 · Primary Care Medicine

What is the first-line treatment for acute gout flare in a patient without contraindications to NSAIDs?

Answer

NSAIDs such as indomethacin 50 mg three times daily or naproxen 500 mg twice daily are first-line for acute gout flares. Colchicine 1.2 mg followed by 0.6 mg one hour later is equally effective. Oral prednisone 30-40 mg daily tapered over 5-7 days is used when NSAIDs and colchicine are contraindicated.

#72 · Primary Care Medicine

What serum uric acid level triggers gout and what is the target uric acid level during prophylactic therapy?

Answer

Uric acid above 6.8 mg/dL exceeds the saturation point and promotes monosodium urate crystal deposition. The target during urate-lowering therapy (allopurinol or febuxostat) is below 6.0 mg/dL (below 5.0 mg/dL in tophaceous gout). Allopurinol is started at 100 mg daily and titrated; check uric acid every 2-4 weeks.

#73 · Primary Care Medicine

What is the most common type of dementia and what medications are used to treat mild-moderate stages?

Answer

Alzheimer's disease accounts for 60-80% of dementia cases. Donepezil (cholinesterase inhibitor) is used for mild to moderate Alzheimer's, starting at 5 mg daily and increasing to 10 mg after 4-6 weeks. Rivastigmine and galantamine are alternatives. Memantine (NMDA receptor antagonist) is added for moderate-severe disease.

#74 · Primary Care Medicine

What are the diagnostic criteria for major depressive disorder?

Answer

DSM-5 requires 5 or more symptoms for at least 2 weeks, including at least one of: depressed mood or anhedonia. Other symptoms: sleep disturbance, fatigue, worthlessness or guilt, concentration difficulty, appetite/weight change, psychomotor changes, recurrent thoughts of death or suicidality. Symptoms must cause functional impairment.

#75 · Primary Care Medicine

What screening tool is used for depression in primary care and what score indicates moderate depression?

Answer

The PHQ-9 (Patient Health Questionnaire-9) is the standard screening tool for depression in primary care. Score 0-4: minimal; 5-9: mild; 10-14: moderate; 15-19: moderately severe; 20-27: severe. A score of 10 or above warrants initiation of treatment. Question 9 (suicidal ideation) requires immediate safety assessment regardless of total score.

#76 · Primary Care Medicine

What is the role of the GAD-7 in primary care and what score suggests generalized anxiety disorder?

Answer

The GAD-7 (Generalized Anxiety Disorder 7-item scale) screens for anxiety disorders in primary care. Scores: 0-4 minimal, 5-9 mild, 10-14 moderate, 15-21 severe. A score of 10 or above suggests GAD and warrants further evaluation and treatment initiation (SSRIs/SNRIs are first-line).

#77 · Primary Care Medicine

What defines a complex febrile seizure vs. a simple febrile seizure?

Answer

Simple febrile seizure: generalized tonic-clonic, lasts below 15 minutes, occurs once in 24 hours in a child aged 6 months to 5 years. Complex febrile seizure: focal, lasts 15 minutes or longer, or recurs within 24 hours. Complex febrile seizures require EEG and neuroimaging. LP is considered if meningitis is suspected.

#78 · Primary Care Medicine

What is the definition of hypertensive crisis and what are examples of end-organ damage?

Answer

Hypertensive crisis is BP above 180/120 mmHg. End-organ damage in hypertensive emergency includes hypertensive encephalopathy, acute stroke, acute MI, acute heart failure, aortic dissection, acute renal failure (creatinine rising), and eclampsia. Retinal hemorrhages and papilledema on fundoscopy also confirm emergency.

#79 · Primary Care Medicine

What vaccines are recommended for all adults aged 65 and older?

Answer

Adults 65 and older should receive: annual influenza vaccine, COVID-19 (updated annually), Tdap (if not given as adult), pneumococcal (PCV21 or PCV20 once), and recombinant zoster vaccine (Shingrix) two doses at least 2-6 months apart. RSV vaccine (Abrysvo or mRESVIA, one dose) is routinely recommended for all adults 75 and older; for adults 60-74 it is given only through shared clinical decision-making for those at increased risk of severe RSV disease, not as a blanket recommendation for everyone 60 and older.

#80 · Primary Care Medicine

What is the USPSTF recommendation for cervical cancer screening?

Answer

USPSTF recommends: Pap smear alone every 3 years for women aged 21-29. For women aged 30-65: Pap smear alone every 3 years, or HPV test alone every 5 years, or Pap plus HPV co-test every 5 years. Screening stops at 65 if prior adequate negative screening. Women with HIV or immunosuppression need more frequent screening.

#81 · Primary Care Medicine

What is the standard diagnostic workup for a new diagnosis of type 2 diabetes at initial visit?

Answer

Initial workup includes: A1c, fasting lipid panel, comprehensive metabolic panel (creatinine, eGFR, liver function), urine albumin-to-creatinine ratio, TSH (if symptomatic), ECG, dilated eye exam, and monofilament foot exam. Blood pressure should be measured and treated to below 130/80 mmHg.

#82 · Primary Care Medicine

What is the mechanism of SGLT-2 inhibitors and what are their cardiovascular and renal benefits?

Answer

SGLT-2 inhibitors (empagliflozin, dapagliflozin, canagliflozin) block glucose reabsorption in the proximal tubule, causing glucosuria and lowering blood glucose. Proven benefits include reduced hospitalization for heart failure (both HFrEF and HFpEF), reduced CKD progression, weight loss, and modest BP reduction. Risk of genital mycotic infections and Fournier gangrene.

#83 · Primary Care Medicine

What is the mechanism of GLP-1 receptor agonists and what are their major clinical benefits?

Answer

GLP-1 agonists (semaglutide, liraglutide, dulaglutide) mimic incretin hormones: they stimulate glucose-dependent insulin secretion, suppress glucagon, slow gastric emptying, and promote satiety. Benefits include significant weight loss (up to 15-20% body weight with semaglutide), reduced ASCVD events, and reduced kidney disease progression.

#84 · Primary Care Medicine

What is the recommended treatment for a patient with type 2 diabetes and established ASCVD?

Answer

In patients with T2DM and established ASCVD, add a GLP-1 receptor agonist (liraglutide, semaglutide) or SGLT-2 inhibitor with proven CV benefit regardless of A1c. Metformin remains the foundational oral agent. Goal A1c is below 7% for most patients. High-intensity statin and BP control below 130/80 mmHg are also essential.

#85 · Primary Care Medicine

What are the classic signs and symptoms of pulmonary embolism?

Answer

Classic PE symptoms include pleuritic chest pain, dyspnea, tachycardia, and hypoxemia. Massive PE may cause hypotension and syncope. ECG may show sinus tachycardia, S1Q3T3 pattern, or new right bundle branch block. CT pulmonary angiography is the gold standard imaging test. D-dimer has high sensitivity to rule out PE in low-probability patients.

#86 · Primary Care Medicine

What is the Wells criteria for PE and what score allows safe D-dimer use to rule out PE?

Answer

Wells PE score: DVT signs (3 pts), no alternative diagnosis (3 pts), tachycardia above 100 (1.5 pts), immobilization or surgery in 4 weeks (1.5 pts), prior DVT/PE (1.5 pts), hemoptysis (1 pt), malignancy (1 pt). Score 0-1 (low probability) with negative D-dimer safely rules out PE without imaging.

#87 · Primary Care Medicine

What is the recommended prostate cancer screening approach per major guidelines?

Answer

USPSTF recommends individual decision-making about PSA screening for men aged 55-69 after discussing potential benefits and harms (overdiagnosis, false positives, biopsy complications). Screening is not recommended for men aged 70 and older. ACS recommends offering PSA at 50 (average risk), 45 (high risk), or 40 (highest risk, multiple first-degree relatives).

#88 · Primary Care Medicine

What is the pharmacologic treatment of choice for fibromyalgia?

Answer

FDA-approved treatments for fibromyalgia include duloxetine (SNRI, 60 mg daily), milnacipran (SNRI, 100-200 mg/day), and pregabalin (300-450 mg/day). Tricyclic antidepressants (amitriptyline 10-50 mg at bedtime) are also effective for pain and sleep. Exercise (aerobic and resistance training) is a cornerstone of non-pharmacologic management.

#89 · Primary Care Medicine

What defines chronic kidney disease and what are the two main markers?

Answer

CKD is defined as kidney abnormalities (structural or functional) persisting for more than 3 months. The two main markers are reduced GFR (below 60 mL/min/1.73m²) and markers of kidney damage such as albuminuria (urine albumin-to-creatinine ratio at or above 30 mg/g), abnormal urinalysis, or structural abnormalities on imaging.

#90 · Primary Care Medicine

What is the recommended HbA1c for an elderly patient (age 75+) with multiple comorbidities?

Answer

For older adults aged 75+ with multiple comorbidities, limited life expectancy, or functional decline, the ADA recommends a less stringent A1c target of below 8.0-8.5%. Strict glycemic control increases hypoglycemia risk in elderly patients, which can cause falls, fractures, and cardiovascular events. Simplifying regimens improves safety.

#91 · Primary Care Medicine

What is the approach to incidentally found adrenal mass (adrenal incidentaloma)?

Answer

An adrenal incidentaloma requires hormonal evaluation: 1 mg overnight dexamethasone suppression test (for cortisol excess), plasma free metanephrines (for pheochromocytoma), and aldosterone-to-renin ratio if hypertensive (for primary aldosteronism). CT characteristics (size above 4 cm, Hounsfield units above 10) raise malignancy concern requiring surgical referral.

#92 · Primary Care Medicine

What are the key features of Cushing syndrome and what is the best initial screening test?

Answer

Cushing syndrome presents with central obesity, purple striae, proximal muscle weakness, moon facies, buffalo hump, hypertension, hyperglycemia, and osteoporosis. The best initial screening tests are 24-hour urine free cortisol, late-night salivary cortisol (two measurements), or 1 mg overnight dexamethasone suppression test (cortisol above 1.8 mcg/dL = abnormal).

#93 · Primary Care Medicine

What are the characteristics of hypothyroidism-related dyslipidemia and how does it respond to treatment?

Answer

Hypothyroidism causes elevated LDL and total cholesterol due to decreased LDL receptor expression and reduced lipoprotein lipase activity. Treating hypothyroidism with levothyroxine often normalizes lipid levels. If lipids remain elevated after achieving euthyroid status, separate statin therapy may be needed per ASCVD risk assessment.

#94 · Primary Care Medicine

What is the recommended treatment for tobacco use disorder in primary care?

Answer

First-line pharmacotherapy includes varenicline (Chantix) 0.5 mg daily x3 days, then 0.5 mg BID x4 days, then 1 mg BID for 12 weeks (most effective), bupropion SR 150 mg daily x3 days then 150 mg BID for 7-12 weeks, and nicotine replacement therapy (patch, gum, lozenge). Combining NRT forms or NRT with varenicline increases efficacy. Behavioral counseling amplifies pharmacotherapy benefit.

#95 · Primary Care Medicine

What is the mechanism and major adverse effect of spironolactone as used in HFrEF?

Answer

Spironolactone is a mineralocorticoid receptor antagonist (MRA) that blocks aldosterone, reducing sodium retention and fibrosis in HFrEF. Major adverse effects include hyperkalemia (especially with ACEi/ARB; avoid if potassium above 5.0 mEq/L or creatinine above 2.5 mg/dL) and gynecomastia in men. Eplerenone causes less gynecomastia.

#96 · Primary Care Medicine

What is the difference between stable and unstable angina?

Answer

Stable angina occurs predictably with exertion, lasts below 20 minutes, and resolves with rest or nitroglycerin. Unstable angina (part of ACS) occurs at rest, with minimal exertion, or is new-onset severe, without troponin elevation. Unstable angina requires urgent evaluation with ECG, serial troponins, and antiplatelet therapy (aspirin + P2Y₁₂ inhibitor).

#97 · Primary Care Medicine

What is the USPSTF recommendation for depression screening in adults?

Answer

USPSTF recommends screening for depression in all adults aged 18 and older, including pregnant and postpartum women. The PHQ-2 is a common initial screener (asking about depressed mood and anhedonia over 2 weeks). A positive PHQ-2 (score at or above 3) should prompt full PHQ-9. Screening should be done when adequate follow-up and treatment are available.

#98 · Primary Care Medicine

What lab tests should be ordered when hypothyroidism is suspected?

Answer

TSH is the best initial screening test for thyroid dysfunction. If TSH is elevated (above 4.5 mIU/L), order free T4 to differentiate overt hypothyroidism (high TSH, low free T4) from subclinical hypothyroidism (high TSH, normal free T4). Thyroid peroxidase (TPO) antibodies confirm Hashimoto thyroiditis as the underlying cause.

#99 · Primary Care Medicine

What medications should be avoided in patients with a known history of angioedema from ACE inhibitor use?

Answer

Patients with ACE inhibitor-induced angioedema should not be rechallenged with any ACE inhibitor. ARBs are generally safe alternatives (cross-reactivity with angioedema is rare, approximately 0.1%). Sacubitril-valsartan (ARNI) should also be avoided due to neprilysin inhibition increasing bradykinin levels, similar to ACEi.

#100 · Primary Care Medicine

What is the recommended approach to isolated systolic hypertension in older adults?

Answer

Isolated systolic hypertension (systolic at or above 130, diastolic below 80) in adults over 60 is treated with thiazide diuretics (chlorthalidone) or dihydropyridine calcium channel blockers (amlodipine) as preferred agents. Target is below 130/80 mmHg per ACC/AHA. Treat cautiously to avoid orthostatic hypotension in frail elderly patients.

#101 · Specialty Systems

What are the diagnostic criteria for rheumatoid arthritis and what is first-line therapy?

Answer

RA diagnosis uses the 2010 ACR/EULAR criteria: joint involvement, serology (RF or anti-CCP), inflammatory markers (CRP or ESR), and symptom duration at or above 6 weeks. Morning stiffness lasting more than 1 hour is characteristic. Methotrexate (MTX) is first-line DMARD, started at 7.5-15 mg weekly and titrated to 20-25 mg weekly.

#102 · Specialty Systems

What labs should be monitored regularly in a patient taking methotrexate for RA?

Answer

Monitor CBC (for myelosuppression, particularly leukopenia), liver function tests (hepatotoxicity), and creatinine (MTX is renally cleared) every 4-8 weeks when starting and every 8-12 weeks when stable. Folic acid 1 mg daily is given to reduce MTX toxicity. MTX is contraindicated in pregnancy (teratogenic) and severe hepatic or renal disease.

#103 · Specialty Systems

What biologic agents are used in RA refractory to methotrexate and what class is most commonly used?

Answer

TNF-alpha inhibitors are the most commonly added biologic class when MTX fails: etanercept, adalimumab, infliximab, certolizumab, and golimumab. IL-6 inhibitors (tocilizumab, sarilumab), CD20 inhibitors (rituximab), and T-cell co-stimulation blockers (abatacept) are alternatives. Screen for latent TB before starting any biologic.

#104 · Specialty Systems

What are the 11 classification criteria for SLE and how many are needed for diagnosis?

Answer

The 1997 ACR criteria include: malar rash, discoid rash, photosensitivity, oral ulcers, arthritis, serositis, renal disorder, neurologic disorder, hematologic disorder (hemolytic anemia, leukopenia, thrombocytopenia), immunologic markers (anti-dsDNA, anti-Sm, antiphospholipid), and positive ANA. A patient must meet 4 of 11 criteria.

#105 · Specialty Systems

What medication is the foundation of SLE treatment and what is its major ocular toxicity?

Answer

Hydroxychloroquine (Plaquenil) 5 mg/kg/day is the cornerstone of SLE therapy, reducing flares, organ damage, and mortality. Major toxicity is chloroquine retinopathy (bull's-eye maculopathy), which can cause permanent vision loss. Annual ophthalmology exams are required after 5 years of use (or sooner in high-risk patients).

#106 · Specialty Systems

What laboratory abnormalities are associated with active systemic lupus erythematosus?

Answer

Active SLE often shows elevated anti-dsDNA antibodies and low complement levels (C3 and C4 are consumed during immune complex formation). CBC may show hemolytic anemia (positive Coombs), leukopenia (below 4,000/uL), and thrombocytopenia (below 100,000/uL). ESR is elevated but CRP is often normal in SLE (unless serositis or infection).

#107 · Specialty Systems

What is the first-line urate-lowering therapy for chronic gout and at what dose should it be started?

Answer

Allopurinol is first-line urate-lowering therapy. Start at 100 mg daily (50 mg daily if CKD stage 3 or above) and titrate every 2-4 weeks to achieve uric acid below 6.0 mg/dL. Maximum dose is 800 mg daily. Febuxostat 40-80 mg daily is an alternative but carries a black box warning for cardiovascular death risk.

#108 · Specialty Systems

What is the T-score threshold for osteoporosis diagnosis and what is the first-line pharmacologic treatment?

Answer

Osteoporosis is diagnosed by DEXA T-score at or below -2.5 at the lumbar spine, femoral neck, or total hip. Osteopenia is T-score -1.0 to -2.5. Bisphosphonates (alendronate 70 mg weekly, risedronate 35 mg weekly, zoledronic acid 5 mg IV annually) are first-line pharmacotherapy. Calcium 1,200 mg/day and vitamin D 800-1,000 IU/day are also recommended.

#109 · Specialty Systems

What does the FRAX tool calculate and when is pharmacotherapy initiated based on FRAX?

Answer

FRAX calculates the 10-year probability of major osteoporotic fracture (hip, spine, humerus, forearm). Pharmacotherapy is recommended when FRAX predicts a 10-year hip fracture probability at or above 3% or major osteoporotic fracture probability at or above 20%. FRAX is used in patients with osteopenia (T-score -1.0 to -2.5) to guide treatment decisions.

#110 · Specialty Systems

What are the antibody findings and clinical features of Hashimoto thyroiditis?

Answer

Hashimoto thyroiditis shows positive thyroid peroxidase (TPO) antibodies in over 90% of patients and positive anti-thyroglobulin antibodies in about 60%. It is the most common cause of hypothyroidism in iodine-sufficient regions. The thyroid may be goitrous (enlarged and firm) or atrophic. Treat with levothyroxine when TSH exceeds 10 mIU/L or when symptomatic.

#111 · Specialty Systems

What are the distinguishing features of Graves disease and what is the first-line medical treatment?

Answer

Graves disease features hyperthyroidism with diffuse goiter, exophthalmos (proptosis), pretibial myxedema, and thyroid bruit. TSH receptor antibodies (TRAb or TSI) are positive. Methimazole is first-line antithyroid drug (preferred over propylthiouracil due to lower hepatotoxicity risk), starting at 10-30 mg daily. PTU is preferred in first trimester of pregnancy.

#112 · Specialty Systems

What are the three treatment options for Graves disease hyperthyroidism?

Answer

The three options are: (1) Antithyroid drugs (methimazole) for 12-18 months with a 30-40% remission rate. (2) Radioactive iodine (I-131) ablation, the most common definitive treatment in the US; leads to hypothyroidism requiring levothyroxine. (3) Thyroidectomy for large goiters, compressive symptoms, or pregnancy planning. All lead to frequent hypothyroidism.

#113 · Specialty Systems

What are the first-line pharmacologic treatments for diabetic peripheral neuropathy pain?

Answer

First-line agents for painful diabetic neuropathy include duloxetine (SNRI, 60-120 mg daily), pregabalin (150-600 mg daily in divided doses), and gabapentin (titrated to 1,800-3,600 mg daily in divided doses). Tricyclic antidepressants (amitriptyline, nortriptyline) are also effective. Topical capsaicin or lidocaine patches can be used adjunctively.

#114 · Specialty Systems

What blood pressure target and drug class are recommended for diabetic nephropathy?

Answer

Target BP is below 130/80 mmHg in diabetic patients with nephropathy. ACE inhibitors or ARBs are first-line because they reduce intraglomerular pressure and proteinuria beyond their antihypertensive effect. Do not combine ACEi and ARB. SGLT-2 inhibitors (dapagliflozin, canagliflozin) are added for additional renoprotection when eGFR is at or above 20 mL/min/1.73m².

#115 · Specialty Systems

What are the key differences between Crohn disease and ulcerative colitis?

Answer

Crohn disease: transmural inflammation, skip lesions, affects any GI tract (mouth to anus), cobblestone mucosa, fistulas and strictures common, granulomas on biopsy. Ulcerative colitis: mucosal-only inflammation, continuous disease starting from the rectum, limited to the colon, pseudopolyps, no fistulas. Crohn has perianal disease; UC does not.

#116 · Specialty Systems

What extraintestinal manifestations are associated with inflammatory bowel disease?

Answer

IBD extraintestinal manifestations include: eyes (uveitis, episcleritis), skin (erythema nodosum, pyoderma gangrenosum), joints (peripheral arthritis, axial spondyloarthritis), liver (primary sclerosing cholangitis, more common in UC), and kidney (nephrolithiasis). PSC with UC carries high risk of cholangiocarcinoma.

#117 · Specialty Systems

What is the hepatitis B serology interpretation for each antigen and antibody?

Answer

HBsAg positive: currently infected. Anti-HBs positive: immune (vaccination or recovery). HBeAg positive: high viral replication and infectivity. Anti-HBe positive: lower infectivity (resolving). Anti-HBc IgM positive: acute hepatitis B. Anti-HBc IgG positive: prior exposure (resolved or chronic). HBV DNA is used to quantify viral load.

#118 · Specialty Systems

When is treatment indicated for chronic hepatitis B and what are the first-line antiviral options?

Answer

Treat chronic HBV when HBV DNA is above 2,000 IU/mL and ALT is elevated, or HBV DNA is above 20,000 IU/mL regardless of ALT, or in advanced fibrosis/cirrhosis. First-line oral antivirals are tenofovir disoproxil fumarate (TDF) 300 mg daily or entecavir 0.5 mg daily (1 mg if decompensated). Avoid nephrotoxins with TDF.

#119 · Specialty Systems

What is the time window for tPA administration in acute ischemic stroke and what are absolute contraindications?

Answer

Alteplase (tPA) can be given within 4.5 hours of symptom onset at 0.9 mg/kg (max 90 mg, 10% as IV bolus, remainder over 60 min). Absolute contraindications include intracranial hemorrhage, recent surgery within 14 days, active internal bleeding, current anticoagulation with INR above 1.7, platelets below 100,000, and systolic BP above 185 or diastolic above 110 mmHg (treat first).

#120 · Specialty Systems

What is the ABCD2 score for TIA and what does a high score indicate?

Answer

ABCD2 predicts 2-day stroke risk after TIA: Age at or above 60 (1 pt), BP at or above 140/90 (1 pt), Clinical features: unilateral weakness (2 pts) or speech impairment without weakness (1 pt), Duration at or above 60 min (2 pts) or 10-59 min (1 pt), Diabetes (1 pt). Score 6-7 = high risk (8.1% 2-day stroke risk); admit for urgent workup.

#121 · Specialty Systems

What are the classic motor features of Parkinson disease and what is the first-line pharmacotherapy?

Answer

The cardinal features of Parkinson disease are resting tremor (pill-rolling at 4-6 Hz), rigidity (cogwheel), bradykinesia, and postural instability. Carbidopa-levodopa is the most effective pharmacotherapy and is first-line when functional impairment occurs. Dopamine agonists (pramipexole, ropinirole) are alternatives in younger patients to delay levodopa-related dyskinesias.

#122 · Specialty Systems

What is the DLB (dementia with Lewy bodies) and what medication must be avoided?

Answer

DLB presents with fluctuating cognition, recurrent visual hallucinations, parkinsonism, and REM sleep behavior disorder. It is the second most common neurodegenerative dementia after Alzheimer's. Antipsychotics (especially typical agents like haloperidol) must be avoided because DLB patients have severe neuroleptic sensitivity reactions (rigidity, fever, death).

#123 · Specialty Systems

What are the migraine acute treatment options and in what patient populations are triptans contraindicated?

Answer

Triptans (sumatriptan 50-100 mg oral or 6 mg subcutaneous, rizatriptan 10 mg) are first-line for moderate-to-severe migraine without contraindications. Triptans are contraindicated in uncontrolled hypertension, CAD, prior MI or stroke, TIA, and hemiplegic or basilar migraine. Lasmiditan and CGRP antagonists (ubrogepant, rimegepant) are alternatives for those with cardiovascular contraindications.

#124 · Specialty Systems

What are the first-line migraine preventive medications?

Answer

Evidence-based migraine preventives include: beta-blockers (propranolol 40-240 mg/day, timolol, metoprolol), antiepileptics (topiramate 50-100 mg/day, valproate), tricyclic antidepressants (amitriptyline 10-75 mg at bedtime), and SNRIs (venlafaxine). CGRP monoclonal antibodies (erenumab, fremanezumab, galcanezumab) are newer agents for chronic migraine.

#125 · Specialty Systems

What are the diagnostic criteria for multiple sclerosis?

Answer

MS requires dissemination in space (lesions in 2+ CNS regions) and dissemination in time (2+ separate attacks or MRI evidence of new lesion activity). MRI brain and spine with gadolinium is the primary diagnostic tool. CSF analysis showing oligoclonal bands and elevated IgG index supports the diagnosis. Evoked potentials detect subclinical lesions.

#126 · Specialty Systems

What is the most common cause of secondary hypertension in young women and how is it diagnosed?

Answer

Fibromuscular dysplasia (FMD) is the most common cause of secondary hypertension in young women, causing renal artery stenosis. It presents as difficult-to-control HTN with an abdominal bruit. Diagnosis is by CT angiography or MR angiography showing a string-of-beads appearance. Percutaneous transluminal angioplasty is the preferred treatment.

#127 · Specialty Systems

What is primary aldosteronism and how is it diagnosed?

Answer

Primary aldosteronism (Conn syndrome) is the most common secondary cause of hypertension, caused by autonomous aldosterone secretion. Suspect in resistant HTN or spontaneous hypokalemia with HTN. Screen with aldosterone-to-renin ratio (ARR): ratio above 30 with aldosterone above 15 ng/dL is positive. Adrenal CT and adrenal vein sampling confirm lateralization.

#128 · Specialty Systems

What are the presenting features of pheochromocytoma and how is it diagnosed?

Answer

Pheochromocytoma presents with episodic hypertension, palpitations, diaphoresis, and headache (the classic triad). Sustained or paroxysmal HTN is common. Diagnosis is by 24-hour urine catecholamines and metanephrines or plasma free metanephrines (most sensitive). CT or MRI adrenals is used for localization. Alpha-blockade before surgery is mandatory.

#129 · Specialty Systems

What is giant cell arteritis (GCA) and what is the major feared complication?

Answer

GCA (temporal arteritis) is a large-vessel vasculitis affecting adults over 50, predominantly women. Classic features include new headache, temporal artery tenderness, jaw claudication, and elevated ESR above 50 mm/hr and CRP. The major feared complication is anterior ischemic optic neuropathy causing permanent blindness. High-dose prednisone (40-60 mg/day) must be started immediately without waiting for biopsy results.

#130 · Specialty Systems

What is the diagnostic test for giant cell arteritis and what lab values support the diagnosis?

Answer

Temporal artery biopsy is the gold standard for GCA diagnosis. A segment at least 1.5 cm long should be excised due to skip lesions. ESR is typically above 50 mm/hr (often above 100), CRP is elevated, and there may be thrombocytosis and normocytic anemia. Start high-dose steroids immediately if GCA is clinically suspected, before biopsy results.

#131 · Specialty Systems

What is polymyalgia rheumatica and how does it relate to giant cell arteritis?

Answer

Polymyalgia rheumatica (PMR) causes proximal muscle aching and stiffness (shoulders, hips, neck) with elevated ESR/CRP but without muscle weakness or elevated creatine kinase. It occurs in adults over 50. Up to 50% of GCA patients have PMR, and 10-15% of PMR patients develop GCA. PMR is treated with lower-dose prednisone (15-20 mg/day).

#132 · Specialty Systems

What is the clinical presentation and treatment of Bell palsy?

Answer

Bell palsy is idiopathic unilateral lower motor neuron facial nerve (CN VII) palsy presenting with sudden-onset facial drooping, inability to close the eye, and loss of forehead wrinkling. Upper motor neuron causes (stroke) spare the forehead. Treatment within 72 hours of onset: prednisone 60-80 mg daily x5 days then tapered, and acyclovir or valacyclovir may be added. Eye care prevents corneal injury.

#133 · Specialty Systems

What is the triad of normal pressure hydrocephalus (NPH) and what is the treatment?

Answer

NPH (Hakim triad) presents with the classic triad of: magnetic gait (wide-based, slow, shuffling), urinary incontinence, and cognitive impairment (dementia). CT or MRI shows enlarged ventricles out of proportion to cortical atrophy. Ventriculoperitoneal (VP) shunt placement is the treatment; a positive response to large-volume LP (30-50 mL removed) predicts shunt response.

#134 · Specialty Systems

What is the mechanism and clinical use of colchicine in gout management?

Answer

Colchicine inhibits microtubule polymerization, blocking neutrophil migration and activation triggered by monosodium urate crystals. For acute gout: 1.2 mg followed by 0.6 mg one hour later. For prophylaxis during urate-lowering therapy initiation: 0.6 mg once or twice daily for 3-6 months. Major toxicities are GI (diarrhea, nausea) and, rarely, myopathy. Reduce dose in CKD; use with caution with CYP3A4 inhibitors.

#135 · Specialty Systems

What labs and imaging are used to evaluate cirrhosis and hepatic decompensation?

Answer

Initial labs: LFTs (AST, ALT, bilirubin, alkaline phosphatase), CBC (thrombocytopenia from hypersplenism), coagulation (INR elevated in cirrhosis), albumin, and ammonia. MELD-Na score (bilirubin, INR, creatinine, sodium) predicts 90-day mortality and guides liver transplant listing. Upper endoscopy screens for varices. Ultrasound plus AFP every 6 months screens for hepatocellular carcinoma.

#136 · Specialty Systems

What is the Child-Pugh classification used for in liver disease?

Answer

Child-Pugh scores liver function using 5 parameters: total bilirubin, serum albumin, PT/INR, ascites, and hepatic encephalopathy. Class A (5-6 pts) = compensated cirrhosis, Class B (7-9 pts) = significant impairment, Class C (10-15 pts) = decompensated cirrhosis. It guides surgical risk assessment and treatment eligibility decisions.

#137 · Specialty Systems

What are the major complications of cirrhosis and their first-line treatments?

Answer

Ascites: sodium restriction plus spironolactone 100 mg ($\pm$ furosemide 40 mg) daily, titrate up. Spontaneous bacterial peritonitis (SBP): cefotaxime IV 2 g q8h empirically; norfloxacin or TMP-SMX prophylaxis. Hepatic encephalopathy: lactulose 30-45 mL BID-TID titrated to 2-3 soft stools daily; rifaximin 550 mg BID added for recurrent episodes. Esophageal varices: nonselective beta-blocker (propranolol or nadolol) for prophylaxis.

#138 · Specialty Systems

What is celiac disease and what diagnostic tests confirm the diagnosis?

Answer

Celiac disease is an immune-mediated enteropathy triggered by gluten (wheat, barley, rye) in genetically susceptible individuals (HLA-DQ2 and DQ8). Serology: tissue transglutaminase IgA (TTG-IgA) is most sensitive and specific (check total IgA level simultaneously as IgA deficiency causes false negatives). Upper endoscopy with duodenal biopsy showing villous atrophy confirms diagnosis.

#139 · Specialty Systems

What are the management principles for a first unprovoked seizure in an adult?

Answer

After a first unprovoked seizure, obtain MRI brain, EEG, basic labs (glucose, electrolytes, CBC), and toxicology screen. Immediate antiepileptic drug (AED) therapy is recommended if EEG shows epileptiform activity, MRI shows a structural lesion, or nocturnal seizure. If these are absent, treatment can be deferred. Patients must not drive for at least 6 months (varies by state).

#140 · Specialty Systems

What is the acute management of status epilepticus?

Answer

Benzodiazepines are first-line: lorazepam 0.1 mg/kg IV (max 4 mg per dose) or diazepam 10 mg IV. If seizure continues after 5-10 minutes, repeat benzodiazepine. Second-line: levetiracetam 60 mg/kg IV (max 4,500 mg), valproate 40 mg/kg IV, or fosphenytoin 20 PE/kg IV. Refractory status requires intubation and propofol or midazolam infusion.

#141 · Specialty Systems

What are the five cardinal features of nephrotic syndrome?

Answer

Nephrotic syndrome: proteinuria above 3.5 g/day (or urine protein-to-creatinine ratio above 3.5), hypoalbuminemia (below 3.5 g/dL), edema (periorbital and peripheral), hyperlipidemia (elevated LDL and triglycerides), and lipiduria (oval fat bodies, fatty casts). The most common cause in adults is focal segmental glomerulosclerosis (FSGS) and membranous nephropathy.

#142 · Specialty Systems

What features distinguish nephritic syndrome from nephrotic syndrome?

Answer

Nephritic syndrome: hematuria with RBC casts, hypertension, oliguria, mild proteinuria (usually below 3.5 g/day), and azotemia. Caused by IgA nephropathy (most common globally), post-streptococcal GN, lupus nephritis, and ANCA vasculitis. Nephrotic: massive proteinuria, edema, hypoalbuminemia, no hematuria. The two syndromes can overlap.

#143 · Specialty Systems

What is the most common cause of acute kidney injury in hospitalized patients?

Answer

Prerenal AKI (volume depletion or reduced perfusion) is the most common cause of hospital-acquired AKI (55-60%). It is distinguished from intrinsic renal causes by FENa below 1%, BUN-to-creatinine ratio above 20:1, and response to IV fluid resuscitation. ATN (acute tubular necrosis) from ischemia or nephrotoxins (aminoglycosides, contrast) is the most common intrinsic cause.

#144 · Specialty Systems

What are the common signs and symptoms of hyperkalemia and what is the immediate treatment for cardiac toxicity?

Answer

Hyperkalemia (K^+ above 5.5 mEq/L) causes muscle weakness, paresthesias, and ECG changes: peaked T waves (earliest), prolonged PR, wide QRS, then sine wave pattern, and ventricular fibrillation. Immediate treatment for cardiac toxicity (K^+ above 6.5 or ECG changes): IV calcium gluconate (membrane stabilizer) 1-2 g over 10 min. Then insulin + dextrose (shifts K^+ intracellularly) and sodium bicarbonate if acidotic.

#145 · Specialty Systems

What are the most common causes of hyponatremia and how is it classified?

Answer

Hyponatremia (Na below 135 mEq/L) is classified by volume status: Hypovolemic (vomiting, diarrhea, diuretics), Euvolemic (SIADH most common, hypothyroidism, adrenal insufficiency), and Hypervolemic (heart failure, cirrhosis, nephrotic syndrome). SIADH criteria: urine osmolality above 100, urine Na above 40, euvolemia, normal adrenal and thyroid function.

#146 · Specialty Systems

What is the correction rate for severe symptomatic hyponatremia and why must overcorrection be avoided?

Answer

Correct severe symptomatic hyponatremia (seizures, coma) with 3% hypertonic saline at 1-2 mL/kg/hr. Target sodium correction: 4-6 mEq/L in the first hour, then no more than 8-10 mEq/L in the first 24 hours. Overcorrection risks osmotic demyelination syndrome (central pontine myelinolysis), causing irreversible quadriplegia and locked-in syndrome.

#147 · Specialty Systems

What is SIADH and what are common causes?

Answer

SIADH (syndrome of inappropriate antidiuretic hormone secretion) causes euvolemic hyponatremia with inappropriately concentrated urine. Common causes include CNS disorders (stroke, meningitis, SAH), pulmonary disease (pneumonia, TB, COPD), malignancy (small cell lung cancer is most common), and medications (SSRIs, carbamazepine, cyclophosphamide, vincristine). Treat with fluid restriction (800-1,000 mL/day).

#148 · Specialty Systems

What is the presentation and management of acute pancreatitis?

Answer

Acute pancreatitis presents with epigastric pain radiating to the back, nausea, vomiting, and elevated lipase (above 3x normal is diagnostic). Most common causes: gallstones (most common) and alcohol. Ranson criteria and BISAP score assess severity. Management: aggressive IV hydration (lactated Ringer preferred), NPO initially, pain control (IV ketorolac or opioids), early enteral nutrition in severe cases. Cholecystectomy planned after recovery for gallstone pancreatitis.

#149 · Specialty Systems

What is the clinical presentation of aortic dissection and what is the immediate management?

Answer

Aortic dissection presents with sudden-onset tearing or ripping chest or back pain that may radiate to the abdomen or legs, pulse differentials between arms, and signs of malperfusion. Type A (ascending) is a surgical emergency; Type B (descending) is typically managed medically. Immediate management: IV labetalol or esmolol to target HR below 60 and SBP 100-120 mmHg. Emergent CT angiography confirms diagnosis.

#150 · Specialty Systems

What is the classic presentation and treatment of bacterial meningitis in an adult?

Answer

Bacterial meningitis presents with fever, severe headache, neck stiffness (meningismus), and photophobia. Kernig and Brudzinski signs may be present. Do not delay antibiotics for LP if papilledema or focal neuro deficits present (do LP after CT). Empiric treatment: dexamethasone 0.15 mg/kg IV (before antibiotics) + ceftriaxone 2 g IV q12h + vancomycin 15-20 mg/kg IV q8-12h + ampicillin 2 g q4h if immunocompromised (covers *Listeria*).

#151 · Specialty Systems

What is the presentation and treatment of temporal lobe epilepsy (complex partial seizures)?

Answer

Temporal lobe epilepsy presents with aura (epigastric rising sensation, déjà vu, or olfactory hallucination), followed by altered awareness, automatisms (lip smacking, hand movements), and postictal confusion. EEG shows temporal spikes. First-line AEDs are carbamazepine or levetiracetam. Drug-resistant cases may benefit from temporal lobectomy with 60-70% seizure freedom rates.

#152 · Specialty Systems

What is Addison disease and what lab pattern is diagnostic?

Answer

Addison disease (primary adrenal insufficiency) presents with fatigue, weight loss, hyperpigmentation, hypotension, and salt craving. Lab findings: hyponatremia, hyperkalemia, low cortisol, and elevated ACTH. The standard diagnostic test is the ACTH stimulation test: cortisol below 18 mcg/dL at 60 minutes after 250 mcg cosyntropin injection. Treatment: hydrocortisone 15-20 mg/day plus fludrocortisone 0.1 mg/day.

#153 · Specialty Systems

What is the difference between primary and secondary adrenal insufficiency?

Answer

Primary adrenal insufficiency (Addison disease): destruction of adrenal cortex (autoimmune most common), causing low cortisol AND low aldosterone, with high ACTH (hyperpigmentation). Secondary: pituitary failure (low ACTH), causing low cortisol only (aldosterone preserved, no hyperpigmentation, no hyperkalemia). Most common cause of secondary is prolonged exogenous glucocorticoid use.

#154 · Specialty Systems

What is the diagnostic criteria for diabetes insipidus and how are central and nephrogenic types differentiated?

Answer

Diabetes insipidus: urine osmolality below 300 mOsm/kg with polyuria (above 3 L/day) and polydipsia. Water deprivation test confirms DI. Differentiate with desmopressin (DDAVP): central DI responds (urine osmolality increases above 50%); nephrogenic DI does not respond. Central DI (pituitary/hypothalamic damage) is treated with intranasal desmopressin.

#155 · Specialty Systems

What is sarcoidosis and what are its common pulmonary and extrapulmonary manifestations?

Answer

Sarcoidosis is a granulomatous disease of unknown etiology most commonly affecting Black adults aged 20-40. Pulmonary: bilateral hilar lymphadenopathy (classic chest X-ray finding), interstitial lung disease.

Extrapulmonary: uveitis, erythema nodosum, lupus pernio, hypercalcemia, cranial nerve palsies (CN VII most common), and cardiac arrhythmias. Diagnosis by tissue biopsy showing non-caseating granulomas.

#156 · Specialty Systems

What are the first-line treatments for active TB and what is the standard treatment regimen?

Answer

Active TB is treated with RIPE therapy: Rifampin, Isoniazid, Pyrazinamide, and Ethambutol for 2 months (intensive phase), followed by Rifampin and Isoniazid alone for 4 more months (continuation phase), totaling 6 months. Pyridoxine (B6) 25-50 mg daily is given with isoniazid to prevent peripheral neuropathy. Test for HIV and drug resistance before and during treatment.

#157 · Specialty Systems

What is latent TB infection (LTBI) and what are the treatment options?

Answer

LTBI is a positive TST or IGRA without active disease (no symptoms, normal CXR). Screen high-risk groups: immigrants from endemic regions, HIV-positive, close TB contacts, and immunocompromised patients. Treatment options: isoniazid 300 mg daily x9 months; rifampin 600 mg daily x4 months; or 3HP (weekly rifapentine + isoniazid x12 weeks). Treat to prevent reactivation.

#158 · Specialty Systems

What is the gold standard diagnostic test for pulmonary embolism and what are alternatives?

Answer

CT pulmonary angiography (CTPA) is the gold standard for PE diagnosis, with sensitivity above 95% and specificity above 96%. V/Q scan is an alternative when CTPA is contraindicated (contrast allergy, renal failure, pregnancy). Echocardiography may show right heart strain in massive PE. Lower extremity venous ultrasound is used when CTPA cannot be performed and DVT would confirm need for anticoagulation.

#159 · Specialty Systems

What is the presentation and management of pneumothorax?

Answer

Spontaneous pneumothorax presents with sudden pleuritic chest pain and dyspnea. Primary spontaneous pneumothorax (no underlying lung disease) occurs in tall, thin young males. Small (below 2 cm apex-to-cupola) and asymptomatic: observe with oxygen. Large or symptomatic: needle aspiration or chest tube insertion. Tension pneumothorax (tracheal deviation, hemodynamic instability) requires immediate needle decompression at the 2nd intercostal space, midclavicular line.

#160 · Specialty Systems

What is the definition and management of community-acquired pneumonia requiring hospitalization?

Answer

Hospitalized non-ICU CAP: beta-lactam (ampicillin-sulbactam 1.5-3 g q6h or ceftriaxone 1-2 g daily) PLUS a respiratory fluoroquinolone (levofloxacin 750 mg daily) or macrolide (azithromycin 500 mg daily). ICU CAP: beta-lactam plus a respiratory fluoroquinolone. If *Pseudomonas* risk, use antipseudomonal beta-lactam (piperacillin-tazobactam). Treat 5-7 days if good response.

#161 · Specialty Systems

What are the features of atypical (walking) pneumonia and common causative organisms?

Answer

Atypical pneumonia presents with gradual onset, dry cough, low-grade fever, headache, and myalgias (patient feels ill but can still walk). Chest X-ray shows interstitial or bilateral patchy infiltrates. Common organisms: *Mycoplasma pneumoniae* (most common in young adults), *Chlamydia pneumoniae*, and *Legionella pneumophila* (associated with water exposure and severe disease). Treat with macrolides, doxycycline, or fluoroquinolones.

#162 · Specialty Systems

What is the definition of heart failure with preserved ejection fraction (HFpEF) and what causes it?

Answer

HFpEF is EF at or above 50% with symptoms of heart failure (dyspnea, edema) due to diastolic dysfunction (impaired ventricular relaxation and increased stiffness). Common causes: hypertension (most common), obesity, atrial fibrillation, diabetes, and aging. Diuretics relieve congestion; treat underlying comorbidities. Empagliflozin and spironolactone reduce hospitalizations in HFpEF.

What is interstitial lung disease (ILD) and what are the most common subtypes?

Answer

ILD encompasses a heterogeneous group of diffuse parenchymal lung diseases causing fibrosis and restrictive physiology. Most common subtypes: Idiopathic pulmonary fibrosis (IPF, bibasilar honeycombing on HRCT), hypersensitivity pneumonitis (antigen exposure history), sarcoidosis, and connective tissue disease-ILD (RA-ILD, SSc-ILD). HRCT of chest is the key diagnostic imaging. PFTs show reduced FVC and DLCO.

What is the treatment for idiopathic pulmonary fibrosis (IPF)?

Answer

IPF is treated with antifibrotic agents: pirfenidone (801 mg three times daily) or nintedanib (150 mg twice daily), which slow progression but do not reverse fibrosis. Lung transplantation is the only curative option. Supplemental oxygen for hypoxemia. Avoid steroids (not effective and worsen outcomes in IPF). Median survival without treatment is 3-5 years from diagnosis.

#165 · Specialty Systems

What is systemic sclerosis (scleroderma) and how are limited vs. diffuse subtypes differentiated?

Answer

Systemic sclerosis causes fibrosis of skin and internal organs. Limited cutaneous SSc (CREST): Calcinosis, Raynaud phenomenon, Esophageal dysmotility, Sclerodactyly, Telangiectasias; anti-centromere antibodies; lower risk of internal organ involvement. Diffuse cutaneous SSc: proximal skin involvement, anti-topoisomerase I (Scl-70) antibodies; higher risk of interstitial lung disease, pulmonary hypertension, and scleroderma renal crisis.

#166 · Specialty Systems

What is the classic triad of antiphospholipid syndrome (APS) and what is the diagnostic test?

Answer

APS (Hughes syndrome) triad: arterial or venous thrombosis, pregnancy morbidity (recurrent miscarriages, fetal loss), and positive antiphospholipid antibodies. Diagnostic antibodies: lupus anticoagulant, anti-cardiolipin IgG/IgM, and anti-beta2-glycoprotein I IgG/IgM, confirmed on two occasions at least 12 weeks apart. Treatment: long-term anticoagulation with warfarin (target INR 2-3 for venous, 2.5-3.5 for arterial).

#167 · Specialty Systems

What is the most common primary bone tumor in adults and what is the most common bone cancer overall?

Answer

Multiple myeloma is the most common primary bone tumor (malignant plasma cell neoplasm) in adults. Metastatic disease is by far the most common cancer to involve bone. Classic myeloma features: CRAB (hyperCalcemia, Renal failure, Anemia, Bone lesions on imaging). Serum protein electrophoresis (SPEP) shows M-protein spike; urine shows Bence Jones proteins.

#168 · Specialty Systems

What is the most common cause of immune thrombocytopenic purpura (ITP) and how is it treated?

Answer

ITP is caused by autoantibodies (typically IgG anti-GPIIb/IIIa) that destroy platelets. It is a diagnosis of exclusion. In adults, treat when platelets are below 30,000/uL or there is significant bleeding. First-line: prednisone 1 mg/kg/day for 2-4 weeks plus IVIG 1 g/kg if rapid response needed. Splenectomy or TPO receptor agonists (eltrombopag, romiplostim) for refractory cases.

#169 · Specialty Systems

What are the diagnostic criteria for polycythemia vera and what is its management?

Answer

PV is a myeloproliferative neoplasm driven by JAK2 V617F mutation (present in above 95%). Diagnostic criteria: hemoglobin above 16.5 g/dL (men) or 16 g/dL (women) or hematocrit above 49%/48%, plus bone marrow biopsy showing panmyelosis, plus low EPO level. Management: phlebotomy to target hematocrit below 45%, low-dose aspirin 81 mg daily, and hydroxyurea for high-risk patients (age above 60 or prior thrombosis).

#170 · Specialty Systems

What are the classic features of vitamin B12 deficiency and what lab findings confirm it?

Answer

B12 deficiency causes macrocytic anemia (MCV above 100 fL), hypersegmented neutrophils on peripheral smear, and neurologic manifestations including subacute combined degeneration (posterior and lateral column involvement) causing paresthesias, gait ataxia, and cognitive changes. Lab: low serum B12, elevated methylmalonic acid and homocysteine. Treat with IM cyanocobalamin 1,000 mcg daily x7 days, then weekly x4, then monthly.

#171 · Specialty Systems

What is the presentation and emergency management of anaphylaxis?

Answer

Anaphylaxis presents within minutes of allergen exposure with urticaria, angioedema, bronchospasm (wheezing), hypotension, and tachycardia. Immediate treatment: epinephrine 0.3-0.5 mg IM (1:1000) in the anterolateral thigh, repeat every 5-15 minutes if needed. Secondary treatments: diphenhydramine, ranitidine, methylprednisolone 125 mg IV. Observe 4-6 hours minimum for biphasic reactions. Prescribe epinephrine auto-injector (EpiPen) upon discharge.

#172 · Specialty Systems

What is the presentation and treatment of acute urticaria?

Answer

Acute urticaria (below 6 weeks) presents with pruritic wheals (hives) that are transient (individual lesions last below 24 hours) and blanching. Most cases are idiopathic or IgE-mediated (food, medication, insect). First-line treatment: non-sedating antihistamines (cetirizine 10 mg or loratadine 10 mg daily). Add H2 blocker (famotidine) for refractory cases. Short-course prednisone for severe episodes. Evaluate for systemic lupus, infection, or thyroid disease in chronic urticaria.

#173 · Specialty Systems

What is the classification and first-line treatment for *Helicobacter pylori* eradication?

Answer

First-line *H. pylori* eradication in areas with low clarithromycin resistance: clarithromycin triple therapy (PPI + clarithromycin 500 mg + amoxicillin 1 g, all BID x14 days). In areas with high clarithromycin resistance or penicillin allergy: bismuth quadruple therapy (PPI + bismuth 524 mg + tetracycline 500 mg + metronidazole 500 mg, all four times daily x10-14 days). Confirm eradication with UBT 4 weeks after treatment.

#174 · Specialty Systems

What is the presentation and workup of a first episode of acute pericarditis?

Answer

Acute pericarditis presents with sharp pleuritic chest pain worse when supine and relieved when leaning forward, pericardial friction rub on exam, diffuse saddle-shaped ST elevation with PR depression on ECG. Workup: ECG, echo (to rule out pericardial effusion and tamponade), CBC, CRP, ESR, troponin (to detect myopericarditis). First-line treatment: aspirin 750-1000 mg TID or ibuprofen 600 mg TID for 2 weeks plus colchicine 0.5 mg BID for 3 months to prevent recurrence.

#175 · Specialty Systems

What is cardiac tamponade and what is the classic physical exam finding (Beck triad)?

Answer

Cardiac tamponade is compression of the heart by pericardial fluid causing impaired diastolic filling and obstructive shock. Beck triad: hypotension, elevated JVP (distended neck veins), and muffled heart sounds. Pulsus paradoxus (decrease in SBP above 10 mmHg with inspiration) is a key finding. ECG shows electrical alternans and low voltage. Immediate pericardiocentesis is life-saving.

#176 · Specialty Systems

What is the pathophysiology and diagnosis of hereditary hemochromatosis?

Answer

Hereditary hemochromatosis is an autosomal recessive disorder (HFE gene mutation, C282Y most common) causing excessive intestinal iron absorption and organ deposition. Classic triad: cirrhosis, diabetes mellitus, and skin bronzing ('bronze diabetes'). Arthropathy and hypogonadism also occur. Diagnosis: elevated transferrin saturation above 45% and elevated ferritin. Genetic testing confirms HFE mutation. Treatment: phlebotomy weekly until ferritin is 50-100 ng/mL.

#177 · Specialty Systems

What are the common drug-drug interactions with warfarin that increase bleeding risk?

Answer

Warfarin (a vitamin K antagonist) has numerous interactions that elevate INR and bleeding risk: antibiotics (fluoroquinolones, metronidazole, azole antifungals, sulfonamides) inhibit CYP2C9 or reduce gut bacteria producing vitamin K. NSAIDs increase GI bleeding risk. Amiodarone, fluconazole, and omeprazole significantly inhibit warfarin metabolism. Vitamin K-rich foods (spinach, kale) in large quantities can decrease INR.

#178 · Specialty Systems

What is the presentation and management of acute angle-closure glaucoma?

Answer

Acute angle-closure glaucoma presents with sudden severe unilateral eye pain, blurry vision with halos around lights, nausea, vomiting, and a mid-dilated non-reactive pupil. Intraocular pressure (IOP) is markedly elevated (above 30-40 mmHg; normal below 21 mmHg). This is an ophthalmologic emergency. Immediate treatment: IV acetazolamide, topical beta-blocker (timolol), and topical pilocarpine. Urgent laser peripheral iridotomy is definitive treatment.

#179 · Specialty Systems

What are the warning signs (red flags) in headache evaluation that require urgent neuroimaging?

Answer

Red flag headache features (SNOOP mnemonic): Systemic symptoms (fever, weight loss), Neurological deficits (focal weakness, vision changes), Onset sudden ('thunderclap', worst headache of life), Older age (above 50, new headache), Positional change (worse with lying down or Valsalva), Papilledema. Thunderclap headache must rule out subarachnoid hemorrhage with non-contrast CT then LP if CT negative.

#180 · Specialty Systems

What is the treatment of choice for cluster headaches acutely and for prophylaxis?

Answer

Acute cluster headache: 100% oxygen 7-12 L/min via non-rebreather mask for 15-20 minutes (highly effective) or sumatriptan 6 mg subcutaneous injection. Intranasal zolmitriptan 5-10 mg is an alternative. Prophylaxis: verapamil 240-480 mg/day (first-line); short-term prednisone burst can be used to break a cluster cycle. Lithium and topiramate are alternatives.

#181 • Women's Health

What are the USPSTF Pap smear screening intervals for women aged 21-29 and 30-65?

Answer

Women aged 21-29: Pap smear alone every 3 years (HPV testing alone is not recommended in this age group). Women aged 30-65: Pap smear alone every 3 years, high-risk HPV test alone every 5 years, or Pap plus HPV co-test every 5 years. Screening stops at age 65 if there has been adequate prior negative screening and no CIN2+ in the past 25 years.

#182 • Women's Health

What is the management of ASCUS (atypical squamous cells of undetermined significance) on Pap smear?

Answer

If HPV co-test is positive with ASCUS: refer for colposcopy. If HPV co-test is negative with ASCUS: repeat co-test in 1 year (not colposcopy). In women aged 21-24 with ASCUS, repeat cytology in 1 year is preferred over reflexive HPV testing. ASCUS with high-risk HPV positive is managed the same as LSIL.

#183 • Women's Health

What is the management of LSIL (low-grade squamous intraepithelial lesion) on Pap smear?

Answer

LSIL in women aged 25 and older: colposcopy is preferred. If HPV co-test was done and is negative in women aged 25-29 (which is unusual since HPV isn't routinely co-tested below 30), repeat co-test in 1 year is an option. In women aged 21-24, LSIL is managed with repeat cytology in 1 year because most LSIL regresses spontaneously in this age group. LSIL is associated with HPV infection in more than 75% of cases.

#184 • Women's Health

What is the management of HSIL (high-grade squamous intraepithelial lesion) on Pap smear?

Answer

HSIL requires immediate colposcopy with directed biopsy regardless of HPV status. HSIL is associated with CIN 2-3 in about 70-75% of cases and with invasive cancer in 1-2%. If colposcopy is inadequate or biopsy shows CIN 2+, loop electrosurgical excision procedure (LEEP) or cold knife conization is performed. Expedited treatment with LEEP without prior biopsy is acceptable in high-risk populations.

#185 · Women's Health

What are the absolute contraindications to combined estrogen-containing contraceptives?

Answer

WHO category 4 (never use) absolute contraindications include: migraine with aura, smoker above age 35 (more than 15 cigarettes/day), personal history of VTE, stroke, or CAD, uncontrolled hypertension (above 160/100 mmHg), current or history of breast cancer, liver disease or benign liver tumor, and diabetes with end-organ damage or more than 20 years duration.

#186 · Women's Health

What are safe contraceptive options for a 38-year-old smoker who has migraines with aura?

Answer

Estrogen-containing methods are absolutely contraindicated (smoker above 35 AND migraines with aura both independently contraindicate). Safe options include: progestin-only pill (mini-pill), depot medroxyprogesterone acetate (DMPA/Depo-Provera), levonorgestrel or etonogestrel subdermal implant (Nexplanon), or hormonal or copper IUD. These methods avoid the stroke risk associated with estrogen.

#187 · Women's Health

What are the Rotterdam criteria for PCOS and how many criteria are required for diagnosis?

Answer

PCOS is diagnosed by meeting 2 of 3 Rotterdam criteria: (1) oligo-ovulation or anovulation (irregular cycles), (2) clinical or biochemical hyperandrogenism (hirsutism, acne, elevated free testosterone), and (3) polycystic ovarian morphology on ultrasound (20 or more follicles per ovary using modern high-frequency transvaginal ultrasound, per the 2023 International Evidence-based PCOS Guideline, or ovarian volume above 10 mL). Other causes of androgen excess (CAH, Cushing, androgen-secreting tumor) must be excluded.

#188 · Women's Health

What is the treatment for PCOS in a woman not seeking fertility?

Answer

Combined oral contraceptive pills (OCP) are first-line for managing irregular cycles and androgen-related symptoms (hirsutism, acne) in PCOS. Spironolactone 50-200 mg/day is added for persistent hirsutism or acne. Metformin 1,500-2,000 mg/day is used when insulin resistance, prediabetes, or obesity is present. Lifestyle modification (diet, weight loss) is cornerstone therapy for all PCOS patients.

#189 · Women's Health

What is the preferred ovulation induction agent for fertility treatment in PCOS and why?

Answer

Letrozole (aromatase inhibitor) 2.5-7.5 mg taken on days 3-7 of the menstrual cycle is preferred over clomiphene citrate for ovulation induction in PCOS. Letrozole achieves higher live birth rates with fewer multiple gestations in PCOS patients. Clomiphene 50-150 mg on days 3-7 is the alternative. Metformin can be added as an adjunct to improve ovulation rates.

#190 · Women's Health

What are the risk factors for ectopic pregnancy and what is the classic presentation?

Answer

Risk factors: prior ectopic pregnancy (highest risk), PID or salpingitis, prior tubal surgery (ligation, reversal), assisted reproduction, IUD use (not causative, but if pregnant with IUD, higher ectopic rate), and cigarette smoking. Classic presentation: pelvic pain, vaginal bleeding, and absence of intrauterine pregnancy on transvaginal ultrasound with positive beta-hCG. Ruptured ectopic causes peritoneal signs and hemodynamic instability.

#191 · Women's Health

What are the criteria for medical (methotrexate) management of ectopic pregnancy?

Answer

Methotrexate 50 mg/m² IM single dose is used for hemodynamically stable ectopic pregnancy with: ectopic mass below 3.5-4 cm, no fetal cardiac activity, beta-hCG below 5,000-10,000 mIU/mL, no severe pain, and no contraindications to MTX (renal, hepatic, or pulmonary disease, immunosuppression). Surgical treatment is required for rupture, hemodynamic instability, or failed MTX therapy.

#192 · Women's Health

What are the diagnostic criteria for preeclampsia?

Answer

Preeclampsia requires: new-onset BP at or above 140/90 mmHg on two occasions at least 4 hours apart after 20 weeks gestation, PLUS proteinuria (at or above 300 mg/24 hours or urine protein-to-creatinine ratio at or above 0.3) OR severe features (BP at or above 160/110, thrombocytopenia below 100,000, renal insufficiency, impaired liver function, pulmonary edema, or new-onset severe headache).

#193 · Women's Health

What is the delivery threshold for preeclampsia without severe features and with severe features?

Answer

Preeclampsia without severe features: delivery at 37 weeks gestation. Preeclampsia with severe features: delivery at 34 weeks (or earlier if maternal or fetal deterioration). Magnesium sulfate is given for seizure prophylaxis during labor and for 24 hours postpartum (dose: 4 g IV loading dose, then 1-2 g/hr maintenance). Antihypertensives (hydralazine IV, labetalol IV, or oral nifedipine) are used when BP is at or above 160/110 mmHg.

#194 · Women's Health

What is HELLP syndrome and how does it relate to preeclampsia?

Answer

HELLP syndrome (Hemolysis, Elevated Liver enzymes, Low Platelets) is a severe variant of preeclampsia. Criteria: microangiopathic hemolytic anemia (LDH above 600, schistocytes on smear), AST above 70 U/L, and platelets below 100,000/uL. Management includes magnesium sulfate, antihypertensives, and delivery (regardless of gestational age in severe cases). Dexamethasone may temporarily improve platelet count.

#195 · Women's Health

What is the diagnostic approach for gestational diabetes mellitus (GDM) and what are the glucose targets?

Answer

Screen all pregnant women for GDM at 24-28 weeks with 1-hour 50 g glucose challenge test (GCT). If GCT is at or above 140 mg/dL (or 130 mg/dL in high-risk), confirm with 3-hour 100 g OGTT. GDM is diagnosed if 2 or more values meet thresholds: fasting at or above 95, 1-hour at or above 180, 2-hour at or above 155, 3-hour at or above 140 mg/dL. Glucose targets: fasting below 95, 1-hour postprandial below 140, 2-hour below 120 mg/dL.

#196 · Women's Health

What is the treatment approach for gestational diabetes and when is insulin initiated?

Answer

Start with MNT (medical nutrition therapy) and exercise. If glucose targets are not met within 1-2 weeks (fasting below 95, 1-hr postprandial below 140, 2-hr below 120 mg/dL), initiate insulin. Insulin is preferred over oral agents in pregnancy per ACOG. Metformin may be used as an alternative but crosses the placenta. Adequate glucose control reduces macrosomia, birth trauma, NICU admission, and neonatal hypoglycemia.

#197 · Women's Health

What is the GBS screening protocol during pregnancy and what is the intrapartum treatment?

Answer

Screen all pregnant women for Group B Streptococcus (GBS) by vaginal-rectal swab culture at 36-37 weeks gestation. If positive (or unknown status with risk factors), give IV penicillin G 5 million units loading dose, then 2.5-3 million units q4h during labor. Ampicillin is an alternative. Cefazolin is given if penicillin-allergic with low anaphylaxis risk. Clindamycin or vancomycin for high anaphylaxis risk (based on sensitivity).

#198 · Women's Health

When is RhoGAM (Rh immune globulin) given during pregnancy?

Answer

Rh-negative mothers who are not sensitized receive RhoGAM 300 mcg IM at 28 weeks gestation (antepartum prophylaxis) and within 72 hours after delivery of an Rh-positive infant. Also given after any sensitizing event: ectopic pregnancy, miscarriage, amniocentesis, chorionic villus sampling, antepartum hemorrhage, or abdominal trauma. Kleihauer-Betke test determines if additional doses are needed after large fetomaternal hemorrhage.

#199 · Women's Health

What defines menopause and what hormones are used for menopausal symptom management?

Answer

Menopause is defined as 12 consecutive months of amenorrhea with FSH above 40 mIU/mL and low estradiol, occurring at a median age of 51. Systemic hormone therapy (HT) is the most effective treatment for vasomotor symptoms. Women with an intact uterus require estrogen plus progestogen (to prevent endometrial hyperplasia). Women post-hysterectomy receive estrogen alone. Use the lowest effective dose for the shortest necessary duration.

#200 · Women's Health

What are the contraindications to menopausal hormone therapy?

Answer

Absolute contraindications to menopausal HT: unexplained vaginal bleeding, history of breast cancer, history of estrogen-sensitive endometrial cancer, active VTE or high VTE risk, active or recent cardiovascular disease, uncontrolled hypertension, active liver disease, and history of stroke. Transdermal estrogen has a lower VTE risk than oral estrogen and may be used in women with VTE risk after risk-benefit discussion.

#201 · Women's Health

What are the BRCA1/2 mutation risk-reduction options for women with confirmed mutations?

Answer

Women with BRCA1/2 mutations have a lifetime breast cancer risk of 50-85% (BRCA1) and 40-60% (BRCA2). Risk-reduction options: prophylactic bilateral salpingo-oophorectomy (BSO) reduces ovarian cancer risk by 80-96% and breast cancer risk when done before natural menopause; prophylactic bilateral mastectomy reduces breast cancer risk by 90-95%; chemoprevention with tamoxifen or raloxifene is less effective but an option for those declining surgery.

#202 · Women's Health

What is the recommended treatment for uncomplicated chlamydia infection?

Answer

Doxycycline 100 mg orally twice daily for 7 days is the preferred treatment for uncomplicated chlamydial urogenital infection (per updated CDC 2021 guidelines, preferred over azithromycin). Azithromycin 1 g orally as a single dose is an alternative when directly observed therapy is preferred. Test for concurrent gonorrhea and treat if positive. Test of cure at 3 weeks for pharyngeal infection only; retest in 3 months for all patients.

#203 · Women's Health

What is the recommended treatment for uncomplicated gonorrhea infection?

Answer

Ceftriaxone 500 mg IM single dose (or 1 g if weight is at or above 150 kg) is first-line for uncomplicated gonorrhea at all sites (urethral, cervical, rectal, and pharyngeal) per 2021 CDC guidelines. No longer co-treat with azithromycin routinely unless chlamydia co-infection is confirmed or not excluded. Doxycycline 100 mg BID x7 days is added if chlamydia testing is not available.

#204 · Women's Health

What is the treatment for primary syphilis and how long after treatment should serologic response be assessed?

Answer

Primary syphilis (painless chancre, positive RPR/VDRL and confirmatory treponemal test) is treated with benzathine penicillin G 2.4 million units IM as a single dose. In penicillin allergy, doxycycline 100 mg twice daily for 14 days is the alternative. Check RPR titers at 6 and 12 months; a 4-fold (2 dilution) decrease confirms adequate treatment. Treatment failure or neurosyphilis requires IV penicillin G.

#205 · Women's Health

What is the treatment for a first episode of genital herpes (HSV)?

Answer

First episode genital herpes is treated with valacyclovir 1 g twice daily for 7-10 days, or acyclovir 400 mg three times daily for 7-10 days. Antiviral therapy reduces duration and severity. Suppressing therapy (valacyclovir 500-1,000 mg daily or acyclovir 400 mg twice daily) is offered for frequent recurrences (at or above 6/year) or to reduce transmission risk to uninfected partners.

#206 · Women's Health

What is bacterial vaginosis (BV) and what are the Amsel criteria for diagnosis?

Answer

BV is a dysbiosis of vaginal flora with loss of *Lactobacillus* and overgrowth of *Gardnerella vaginalis* and anaerobes. Amsel criteria (3 of 4 required): thin, grey-white homogeneous discharge; vaginal pH above 4.5; positive whiff test (fishy odor with KOH); and clue cells on wet prep (above 20% of epithelial cells). Treatment: metronidazole 500 mg twice daily for 7 days or 0.75% gel intravaginal once daily for 5 days.

#207 · Women's Health

What is the treatment for recurrent vulvovaginal candidiasis (3 or more episodes per year)?

Answer

Recurrent VVC (at or above 3 episodes/year) is treated with an induction phase: fluconazole 150 mg every 72 hours for 3 doses, followed by maintenance therapy: weekly fluconazole 150 mg for 6 months. This suppressive regimen reduces recurrence in about 90% of women. Evaluate for underlying diabetes, HIV, or immunosuppression. Oral fluconazole is preferred over topical azoles for systemic suppression.

#208 · Women's Health

What are the diagnostic criteria and treatment for pelvic inflammatory disease (PID)?

Answer

PID clinical criteria: uterine tenderness, adnexal tenderness, or cervical motion tenderness in a sexually active woman without other clear cause. Empiric treatment should be started. Outpatient regimen: ceftriaxone 500 mg IM once PLUS doxycycline 100 mg BID x14 days WITH metronidazole 500 mg BID x14 days. Inpatient criteria: TOA, pregnancy, severe illness, inability to tolerate oral, or failure of outpatient treatment.

#209 · Women's Health

What are the emergency contraception options and their effective time windows?

Answer

Copper IUD (Paragard): most effective emergency contraception, reduces pregnancy risk by above 99%, effective up to 5 days after unprotected sex. Ella (ulipristal acetate) 30 mg: effective up to 120 hours (5 days). Plan B (levonorgestrel) 1.5 mg: effective up to 72 hours but can be used up to 120 hours with reduced efficacy. BMI above 30 kg/m² reduces levonorgestrel EC efficacy; copper IUD or ella preferred in obese patients.

#210 · Women's Health

What are the most common causes of postmenopausal bleeding and what is the evaluation?

Answer

Any postmenopausal bleeding (PMB) must be evaluated to exclude endometrial cancer (most concerning cause). Common causes: endometrial atrophy (most common), endometrial polyp, endometrial hyperplasia, and endometrial carcinoma. Evaluation: transvaginal ultrasound (endometrial stripe below 4-5 mm makes cancer less likely) followed by endometrial biopsy if stripe is above 4 mm, bleeding persists, or high-risk features are present.

#211 · Women's Health

What are the risk factors for endometrial cancer and how does it present?

Answer

Risk factors: unopposed estrogen exposure (obesity, PCOS, tamoxifen use, estrogen-only HT, anovulation), late menopause, nulliparity, diabetes, and hereditary syndromes (Lynch syndrome). Endometrial cancer presents with postmenopausal bleeding (most common) or abnormal uterine bleeding in premenopausal women. Endometrial biopsy is the primary diagnostic test; D&C; under hysteroscopy is used when biopsy is insufficient.

#212 · Women's Health

What is the intrauterine device (IUD) selection guide for different clinical scenarios?

Answer

Copper IUD (Paragard): non-hormonal, preferred for women who want to avoid hormones, lasts up to 10-12 years, most effective EC; may worsen dysmenorrhea. Levonorgestrel IUDs (Mirena 52 mg: 8 years; Kyleena 19.5 mg: 5 years; Skyla 13.5 mg: 3 years): reduce menstrual bleeding and cramping, preferred in endometriosis or heavy menstrual bleeding. Both are safe in nulliparous women and do not affect future fertility.

#213 · Women's Health

What are the common causes and treatment of dysmenorrhea?

Answer

Primary dysmenorrhea: painful menses without pelvic pathology, due to excess prostaglandins. Treat with NSAIDs (ibuprofen 600-800 mg or naproxen 500 mg TID, starting 1-2 days before onset and continuing through day 2-3). Combined OCP reduces prostaglandin production. Secondary dysmenorrhea: due to endometriosis, fibroids, adenomyosis, or PID. Laparoscopy may be needed to diagnose endometriosis.

#214 · Women's Health

What is endometriosis and what are the first-line pharmacologic treatments?

Answer

Endometriosis is endometrial glands and stroma outside the uterus, causing dysmenorrhea, dyspareunia, chronic pelvic pain, and subfertility. First-line hormonal treatments: combined OCP (cyclic or continuous), progestin-only methods (norethindrone acetate, DMPA, LNG-IUD), and NSAIDs for pain. Second-line: GnRH agonists (leuprolide depot) with add-back hormone therapy. Definitive diagnosis is by laparoscopy with biopsy.

#215 · Women's Health

What is the workup for secondary amenorrhea?

Answer

Secondary amenorrhea (absence of menses for 3 months in previously regular cycles or 6 months in irregular) requires: pregnancy test first (always), then TSH, prolactin, FSH, and LH. High FSH: premature ovarian insufficiency. Low FSH/LH: hypogonadotropic hypogonadism (hypothalamic amenorrhea, pituitary pathology). High prolactin: pituitary adenoma (MRI brain). Progesterone challenge test assesses estrogen status and outflow tract.

#216 · Women's Health

What is premature ovarian insufficiency (POI) and how is it distinguished from menopause?

Answer

POI (formerly premature menopause) is loss of ovarian function before age 40, defined by FSH above 40 mIU/mL on two occasions at least 4 weeks apart with amenorrhea for at least 4 months. Unlike menopause, POI may have intermittent ovarian function and spontaneous pregnancy is possible (5-10%). Causes: autoimmune (most common), Turner syndrome, fragile X premutation, and iatrogenic. Hormone replacement until age 50 is recommended to prevent osteoporosis and cardiovascular effects.

#217 · Women's Health

What is the USPSTF recommendation for ovarian cancer screening?

Answer

USPSTF recommends against routine screening for ovarian cancer in average-risk women (D recommendation). Transvaginal ultrasound and CA-125 testing do not reduce mortality and lead to false positives with unnecessary surgery. Women with BRCA1/2 mutations, Lynch syndrome, or strong family history of ovarian cancer should be referred to genetic counseling and high-risk surveillance programs.

#218 · Women's Health

What are the USPSTF screening recommendations for domestic violence/intimate partner violence?

Answer

USPSTF recommends screening all women of reproductive age (14-46) for intimate partner violence (IPV) and referring those who screen positive to ongoing support services. Validated screening tools include HITS (Hurt, Insult, Threaten, Scream) and WAST (Woman Abuse Screening Tool). Screen in a private setting without the partner present. Document findings using the patient's own words.

#219 · Women's Health

What is vulvodynia and what are the treatment options?

Answer

Vulvodynia is chronic vulvar pain lasting at least 3 months without an identifiable cause (not caused by infection, inflammation, or skin disease). Provoked vestibulodynia (pain with touch or insertion) is the most common type. Treatments include topical lidocaine 2% gel or ointment, topical amitriptyline/baclofen creams, oral tricyclic antidepressants (amitriptyline 10-75 mg), gabapentin 300-1,800 mg/day, pelvic floor physical therapy, and vestibulectomy for refractory cases.

#220 · Women's Health

What is the treatment for postpartum hemorrhage (PPH) and what are the most common causes?

Answer

The 4 Ts of PPH causes: Tone (uterine atony, most common cause of 80%), Trauma (lacerations, uterine rupture), Tissue (retained placenta or membranes), Thrombin (coagulopathy). Treatment of uterine atony: uterine massage, oxytocin 10-40 units IV infusion, methylergonovine 0.2 mg IM (avoid in HTN), carboprost (Hemabate) 0.25 mg IM, misoprostol 600-1,000 mcg PR/SL. Bimanual compression and surgical intervention if bleeding continues.

#221 · Women's Health

What are the indications for cervical cerclage during pregnancy?

Answer

Cervical cerclage is a surgical procedure placing a stitch around the cervix to prevent preterm birth. Indications: (1) history-indicated: at least 3 prior second-trimester losses or preterm deliveries, placed at 12-14 weeks; (2) ultrasound-indicated: cervical length below 25 mm before 24 weeks in a patient with prior preterm birth; (3) physical exam-indicated (emergency): dilated cervix with membranes visible. Progesterone supplementation is also used for cervical shortening.

#222 · Women's Health

What is the recommended prenatal lab panel at the first obstetric visit?

Answer

First prenatal visit labs: blood type and Rh factor, antibody screen, CBC, rubella IgG titer, varicella immunity, hepatitis B surface antigen, syphilis (RPR or VDRL), HIV (opt-out testing), gonorrhea and chlamydia NAAT, urinalysis and urine culture, Pap smear (if due), TSH (if symptomatic or high risk), and hemoglobin electrophoresis (if at risk for hemoglobinopathy). Genetic carrier screening is offered to all patients.

#223 · Women's Health

What supplements are recommended for pregnant women and in what doses?

Answer

Folic acid 400-800 mcg daily beginning at least 1 month before conception and through the first trimester reduces neural tube defects by 50-70%. High-risk women (prior NTD, on antiepileptic drugs) need 4,000 mcg (4 mg) daily. Iron 27 mg/day elemental iron. Calcium 1,000 mg/day. Iodine 150-220 mcg/day. Omega-3 DHA 200-300 mg/day. Avoid vitamin A above 10,000 IU (teratogenic in excess).

#224 · Women's Health

What are the indications for elective cesarean delivery and what are the key risks?

Answer

Indications for C-section: prior classical (vertical) uterine incision (VBAC contraindicated), active genital herpes at labor onset, certain placenta previa or accreta spectrum disorders, malpresentation at term, and certain maternal cardiac or neurologic conditions. C-section risks include increased bleeding, infection, injury to adjacent organs, thromboembolism, and future placenta accreta risk. VBAC success rate is 60-80% in appropriate candidates.

#225 · Women's Health

What is placenta previa and how is it managed?

Answer

Placenta previa is placental implantation over or adjacent to the internal cervical os, presenting with painless bright red vaginal bleeding in the second or third trimester. Diagnosed by transvaginal ultrasound. Management: pelvic rest (no vaginal exams, intercourse, or tampons), hospitalization if bleeding is significant, antenatal corticosteroids if below 34 weeks, and planned C-section at 36-37 weeks. Avoid digital cervical examination.

#226 · Women's Health

What is placental abruption and how does it differ from placenta previa?

Answer

Placental abruption is premature separation of the normally implanted placenta from the uterine wall, causing painful vaginal bleeding and uterine rigidity (board-like abdomen). Risk factors: hypertension, cocaine use, trauma, prior abruption. Unlike placenta previa (painless bleeding), abruption is painful. Severe abruption causes DIC and fetal distress. Management depends on gestational age and severity; may require emergency delivery.

#227 · Women's Health

What is postpartum depression and how does it differ from postpartum blues?

Answer

Postpartum blues (baby blues): transient mood lability, tearfulness, anxiety lasting 2-3 days to 2 weeks postpartum; affects 50-80% of women; resolves spontaneously. Postpartum depression (PPD): major depressive episode occurring within 4 weeks to 1 year postpartum; affects 10-15% of women; requires treatment. Screen with Edinburgh Postnatal Depression Scale (EPDS); score at or above 13 indicates PPD. First-line treatment: sertraline (preferred as minimal breast milk transfer) or psychotherapy.

#228 · Women's Health

What are the USPSTF recommendations for breast cancer prevention with chemoprevention?

Answer

USPSTF recommends clinicians offer risk-reducing medications to women at increased breast cancer risk (5-year risk at or above 3% by Tyrer-Cuzick or similar model) after risk-benefit discussion. Options: tamoxifen 20 mg/day x5 years (premenopausal or postmenopausal), raloxifene 60 mg/day x5 years (postmenopausal only), or exemestane 25 mg/day x5 years (postmenopausal). These reduce risk by 30-50% but have side effects including VTE and hot flashes.

#229 · Women's Health

What are the characteristics and management of gestational hypertension?

Answer

Gestational hypertension: BP at or above 140/90 mmHg after 20 weeks gestation without proteinuria or other severe features. It resolves by 12 weeks postpartum. Monitor weekly for development of preeclampsia features. Antihypertensives are started if BP is at or above 160/110 mmHg (labetalol or nifedipine oral). Deliver at 37 weeks. Gestational HTN resolving postpartum does not require long-term antihypertensive therapy.

#230 · Women's Health

What STI screening is recommended for all sexually active women under 25 per USPSTF?

Answer

USPSTF recommends annual chlamydia and gonorrhea screening for all sexually active women aged 24 and younger, and for older women who are at increased risk (new or multiple partners, inconsistent condom use, prior STI, commercial sex work). Syphilis screening is recommended for women at high risk. HIV testing is recommended at least once for all adults aged 15-65, and more frequently for high-risk individuals.

#231 · Women's Health

What is the difference between primary and secondary dyspareunia?

Answer

Primary dyspareunia: painful intercourse since first attempt at penetration. Secondary dyspareunia: pain develops after a period of pain-free intercourse. Common causes by location: superficial/entry dyspareunia (vulvodynia, vaginismus, skin conditions); deep dyspareunia (endometriosis, PID, ovarian cysts, uterine fibroids). Evaluation includes pelvic exam, culture, hormonal assessment, and imaging. Treatment targets underlying etiology.

#232 · Women's Health

What is vaginismus and how is it treated?

Answer

Vaginismus (now termed genito-pelvic pain/penetration disorder) is involuntary contraction of the pelvic floor muscles preventing vaginal penetration, causing pain or inability to complete intercourse. Treatment includes pelvic floor physical therapy with progressive vaginal dilator therapy, psychosexual therapy or CBT, and topical lidocaine for pain relief. Botulinum toxin injection into the pubococcygeus muscle is an option for refractory cases.

#233 · Women's Health

What is the recommended screening for ovarian cancer in women with BRCA mutations?

Answer

Women with BRCA1 or BRCA2 mutations do not have a proven effective screening method for ovarian cancer. USPSTF, ACOG, and NCCN acknowledge no survival benefit for transvaginal ultrasound or CA-125 screening. Prophylactic bilateral salpingo-oophorectomy (BSO) at age 35-40 (BRCA1) or 40-45 (BRCA2) after completion of childbearing is recommended as the most effective risk-reduction strategy.

#234 · Women's Health

What is the normal CA-125 level and when is it used clinically?

Answer

Normal CA-125 is below 35 U/mL. It is used to monitor response to treatment in known ovarian cancer, not for primary screening. It can be elevated in benign conditions (endometriosis, fibroids, PID, liver disease, menstruation) reducing its specificity. ROMA score (combining CA-125 and HE4) is used preoperatively to assess malignancy risk in women with pelvic masses before surgical planning.

#235 · Women's Health

What are the effects of the progestin-only pill (mini-pill) and what is the main concern with missing a dose?

Answer

The progestin-only pill (norethindrone 0.35 mg daily) works primarily by thickening cervical mucus and suppressing ovulation in about 50% of cycles. It must be taken at the same time daily (within 3-hour window). If more than 3 hours late, use backup contraception for 48 hours. It is safe for breastfeeding women and those with estrogen contraindications. The newer norethindrone 0.075 mg pill (Slynd) has a 24-hour missed-pill window.

#236 · Women's Health

What is depot medroxyprogesterone acetate (DMPA) and what are its advantages and disadvantages?

Answer

DMPA (Depo-Provera) 150 mg IM every 12-13 weeks (approximately every 3 months) or subcutaneous 104 mg every 12-13 weeks is a highly effective progestin-only contraceptive (failure rate below 1% with perfect use). Advantages: no daily pill, good for women who cannot use estrogen, reduces sickle cell crises. Disadvantages: bone mineral density loss (reversible after stopping), irregular bleeding or amenorrhea, weight gain, and delayed return of fertility (median 10 months after last injection).

#237 · Women's Health

What is the BRCA gene testing indication and referral criteria?

Answer

Refer for BRCA genetic counseling and testing when: personal history of breast cancer below age 50, bilateral breast cancer, triple-negative breast cancer below 60, breast and ovarian cancer in same patient, male breast cancer, two first-degree relatives with breast or ovarian cancer, Ashkenazi Jewish ancestry with family breast/ovarian cancer history, or family member with known BRCA mutation. USPSTF recommends risk assessment tool (Ontario BRCA or equivalent) before referral.

#238 · Women's Health

What is the recommended timing and type of fetal surveillance in a post-term pregnancy?

Answer

Post-term pregnancy is beyond 42 weeks (41 0/7 to 42 6/7 weeks is late-term). ACOG recommends initiating fetal surveillance (non-stress test twice weekly or biophysical profile) starting at 41 weeks and offering membrane sweeping at 39-40 weeks. Induction of labor is offered at 41 weeks to reduce risks of meconium aspiration, macrosomia, shoulder dystocia, placental insufficiency, and stillbirth. Cervical ripening with prostaglandins or mechanical methods may be needed.

#239 · Women's Health

What is the first-line treatment for urge urinary incontinence in women?

Answer

Pelvic floor muscle training (Kegel exercises) and bladder training are first-line non-pharmacologic treatments. First-line pharmacotherapy: antimuscarinics (oxybutynin extended-release 5-30 mg daily, solifenacin 5-10 mg daily, tolterodine ER 4 mg daily). Beta-3 agonist mirabegron 25-50 mg daily is equally effective with fewer anticholinergic effects (preferred in older adults to avoid cognitive side effects). Combination oxybutynin + mirabegron is more effective than monotherapy.

#240 · Women's Health

What is stress urinary incontinence and what are the treatment options?

Answer

Stress urinary incontinence (SUI) is leakage of urine with increased intraabdominal pressure (coughing, sneezing, laughing, exercise). Caused by urethral hypermobility or intrinsic sphincter deficiency. First-line: pelvic floor physical therapy (Kegel exercises), pessary for urethral support. Topical vaginal estrogen if atrophic vaginitis is present. Surgical options: midurethral sling (synthetic mesh) is most effective. No FDA-approved pharmacotherapy specifically for SUI in the US.

#241 · Women's Health

What are the antenatal corticosteroid indications and regimen during preterm labor?

Answer

Antenatal corticosteroids are given to women at risk for preterm delivery between 24 0/7 and 34 6/7 weeks gestation. Regimen: betamethasone 12 mg IM every 24 hours for 2 doses, or dexamethasone 6 mg IM every 12 hours for 4 doses. Benefits: reduces RDS, IVH, NEC, and neonatal death. A single rescue course may be considered if prior course was given more than 14 days ago and delivery remains imminent at less than 34 weeks.

#242 · Women's Health

What is the recommended antibiotic prophylaxis for Cesarean section and when should it be given?

Answer

Cefazolin is the preferred prophylactic antibiotic for C-section, given as a single IV dose within 60 minutes before skin incision. Dosing: 2 g IV (or 3 g IV if weight is at or above 120 kg). Azithromycin 500 mg IV may be added for unscheduled C-sections after labor or rupture of membranes to reduce endometritis. Clindamycin plus gentamicin is used in penicillin-allergic patients.

#243 · Women's Health

What is the ACOG recommendation for folic acid supplementation to prevent neural tube defects?

Answer

All women capable of becoming pregnant should take 400-800 mcg of folic acid daily. Women with a prior NTD-affected pregnancy or on valproate, carbamazepine, or methotrexate should take 4,000 mcg (4 mg) daily beginning 1-3 months before conception. Folic acid supplementation reduces NTD risk by 50-70%. NTDs occur in the first 28 days post-conception, often before women know they are pregnant.

#244 · Women's Health

What is the clinical management of a spontaneous abortion (miscarriage) in the first trimester?

Answer

Options for complete management of first trimester miscarriage: (1) Expectant management (natural expulsion) within 2-6 weeks; (2) Medical management: misoprostol 800 mcg vaginally or sublingually (85-90% complete within 1 week); (3) Surgical management: manual or electric vacuum aspiration, most rapid and definitive. Rh-negative patients should receive RhoGAM. Emotional support and follow-up beta-hCG to confirm completion are essential.

#245 · Women's Health

What are the key differences between primary and secondary infertility workup?

Answer

Primary infertility: no prior pregnancies.
Secondary infertility: inability to conceive after a prior pregnancy. Workup is similar for both: semen analysis (male factor accounts for 30-40% of infertility), ovarian reserve testing (AMH, day 3 FSH and estradiol, antral follicle count), hysterosalpingography (HSG) to assess tubal patency and uterine cavity, and evaluation of ovulation (day 21 progesterone, LH surge monitoring). Investigate after 12 months of unprotected sex (6 months if above age 35).

#246 · Women's Health

What is the recommended timing for a follow-up Pap smear after treatment for CIN 2 or CIN 3?

Answer

After treatment for CIN 2 or CIN 3 (LEEP, cold knife cone, or cryotherapy), patients require surveillance co-testing (Pap plus HPV) at 6 months, 12 months, and 24 months. If all three are negative, return to routine co-testing every 3 years for 25 years (total). Persistent positive HPV with any abnormal cytology requires colposcopy. CIN 2/3 recurrence rates are 5-10% within 2 years of treatment.

#247 · Women's Health

What is lichen sclerosus and how is it treated?

Answer

Lichen sclerosus is a chronic inflammatory dermatosis most commonly affecting the vulva in postmenopausal women (but can occur in any age or sex). Presents with white, crinkled, atrophic skin, severe pruritus, and dyspareunia. Thin, fragile skin may fuse the labia minora or clitoral hood. First-line treatment: ultrapotent topical corticosteroid (clobetasol propionate 0.05% ointment) daily for 3 months, then tapered to 2-3x/week for maintenance. Biopsy if suspicious for malignancy.

#248 · Women's Health

What is the USPSTF recommendation for STI counseling for sexually active adolescents and adults?

Answer

USPSTF recommends intensive behavioral counseling for all sexually active adolescents and for adults at increased risk for STIs (multiple partners, inconsistent condom use, prior STI, illicit drug use). Counseling covers condom use, risk reduction, STI testing, and partner notification. High-intensity interventions (multiple sessions, individualized) reduce STI incidence. Brief counseling at every visit is also recommended.

#249 · Women's Health

What is trichomonas vaginalis infection and how is it treated?

Answer

Trichomonas vaginalis is the most common curable STI worldwide. Presentation: profuse, frothy, yellow-green vaginal discharge with a 'strawberry cervix' (punctate hemorrhages) and vaginal pH above 4.5. Motile trichomonads on wet prep confirm diagnosis; NAAT is more sensitive. Treatment is sex-specific per CDC STI Treatment Guidelines: women receive metronidazole 500 mg PO twice daily for 7 days (preferred, higher cure rate in women), while men receive metronidazole 2 g PO single dose. Tinidazole 2 g PO single dose is an alternative. Treat partners concurrently. Test-of-cure at 2-3 weeks is recommended due to reinfection and treatment failure risk.

#250 · Women's Health

What is the management of a woman found to have HPV 16 or 18 on co-testing with a negative Pap smear?

Answer

HPV 16 and 18 are the highest-risk genotypes, responsible for approximately 70% of cervical cancers. Per ASCCP 2019 guidelines, women with HPV 16 or 18 detected on co-testing (even with negative cytology) should proceed directly to colposcopy rather than returning for repeat co-testing in 1 year. This differs from 'other high-risk HPV' genotypes (not 16/18) with negative cytology, which can be repeated in 1 year.

#251 · Women's Health

What are the USPSTF recommendations for breast cancer screening with mammography?

Answer

The finalized 2024 USPSTF recommendation calls for biennial mammography starting at age 40 for all average-risk women (B recommendation), replacing the older 2016 stance that started routine screening at 50. Women with high-risk factors such as BRCA mutations may need earlier or more frequent screening.

#252 · Women's Health

What does postmenopausal bleeding always require in clinical evaluation?

Answer

Postmenopausal bleeding always requires endometrial biopsy to rule out endometrial cancer. Any vaginal bleeding occurring more than 12 months after the last menstrual period is considered postmenopausal. Transvaginal ultrasound showing endometrial thickness greater than 4 mm also warrants biopsy.

#253 · Women's Health

Which HPV strains cause most cervical cancers, and what is the recommended vaccination age range?

Answer

HPV types 16 and 18 cause approximately 70% of all cervical cancers. The HPV vaccine is routinely recommended for individuals ages 9-26. For adults ages 27-45, shared decision-making is used, as benefit decreases with age and prior HPV exposure.

#254 · Women's Health

What is the most common type of uterine fibroid, and what medical treatments are available for heavy bleeding?

Answer

Intramural fibroids are the most common type of uterine fibroid, located within the muscular wall of the uterus. Medical treatments for heavy uterine bleeding include progestin-releasing IUDs (such as levonorgestrel IUD), combined oral contraceptives, and GnRH agonists such as leuprolide. Surgical options include myomectomy or hysterectomy.

#255 · Women's Health

What are the classic symptoms of endometriosis and how is it definitively diagnosed?

Answer

Endometriosis presents with cyclic pelvic pain, dysmenorrhea (painful periods), dyspareunia (painful intercourse), and infertility. First-line medical treatment includes combined oral contraceptives or progestins. Definitive diagnosis requires laparoscopy with direct visualization and biopsy of endometrial tissue outside the uterus.

#256 · Women's Health

What are the four main types of endometriosis lesions and which location causes the most severe symptoms?

Answer

The four types of endometriosis lesions are peritoneal implants, ovarian endometriomas (chocolate cysts), deep infiltrating endometriosis, and rectovaginal nodules. Deep infiltrating endometriosis involving the rectovaginal septum tends to cause the most severe pain, including dyschezia (painful bowel movements). Ovarian endometriomas greater than 4 cm are typically managed surgically.

#257 · Women's Health

What is the ASCCP guideline for managing a patient with LSIL on Pap smear who is 25 years old?

Answer

For a patient age 25 or older with LSIL (low-grade squamous intraepithelial lesion), colposcopy is recommended. For patients ages 21-24, a 12-month repeat Pap smear is preferred over immediate colposcopy, as most LSIL in this age group regresses spontaneously. HPV co-testing guides further management in women 30 and older.

#258 · Women's Health

What is the standard treatment for bacterial vaginosis, and how does it differ in pregnant patients?

Answer

Bacterial vaginosis is treated with metronidazole 500 mg orally twice daily for 7 days, or metronidazole 0.75% vaginal gel once daily for 5 days, or clindamycin 2% vaginal cream at bedtime for 7 days. In pregnant patients, oral metronidazole is preferred as intravaginal clindamycin cream has been associated with preterm birth. All symptomatic pregnant women with BV should be treated.

#259 · Women's Health

What are the diagnostic criteria and first-line treatment for polycystic ovary syndrome (PCOS)?

Answer

PCOS is diagnosed using the Rotterdam criteria, requiring 2 of 3 features: oligomenorrhea or anovulation, clinical or biochemical hyperandrogenism (elevated free testosterone, hirsutism, acne), and polycystic ovaries on ultrasound (20 or more follicles in one ovary (per the 2023 International Evidence-based PCOS Guideline) or ovarian volume over 10 mL). First-line treatment is combined oral contraceptives for menstrual regulation and hirsutism, plus metformin for insulin resistance and metabolic concerns.

#260 · Women's Health

What are the contraindications to combined oral contraceptive use (COC)?

Answer

Absolute contraindications to COCs include age 35 or older plus smoking 15 or more cigarettes per day, uncontrolled hypertension (systolic above 160 or diastolic above 100), personal history of deep vein thrombosis or pulmonary embolism, migraine with aura at any age, history of breast cancer, and active liver disease. Women with migraine with aura have a 2-fold increased stroke risk with estrogen-containing contraceptives.

#261 · Mental Health

What are the DSM-5 diagnostic criteria for major depressive disorder (MDD)?

Answer

MDD requires 5 or more of 9 SIGECAPS symptoms present for at least 2 weeks, and must include either depressed mood or anhedonia (loss of interest). SIGECAPS stands for Sleep changes, Interest loss, Guilt, Energy loss, Concentration problems, Appetite changes, Psychomotor changes, and Suicidal ideation. Symptoms must cause significant impairment and not be caused by substances or another medical condition.

#262 · Mental Health

What are the DSM-5 criteria for generalized anxiety disorder (GAD)?

Answer

GAD requires excessive, uncontrollable worry occurring more days than not for at least 6 months, accompanied by at least 3 of 6 symptoms: restlessness, easy fatigability, difficulty concentrating, irritability, muscle tension, and sleep disturbance. First-line treatment is an SSRI or SNRI combined with cognitive behavioral therapy (CBT). Common first-line SSRIs include sertraline (50-200 mg) and escitalopram (10-20 mg).

#263 · Mental Health

What distinguishes Bipolar I from Bipolar II disorder?

Answer

Bipolar I disorder requires at least one manic episode lasting 7 or more days (or of any duration if hospitalization is required). Bipolar II disorder is characterized by at least one hypomanic episode (lasting 4 or more days, less severe than mania) plus at least one major depressive episode, with no history of full mania. First-line mood stabilizers for Bipolar I include lithium and valproate; lamotrigine is preferred for the depressive phase of Bipolar II.

#264 · Mental Health

What are the diagnostic criteria for schizophrenia, and what is the first-line treatment for a first episode?

Answer

Schizophrenia requires 2 or more of the following for at least 1 month: hallucinations, delusions, disorganized speech, grossly disorganized or catatonic behavior, and negative symptoms (flat affect, avolition, alogia). Total duration including prodrome must be at least 6 months. For a first episode, second-generation antipsychotics (SGAs) such as risperidone or aripiprazole are first-line due to a more favorable side effect profile than first-generation agents.

#265 · Mental Health

What are the DSM-5 criteria for PTSD, and which medications are FDA-approved for its treatment?

Answer

PTSD requires exposure to actual or threatened death, serious injury, or sexual violence, followed by symptoms in four clusters lasting more than 1 month: intrusion (flashbacks, nightmares), avoidance, negative alterations in cognition or mood, and alterations in arousal or reactivity. The only two FDA-approved medications for PTSD are sertraline (Zoloft) and paroxetine (Paxil), both SSRIs. Trauma-focused CBT is the gold-standard psychotherapy.

#266 · Mental Health

What is the first-line treatment for obsessive-compulsive disorder (OCD)?

Answer

First-line treatment for OCD is exposure and response prevention (ERP) therapy combined with an SSRI. SSRIs for OCD typically require higher doses than for depression, such as fluoxetine 40-80 mg, sertraline 100-200 mg, or fluvoxamine 100-300 mg. Response may take 8-12 weeks. For treatment-resistant OCD, augmentation with low-dose antipsychotics such as risperidone may be added.

#267 · Mental Health

What are the diagnostic criteria for panic disorder, and what is the recommended first-line treatment?

Answer

Panic disorder requires recurrent unexpected panic attacks plus at least 1 month of persistent concern about future attacks, their consequences, or significant behavioral change to avoid them. First-line treatment is an SSRI (such as sertraline 25-200 mg or escitalopram 10-20 mg) combined with CBT. Long-term benzodiazepine use should be avoided due to dependence risk, though they may be used short-term for acute symptom relief.

#268 · Mental Health

What is the first-line treatment for ADHD in children under age 6 versus children age 6 and older?

Answer

For children under age 6 with ADHD, behavioral therapy alone is the first-line treatment, as the risks of stimulant medications outweigh benefits in very young children. For children age 6 and older, both behavioral therapy and stimulant medications are first-line, with methylphenidate (Ritalin, Concerta) and amphetamine salts (Adderall) being the most commonly used agents. Starting doses of methylphenidate are typically 5 mg twice daily.

#269 · Mental Health

What are the first-line pharmacologic options for ADHD in adults?

Answer

First-line pharmacologic treatment for adult ADHD includes stimulants: mixed amphetamine salts (Adderall, starting at 5-10 mg daily) and methylphenidate formulations (Concerta, Ritalin, starting at 18-36 mg daily). Atomoxetine (Strattera), a selective norepinephrine reuptake inhibitor, is the primary non-stimulant option and preferred when there is a history of substance use disorder or significant anxiety. Bupropion and viloxazine are additional non-stimulant alternatives.

#270 · Mental Health

What is the therapeutic lithium level, and what signs indicate toxicity?

Answer

The therapeutic lithium level for maintenance is 0.6-1.2 mEq/L (acute mania 0.8-1.2 mEq/L). Mild-to-moderate toxicity occurs at levels above 1.5 mEq/L and presents with tremor, diarrhea, nausea, and confusion. Severe toxicity occurs above 2.0 mEq/L with seizures, cardiac arrhythmias, and can be life-threatening. TSH, BMP (renal function), and lithium levels should be monitored every 6-12 months in stable patients.

#271 · Mental Health

What serious adverse effect is associated with clozapine, and how is it monitored?

Answer

Clozapine carries a risk of agranulocytosis, a potentially fatal reduction in white blood cells that occurs in approximately 1% of patients. Because of this risk, clozapine requires enrollment in the REMS (Risk Evaluation and Mitigation Strategy) program with mandatory absolute neutrophil count (ANC) monitoring weekly for 6 months, then every 2 weeks for 6 months, and then monthly thereafter. Clozapine is reserved for treatment-resistant schizophrenia.

#272 · Mental Health

What is tardive dyskinesia, what causes it, and how is it treated?

Answer

Tardive dyskinesia (TD) is an involuntary movement disorder characterized by repetitive, purposeless movements (lip smacking, tongue protrusion, choreiform limb movements) caused by long-term use of dopamine receptor antagonists, particularly first-generation antipsychotics. FDA-approved treatments include valbenazine (Ingrezza, 40-80 mg daily) and deutetrabenazine (Austedo, 12-48 mg daily), both of which are VMAT2 inhibitors that reduce dopamine release.

#273 · Mental Health

What are the extrapyramidal side effects (EPS) of antipsychotics and how are they managed?

Answer

EPS from antipsychotics include akathisia (subjective restlessness), acute dystonia (painful involuntary muscle spasm), and pseudoparkinsonism (rigidity, bradykinesia, tremor). Acute dystonia is treated with benztropine 1-2 mg IM or IV, or diphenhydramine 25-50 mg IM. Akathisia responds best to propranolol 20-40 mg or mirtazapine. Pseudoparkinsonism is managed with benztropine or amantadine, or by switching to a lower-potency antipsychotic.

#274 · Mental Health

What is serotonin syndrome and which drug combinations commonly cause it?

Answer

Serotonin syndrome is characterized by a clinical triad of mental status changes (agitation, confusion), autonomic instability (hyperthermia, tachycardia, diaphoresis, hypertension), and neuromuscular abnormalities (clonus, hyperreflexia, tremor). Common causative combinations include SSRIs with tramadol, fentanyl, triptans, linezolid, or MAOIs. Onset is typically rapid, within hours of exposure. Clonus is the hallmark neuromuscular finding.

#275 · Mental Health

How does neuroleptic malignant syndrome (NMS) differ from serotonin syndrome?

Answer

NMS is caused by dopamine antagonists (antipsychotics) and presents with lead-pipe muscle rigidity, hyperthermia, diaphoresis, and autonomic instability, typically developing over days. Serotonin syndrome is caused by serotonergic drugs and presents with clonus, hyperreflexia, and develops within hours. NMS does not produce clonus. Treatment for NMS includes stopping the offending agent, dantrolene (a muscle relaxant), and bromocriptine (a dopamine agonist); ICU admission is often required.

#276 · Mental Health

What is the CIWA-Ar scale used for, and what scores guide management of alcohol withdrawal?

Answer

The Clinical Institute Withdrawal Assessment for Alcohol (CIWA-Ar) measures severity of alcohol withdrawal using 10 items including tremor, diaphoresis, anxiety, agitation, and perceptual disturbances. A score of 8-14 indicates moderate withdrawal warranting medication. A score of 15 or more indicates severe withdrawal requiring inpatient management. Benzodiazepines are first-line pharmacotherapy, with lorazepam preferred in liver disease and diazepam used for longer-acting coverage.

#277 · Mental Health

What does a positive CAGE questionnaire indicate?

Answer

The CAGE questionnaire screens for alcohol use disorder with four questions: Have you felt you should Cut down? Have people Annoyed you by criticizing your drinking? Have you felt Guilty about drinking? Have you had an Eye-opener drink in the morning? A score of 2 or more positive responses indicates likely alcohol use disorder and warrants further evaluation. Sensitivity is approximately 71-85% for AUD.

#278 · Mental Health

What is buprenorphine, how does it work, and when should it be initiated?

Answer

Buprenorphine is a partial mu-opioid agonist and kappa-opioid antagonist used for opioid use disorder (OUD). It should be initiated when the patient has a Clinical Opioid Withdrawal Scale (COWS) score of 8-12 or greater, indicating mild-to-moderate withdrawal, to prevent precipitating severe withdrawal. It is given sublingually (Suboxone combines buprenorphine with naloxone). Typical starting dose is 2-4 mg sublingually, titrating to 8-24 mg daily.

#279 · Mental Health

What is naltrexone, and what precautions are needed before starting it for opioid use disorder?

Answer

Naltrexone is a full opioid receptor antagonist used for both opioid use disorder and alcohol use disorder (AUD). Before initiating naltrexone for OUD, the patient must be completely opioid-free for at least 7-10 days (10-14 days for methadone) to avoid precipitating severe withdrawal. Naloxone challenge testing may be done before the first dose. Monthly injectable naltrexone (Vivitrol 380 mg IM) improves adherence over oral formulations.

#280 · Mental Health

What is acamprosate and what makes it unique among alcohol use disorder medications?

Answer

Acamprosate (Campral) reduces alcohol cravings by modulating glutamate and GABA neurotransmitter systems in patients who have already achieved abstinence. A key advantage is that it is safe to use in patients with liver disease, as it is renally cleared rather than hepatically metabolized. The standard dose is 666 mg (two 333 mg tablets) three times daily, for a total of 6 tablets per day (1,998 mg/day total). It does not prevent intoxication if the patient drinks.

#281 · Mental Health

How does disulfiram work for alcohol use disorder, and what is the most important patient education point?

Answer

Disulfiram (Antabuse) inhibits aldehyde dehydrogenase, the enzyme that breaks down acetaldehyde, a toxic byproduct of alcohol metabolism. If alcohol is consumed while taking disulfiram, acetaldehyde accumulates causing flushing, nausea, vomiting, diaphoresis, and tachycardia. The most critical patient education point is that the patient must maintain complete abstinence from all alcohol-containing products, including mouthwash, sauces, and topical preparations, as even small amounts trigger the reaction.

#282 · Mental Health

What combination of factors defines high suicide risk, and what is the key intervention strategy?

Answer

High suicide risk is defined by the presence of a specific plan (how), access to means (what), and stated intent (when/will do it), requiring inpatient psychiatric admission. Lethal means counseling and means restriction is the single most evidence-based intervention, particularly securing or removing firearms from the home, which are involved in approximately 50% of suicide deaths in the US. Safety planning is an essential outpatient tool.

#283 · Mental Health

What is the PHQ-9, and what scores correlate with MDD severity?

Answer

The PHQ-9 (Patient Health Questionnaire-9) is a validated 9-item depression screening and severity tool based on the DSM criteria for MDD, with each item scored 0-3. Scores of 5-9 indicate mild depression, 10-14 indicate moderate depression, 15-19 indicate moderately severe depression, and 20-27 indicate severe depression. A score of 10 or above is the standard threshold for initiating treatment.

#284 · Mental Health

What are the first-line SSRIs for depression and their typical starting and target doses?

Answer

Common first-line SSRIs for MDD include sertraline (start 25-50 mg, target 50-200 mg daily), escitalopram (start 5-10 mg, target 10-20 mg daily), and fluoxetine (start 10-20 mg, target 20-60 mg daily). An adequate trial requires at least 4-8 weeks at a therapeutic dose before declaring treatment failure. Sertraline and escitalopram have the most favorable drug interaction profiles.

#285 · Mental Health

What is the mechanism and clinical use of venlafaxine, and when is it preferred over SSRIs?

Answer

Venlafaxine (Effexor) is an SNRI (serotonin-norepinephrine reuptake inhibitor) that inhibits both serotonin and norepinephrine transporters. It is preferred when a patient has comorbid anxiety and depression, neuropathic pain, or fibromyalgia. Target doses for depression are 75-225 mg daily. At doses below 150 mg, it acts predominantly as an SSRI; norepinephrine reuptake inhibition becomes clinically significant at 150 mg and above.

#286 · Mental Health

What are the unique properties of bupropion (Wellbutrin) that distinguish it from SSRIs?

Answer

Bupropion inhibits dopamine and norepinephrine reuptake and has no serotonergic activity. It does not cause sexual dysfunction or weight gain, making it preferred when these SSRI side effects are a concern. It is also FDA-approved for smoking cessation (Zyban) at 150 mg twice daily. Bupropion lowers the seizure threshold and is contraindicated in patients with eating disorders (bulimia, anorexia), seizure disorders, and abrupt alcohol or benzodiazepine withdrawal.

#287 · Mental Health

What is mirtazapine and when is it particularly useful in depression management?

Answer

Mirtazapine (Remeron) is a noradrenergic and specific serotonergic antidepressant (NaSSA) that blocks alpha-2 adrenergic receptors and histamine receptors. It is particularly useful in patients with depression complicated by insomnia, poor appetite, or significant weight loss, as it promotes sedation and weight gain. Standard dosing is 15-45 mg at bedtime. Paradoxically, lower doses (7.5-15 mg) are more sedating than higher doses.

#288 · Mental Health

What is the black box warning for SSRIs in children and young adults?

Answer

SSRIs carry an FDA black box warning for increased risk of suicidal ideation and behavior in children, adolescents, and young adults up to age 24. This risk is highest in the first 1-4 weeks of treatment. All patients started on SSRIs should be seen within 1-2 weeks of initiation, and more frequently in pediatric and young adult patients. The absolute risk increase is small (4% vs 2% placebo), and untreated depression carries a higher overall suicide risk.

#289 · Mental Health

What is the clinical presentation and treatment of social anxiety disorder?

Answer

Social anxiety disorder (social phobia) presents with marked fear or anxiety about social situations where the individual may be scrutinized, lasting 6 or more months and causing significant functional impairment. First-line treatment is CBT (specifically, exposure-based therapy) and SSRIs, with paroxetine, sertraline, and fluvoxamine CR all FDA-approved. Beta-blockers such as propranolol 10-40 mg can be used situationally for performance anxiety but are not effective for generalized social anxiety.

#290 · Mental Health

What are the DSM-5 criteria for alcohol use disorder (AUD) and how is severity classified?

Answer

AUD is diagnosed when 2 or more of 11 criteria are met within a 12-month period, including tolerance, withdrawal, using more than intended, failed attempts to cut down, time spent obtaining/using/recovering, craving, role failure, interpersonal problems, giving up activities, hazardous use, and continued use despite knowledge of harm. Mild AUD = 2-3 criteria, moderate = 4-5 criteria, severe = 6 or more criteria.

#291 · Mental Health

What are the signs and symptoms of opioid withdrawal and how is severity assessed?

Answer

Opioid withdrawal symptoms include anxiety, restlessness, lacrimation, rhinorrhea, yawning, diaphoresis, piloerection (goosebumps), nausea, vomiting, diarrhea, muscle aches, and insomnia. The Clinical Opioid Withdrawal Scale (COWS) scores 11 items on a 0-48 scale: scores 5-12 = mild, 13-24 = moderate, 25-36 = moderately severe, above 36 = severe. Buprenorphine initiation is recommended at COWS 8-12 or above.

#292 · Mental Health

What is the role of quetiapine in bipolar disorder and what metabolic monitoring is required?

Answer

Quetiapine (Seroquel) is FDA-approved for bipolar mania, bipolar depression, and as adjunctive therapy in MDD. It is one of the few agents effective for both the manic and depressive phases of bipolar disorder. Metabolic monitoring includes fasting glucose, lipid panel, weight, BMI, and waist circumference at baseline, at 3 months, and then annually. Weight gain and metabolic syndrome are the most common significant adverse effects.

#293 · Mental Health

What is the difference between specific and social phobias, and how are they treated?

Answer

Specific phobias involve intense, irrational fear of a specific object or situation (animals, heights, blood, injections, flying) that is out of proportion to the actual danger. Social phobia (social anxiety disorder) specifically involves fear of negative evaluation in social situations. For specific phobias, exposure-based CBT (graduated exposure) is the primary treatment and is highly effective. Medications play a minimal role in specific phobia management, though benzodiazepines may be used short-term.

#294 · Mental Health

What is duloxetine (Cymbalta) and what conditions beyond depression is it approved to treat?

Answer

Duloxetine is an SNRI with FDA approvals for major depressive disorder, GAD, diabetic peripheral neuropathy, fibromyalgia, and chronic musculoskeletal pain. The typical dose range is 30-120 mg daily. It is a moderate inhibitor of CYP2D6, which can raise levels of drugs metabolized by that enzyme, such as tricyclic antidepressants. Duloxetine should not be abruptly discontinued due to discontinuation syndrome.

#295 · Mental Health

What is antidepressant discontinuation syndrome and how is it managed?

Answer

Discontinuation syndrome occurs when SSRIs or SNRIs are abruptly stopped after regular use, presenting with flu-like symptoms, dizziness, insomnia, irritability, and electric shock sensations (brain zaps). Paroxetine and venlafaxine have the highest risk due to their short half-lives. Fluoxetine has the lowest risk due to its long half-life of 1-4 days (active metabolite 4-16 days). Management involves gradual tapering over weeks and, if needed, restarting the medication and tapering more slowly.

#296 · Mental Health

What medications are used for the acute management of mania in bipolar disorder?

Answer

Acute mania is managed with lithium (targeting a level of 0.8-1.2 mEq/L), valproate (targeting a level of 50-125 mcg/mL), or second-generation antipsychotics (SGAs) such as olanzapine, quetiapine, or risperidone. Combination therapy (lithium or valproate plus an SGA) is often used for severe mania. Benzodiazepines may be added short-term for sedation and behavioral control. Antidepressants should be avoided during acute mania as they may trigger cycling.

#297 · Mental Health

What is the treatment for ADHD in patients with comorbid substance use disorder?

Answer

For patients with ADHD and comorbid substance use disorder (SUD), non-stimulant medications are preferred to reduce misuse risk. Atomoxetine (Strattera, 40-100 mg daily), bupropion (150-300 mg daily), and viloxazine (Qelbree) are first-line non-stimulant options. If stimulants are needed, long-acting formulations are preferred over short-acting ones. The SUD should be stabilized before initiating ADHD pharmacotherapy.

#298 · Mental Health

What is the clinical presentation of borderline personality disorder (BPD) and how is it managed?

Answer

BPD is characterized by a pervasive pattern of instability in interpersonal relationships, self-image, affects, and marked impulsivity. DSM-5 requires 5 or more of 9 criteria including fear of abandonment, unstable relationships, identity disturbance, impulsivity, suicidal or self-harming behavior, emotional instability, chronic emptiness, anger problems, and stress-related paranoia. The gold-standard treatment is dialectical behavior therapy (DBT). Medications target specific symptoms rather than the diagnosis itself.

#299 · Mental Health

What screening tool is used for alcohol misuse in primary care, and what scores indicate hazardous drinking?

Answer

The AUDIT (Alcohol Use Disorders Identification Test) is the WHO-recommended 10-item tool for screening alcohol misuse in primary care settings. A score of 8-15 indicates hazardous or harmful drinking requiring brief counseling. A score of 16-19 indicates harmful drinking warranting brief counseling and monitoring. A score of 20 or above indicates likely dependence, requiring referral for further diagnostic evaluation and specialized treatment.

#300 · Mental Health

What is the Columbia Suicide Severity Rating Scale (C-SSRS) used for in clinical practice?

Answer

The C-SSRS is a validated tool used to assess suicidal ideation and behavior in clinical and research settings. It measures the intensity of ideation (passive ideation, active ideation with or without plan or intent) and behavior (preparatory actions, interrupted attempts, aborted attempts, actual attempts). It helps standardize documentation and guide clinical decision-making about level of care. A patient with active suicidal ideation with a specific plan and intent requires inpatient evaluation.

#301 · Mental Health

What is the mechanism of buspirone and when is it used for anxiety?

Answer

Buspirone is a partial agonist at serotonin 5-HT_{1A} receptors and a weak dopamine D₂ antagonist. It is used for generalized anxiety disorder (GAD) and has no addiction potential, making it a good option in patients with substance use disorder. The typical dose is 15-60 mg daily in divided doses (7.5-15 mg twice or three times daily). A major limitation is the delayed onset of 2-4 weeks, and it does not work for acute or as-needed anxiety relief.

#302 · Mental Health

What are the indications for electroconvulsive therapy (ECT) in psychiatric care?

Answer

ECT is indicated for severe, treatment-resistant MDD that has not responded to 2 or more adequate antidepressant trials, severe MDD with psychotic features, severe suicidality requiring rapid response, and catatonia unresponsive to benzodiazepines. ECT is also indicated in bipolar disorder with severe mania or depression, and in NMS. It has a response rate of approximately 70-90% in treatment-resistant depression, often within 2-3 weeks of treatment.

#303 · Mental Health

What are the cognitive effects of benzodiazepines in elderly patients, and what guideline addresses this?

Answer

Benzodiazepines in elderly patients significantly increase risks of sedation, cognitive impairment, delirium, falls, and hip fractures. Long-acting benzodiazepines (diazepam, chlordiazepoxide) are particularly hazardous in the elderly due to prolonged half-lives. The American Geriatrics Society Beers Criteria explicitly lists all benzodiazepines as potentially inappropriate in adults 65 and older. Non-pharmacologic approaches (sleep hygiene, CBT-I for insomnia) and shorter-acting alternatives should be prioritized.

#304 · Mental Health

What is the clinical presentation of delirium and how does it differ from dementia?

Answer

Delirium presents with acute onset (hours to days) of fluctuating altered attention, awareness, and cognition with an identifiable medical cause such as infection, medication toxicity, or metabolic disturbance. Key features are acute onset and fluctuating course. Dementia has an insidious onset over months to years with stable or slowly progressive course, and consciousness is preserved until late stages. The mnemonic DELIRIUM helps identify causes: Drugs, Electrolytes, Low O₂, Infection, Retention (urinary/fecal), Ictal, Underhydration, Metabolic.

#305 · Mental Health

What is the first-line pharmacologic treatment for insomnia disorder in adults?

Answer

Cognitive behavioral therapy for insomnia (CBT-I) is the gold-standard first-line treatment for chronic insomnia, more effective and more durable than medication. For pharmacologic treatment, FDA-approved options include doxepin 3-6 mg (for sleep maintenance), suvorexant (Belsomra, orexin receptor antagonist, 5-20 mg), and lemborexant. Zolpidem (Ambien) is commonly used but carries risks of dependence, parasomnia (sleepwalking), and cognitive impairment, particularly in women and older adults.

#306 · Mental Health

What is the difference between a hypomanic episode and a manic episode in bipolar spectrum disorders?

Answer

Hypomania lasts at least 4 consecutive days, does not require hospitalization, does not include psychotic features, and does not cause marked functional impairment. Mania lasts at least 7 days (or any duration requiring hospitalization), may include psychotic features, and causes marked functional impairment or necessitates inpatient care. Both require at least 3 additional DIGFAST symptoms: Distractibility, Impulsivity, Grandiosity, Flight of ideas, Activity increase, Sleep deficit, and Talkativeness.

#307 · Mental Health

What is the risk of valproate in women of childbearing age with bipolar disorder?

Answer

Valproate carries the highest teratogenic risk among mood stabilizers and antiepileptic drugs. It causes neural tube defects (spina bifida) in approximately 1-2% of exposed pregnancies, and is also associated with cognitive impairment, autism spectrum features, and craniofacial abnormalities in children exposed in utero. Valproate should be avoided in women of childbearing potential unless other treatments have failed and reliable contraception is in place. Folic acid 4 mg daily should be taken if valproate cannot be avoided.

#308 · Mental Health

What are the symptoms of cannabis use disorder and how is it treated?

Answer

Cannabis use disorder is diagnosed when cannabis use leads to significant impairment including tolerance, withdrawal (irritability, sleep problems, decreased appetite, anxiety), using more than intended, failed attempts to quit, and continued use despite consequences. At least 2 of 11 criteria are required within 12 months. There are no FDA-approved medications for cannabis use disorder. CBT and motivational enhancement therapy are the most effective treatments. N-acetylcysteine has shown some benefit in adolescents.

#309 · Mental Health

What is adjustment disorder and how does it differ from MDD?

Answer

Adjustment disorder is an emotional or behavioral reaction to an identifiable stressor occurring within 3 months of the stressor, that is out of proportion to the stressor's severity and causes significant impairment. Symptoms resolve within 6 months after the stressor ends. It does not meet full criteria for MDD, anxiety disorder, or PTSD. Treatment is primarily psychotherapy (supportive or CBT). Medications may be used short-term for specific symptoms such as sleep or anxiety, but are not the primary intervention.

#310 · Mental Health

What is the role of lithium in suicide prevention?

Answer

Lithium has the most robust evidence among all psychotropic medications for reducing suicide and suicidal behavior, with meta-analyses showing a 5-fold reduction in suicide risk in patients with bipolar disorder and MDD. The anti-suicidal effect appears independent of its mood-stabilizing properties and may be related to its effects on serotonin and impulsivity. This anti-suicidal effect is a major clinical reason to choose lithium over other mood stabilizers in high-risk patients.

#311 · Mental Health

What is the DAST-10 and what score warrants further assessment?

Answer

The DAST-10 (Drug Abuse Screening Test) is a validated 10-item questionnaire screening for problematic drug use (excluding alcohol and tobacco). Scores range from 0-10: 1-2 = low level, warrants monitoring; 3-5 = moderate level, warrants further assessment and intervention; 6-8 = substantial level, warrants assessment and referral; 9-10 = severe level, warrants immediate referral. It is widely used in primary care for annual substance use screening.

#312 · Mental Health

What is the recommended duration of antidepressant treatment after a first episode of MDD?

Answer

After achieving remission from a first episode of MDD, antidepressants should be continued for a minimum of 6-12 months before considering tapering, to prevent relapse. Patients with a history of 2 or more episodes, severe first episodes, or strong family history of MDD should be counseled about indefinite maintenance therapy. Tapering should be gradual over weeks to months, and the patient should be monitored closely for return of symptoms. Abrupt discontinuation significantly increases the relapse rate.

#313 · Mental Health

What is motivational interviewing (MI) and in what clinical situations is it most useful?

Answer

Motivational interviewing is a collaborative, patient-centered communication style designed to strengthen intrinsic motivation for behavior change by exploring and resolving ambivalence. It is most useful in managing substance use disorders, medication non-adherence, lifestyle changes for chronic disease (diabetes, obesity, hypertension), and smoking cessation. Core MI skills are summarized by OARS: Open-ended questions, Affirmations, Reflective listening, and Summaries. The provider avoids arguing or confronting the patient.

#314 · Mental Health

What are the clinical features that distinguish persistent depressive disorder (dysthymia) from MDD?

Answer

Persistent depressive disorder (dysthymia) requires depressed mood for most of the day, more days than not, for at least 2 years in adults (1 year in children and adolescents), with at least 2 additional symptoms from: poor appetite or overeating, insomnia or hypersomnia, low energy, low self-esteem, poor concentration, and hopelessness. The depressed mood is less severe than MDD but more chronic. Patients may have symptom-free periods of no more than 2 months. Treatment includes SSRIs and psychotherapy.

#315 · Mental Health

What is the treatment for attention deficit hyperactivity disorder when a patient has hypertension?

Answer

Stimulant medications (methylphenidate and amphetamine salts) can raise heart rate by 3-6 beats per minute and blood pressure by 2-4 mmHg, which is concerning in patients with uncontrolled hypertension. In patients with ADHD and hypertension, non-stimulant options such as atomoxetine, extended-release guanfacine (Intuniv 1-4 mg), or extended-release clonidine (Kapvay) are preferred. Guanfacine and clonidine are alpha-2 agonists that can actually lower blood pressure and address ADHD symptoms simultaneously.

#316 · Mental Health

What defines a psychotic disorder versus a mood disorder with psychotic features?

Answer

In a primary psychotic disorder (schizophrenia, schizoaffective disorder), psychotic symptoms are the central feature and persist beyond mood episodes. In mood disorders with psychotic features (psychotic depression, mania with psychosis), psychotic symptoms occur exclusively during mood episodes and resolve when the mood episode resolves. Mood-congruent psychotic features (grandiose delusions in mania, nihilistic delusions in depression) are more common in mood disorders, while bizarre delusions and prominent negative symptoms suggest primary psychosis.

#317 · Mental Health

What is the pharmacologic treatment for generalized anxiety disorder in elderly patients?

Answer

In elderly patients with GAD, SSRIs (sertraline or escitalopram) and SNRIs (duloxetine) are preferred first-line pharmacologic options. Buspirone is also appropriate given its lack of dependence potential. Benzodiazepines should be avoided due to risks of falls, cognitive impairment, and delirium per the Beers Criteria. Pregabalin (75-300 mg daily) is an effective option, particularly for patients with comorbid pain. CBT adapted for older adults is effective and should be combined with medication.

#318 · Mental Health

What is the GAD-7 and how are scores interpreted?

Answer

The GAD-7 (Generalized Anxiety Disorder 7-item scale) is a validated 7-item self-report screening and severity tool for GAD and anxiety in general. Scores of 5-9 indicate mild anxiety, 10-14 indicate moderate anxiety, and 15 or above indicate severe anxiety. A score of 10 or above is the standard threshold for further evaluation and treatment. The GAD-7 also screens for panic disorder, social anxiety disorder, and PTSD, with a lower threshold of 8 used for these conditions.

#319 · Mental Health

What is the clinical presentation of conduct disorder versus oppositional defiant disorder (ODD) in children?

Answer

ODD is characterized by a persistent pattern of angry/irritable mood, argumentative/defiant behavior, and vindictiveness toward authority figures, lasting 6 or more months. Conduct disorder (CD) is more severe and involves repetitive violation of others' rights or social rules, including aggression to people or animals, destruction of property, deceitfulness/theft, and serious rule violations. CD may progress to antisocial personality disorder in adulthood. Treatment for both includes parent management training and individual psychotherapy.

#320 · Mental Health

What are the CRAFFT screening questions and who should be screened?

Answer

The CRAFFT is a 6-item validated substance use screening tool for adolescents ages 12-21 used in primary care. Questions ask: Car (ridden with someone high), Relax (used to relax), Alone (used alone), Forget (forgotten things while using), Friends/Family (told you should cut down), Trouble (gotten into trouble while using). A score of 2 or more indicates high-risk use warranting brief intervention and possible referral. The AAP recommends annual CRAFFT screening at all adolescent well visits.

#321 · Mental Health

What is the mechanism and clinical use of lamotrigine in bipolar disorder?

Answer

Lamotrigine (Lamictal) stabilizes neuronal membranes by blocking voltage-gated sodium channels and inhibiting glutamate release. It is FDA-approved for the maintenance treatment of Bipolar I disorder and is the preferred agent for the depressive phase of Bipolar II. It is generally weight-neutral and does not cause cognitive dulling. The major adverse effect is Stevens-Johnson syndrome (SJS), which is avoided by starting at 25 mg daily and titrating slowly over 6-8 weeks. Concurrent valproate doubles lamotrigine levels, requiring starting at 12.5 mg every other day.

#322 · Mental Health

What is the screening tool for PTSD in primary care, and what score indicates a positive screen?

Answer

The PC-PTSD-5 (Primary Care PTSD Screen for DSM-5) is a validated 5-item tool recommended for PTSD screening in primary care. A score of 3 or more (answering yes to 3 of 5 questions about nightmares, avoidance, on guard, feeling numb, and guilt/blame) is a positive screen requiring further evaluation. The PCL-5 (PTSD Checklist for DSM-5) is a 20-item tool used to assess PTSD symptom severity, with a score of 31-33 or above suggesting likely PTSD diagnosis.

#323 · Mental Health

What is harm reduction in the context of substance use disorders, and what are examples of harm reduction strategies?

Answer

Harm reduction is a public health approach that aims to minimize the negative health, social, and economic consequences of substance use without requiring abstinence as a precondition for services. Examples include needle exchange programs (reducing HIV/HCV transmission), naloxone distribution to people who use opioids (reversing overdose), fentanyl test strips, supervised consumption sites, and providing clean smoking supplies. Medication-assisted treatment (buprenorphine, methadone) is itself considered a harm reduction strategy.

#324 · Mental Health

What is the difference between opioid tolerance, physical dependence, and addiction?

Answer

Tolerance is a physiologic adaptation where the same dose produces less effect over time, requiring dose increases for the same result. Physical dependence is a physiologic state where abrupt discontinuation produces withdrawal symptoms; it occurs with any regular opioid use and does not indicate addiction. Opioid use disorder (addiction) involves compulsive use despite negative consequences, loss of control, and craving. Patients with cancer pain on opioids may be tolerant and physically dependent without having OUD.

#325 · Mental Health

What are the diagnostic features of somatic symptom disorder (SSD)?

Answer

Somatic symptom disorder (SSD) requires one or more somatic symptoms causing significant distress or functional impairment, plus excessive thoughts, feelings, or behaviors related to those symptoms for at least 6 months. The symptoms may or may not have a clear medical explanation; the diagnosis is not one of exclusion. SSD replaces DSM-IV diagnoses of somatization disorder, hypochondriasis, and undifferentiated somatoform disorder. First-line treatment is CBT and regularly scheduled primary care visits, avoiding reactive testing and referrals.

#326 · Mental Health

What is the first-line treatment for tobacco use disorder (nicotine dependence)?

Answer

The most effective treatment for tobacco cessation is combination pharmacotherapy plus behavioral counseling. Varenicline (Chantix, 0.5 mg daily for 3 days, then 0.5 mg twice daily for 4 days, then 1 mg twice daily for 12 weeks) is the most effective single agent. Combination nicotine replacement therapy (patch plus lozenge or gum) is also highly effective. Bupropion 150 mg twice daily is a second-line agent. All patients who smoke should be counseled using the 5 A's: Ask, Advise, Assess, Assist, Arrange follow-up.

#327 · Mental Health

What are the features of narcissistic personality disorder and how does it present in clinical encounters?

Answer

Narcissistic personality disorder (NPD) is characterized by a pervasive pattern of grandiosity, need for admiration, and lack of empathy, with 5 or more of 9 DSM-5 criteria including grandiose sense of self-importance, preoccupation with fantasies of unlimited success, sense of entitlement, interpersonal exploitation, and arrogant behaviors. In clinical encounters, patients may be dismissive of providers, demand special treatment, or become angry when given unfavorable diagnoses. Psychotherapy, particularly transference-focused therapy, is the primary treatment.

#328 · Mental Health

What is the pharmacologic management of acute alcohol withdrawal requiring hospitalization?

Answer

Hospitalized patients with alcohol withdrawal are managed with symptom-triggered benzodiazepine protocols guided by CIWA-Ar scores. Long-acting benzodiazepines such as diazepam (5-20 mg IV/PO q1-2h PRN) are preferred for self-tapering via redistribution. In patients with liver disease, lorazepam (1-2 mg q4-6h) or oxazepam (which avoid hepatic metabolism) are preferred. Thiamine 100-500 mg IV must be given before glucose administration to prevent Wernicke encephalopathy. Seizure prophylaxis with phenobarbital or valproate may be added in severe withdrawal.

#329 · Mental Health

What is premenstrual dysphoric disorder (PMDD) and how is it treated?

Answer

PMDD is a severe form of PMS characterized by marked mood symptoms (depression, anxiety, irritability, mood swings) plus physical symptoms in the luteal phase of the menstrual cycle, resolving within a few days of menstruation onset. At least 5 symptoms must be present, including at least 1 mood symptom, causing significant functional impairment. First-line treatment is SSRIs, which can be given continuously or only during the luteal phase (day 14 until menses). Oral contraceptives (drospirenone-containing pills such as Yaz) are also FDA-approved for PMDD.

#330 · Mental Health

What is the recommended monitoring for patients taking antipsychotics for metabolic complications?

Answer

All patients on antipsychotic medications (particularly second-generation antipsychotics like clozapine and olanzapine) should be monitored with: weight and BMI at baseline, 4 weeks, 8 weeks, 12 weeks, and then quarterly; fasting glucose and lipid panel at baseline, 12 weeks, and then annually. Blood pressure should be measured at baseline and 12 weeks. A 5% or greater weight gain warrants switching to a more weight-neutral antipsychotic such as aripiprazole or ziprasidone. Prolactin levels should be checked if galactorrhea or sexual dysfunction occurs.

#331 · Pediatric Care

What developmental milestones are expected at 2 months of age?

Answer

At 2 months, expected milestones include social smiling in response to a caregiver's face, visual tracking past the midline, and cooing (making soft vowel sounds). The infant should be able to briefly lift their head while prone. Failure to achieve social smile by 2-3 months warrants developmental evaluation. Red flags include absent response to loud sounds or visual stimuli.

#332 · Pediatric Care

What developmental milestones are expected at 4 months of age?

Answer

At 4 months, infants should roll from front to back, laugh and squeal, and actively reach for objects. They should hold their head steadily when upright, push up on arms during tummy time, and visually track objects through a full 180-degree arc. Babbling with consonant sounds (ba, ma, ga) typically begins around 4-6 months. Failure to reach for objects or respond to faces is a red flag.

#333 · Pediatric Care

What developmental milestones are expected at 6 months of age?

Answer

At 6 months, infants should sit with support, transfer objects from one hand to the other, and produce repetitive consonant-vowel babbling such as 'ba-ba' or 'da-da' (without meaning). They demonstrate stranger anxiety beginning around this age and enjoy social play. Rolling both ways (front to back and back to front) is typically achieved by 5-6 months. Absence of babbling or failure to respond to own name by 6 months is a developmental red flag.

#334 · Pediatric Care

What developmental milestones are expected at 9 months of age?

Answer

At 9 months, the pincer grasp (using thumb and forefinger) is beginning to emerge. Infants wave bye-bye, show clear stranger anxiety, and use gestures to communicate. They pull to stand and may cruise along furniture. 'Mama' and 'dada' are said but without specific meaning. Failure to show stranger anxiety, failure to babble, or inability to bear weight on legs are red flags at this age.

#335 · Pediatric Care

What developmental milestones are expected at 12 months of age?

Answer

At 12 months, children typically take their first independent steps (normal range 9-15 months), use 'mama' and 'dada' with meaning, say at least one other word, and finger-feed themselves. They wave bye-bye, follow simple one-step commands with gestures, and can put objects in a container. No single words by 12 months or no pointing by 12 months are red flags requiring developmental evaluation.

#336 · Pediatric Care

What developmental milestones are expected at 18 months of age?

Answer

At 18 months, children should have a vocabulary of 10-25 words, point to at least 2-4 body parts when asked, walk well independently, and feed themselves with a spoon (with spilling). They carry out 2-step commands and demonstrate pretend play (pretending to feed a doll). Fewer than 10 words, inability to walk independently, or not pointing to show interest are red flags at 18 months.

#337 · Pediatric Care

What developmental milestones are expected at 2 years of age?

Answer

At 24 months (2 years), children should use at least 50 words and combine 2-word phrases (such as 'more milk' or 'daddy go'). They run (though falls occur), kick a ball, engage in parallel play (playing alongside but not with other children), and point to objects when asked. An unfamiliar listener should understand approximately 50% of the child's speech. Fewer than 50 words or no 2-word combinations by age 2 are referral-warranting red flags.

#338 · Pediatric Care

What developmental milestones are expected at 3 years of age?

Answer

At 3 years, children produce 3-word sentences (subject-verb-object), pedal a tricycle, name colors, and engage in associative play with other children. Strangers should understand approximately 75% of the child's speech. They can draw a circle, climb stairs with alternating feet, and dress/undress with assistance. Red flags at 3 years include unintelligible speech, inability to speak in short sentences, or persistent solitary play with no interest in peers.

#339 · Pediatric Care

What developmental milestones are expected at 4 years of age?

Answer

At 4 years, children tell stories with beginning, middle, and end; hop on one foot; draw a person with at least 4 body parts; and engage in cooperative play. Speech should be almost fully intelligible to strangers. They can button large buttons, catch a bounced ball, and follow 3-step commands. Red flags at age 4 include inability to draw a circle, use 3-word sentences, or engage in fantasy play.

#340 · Pediatric Care

What developmental milestones are expected at 5 years of age?

Answer

At 5 years, children use complete 5-word sentences, skip on alternating feet, print some letters and their own name, and know their home address. They count to 10, draw a person with 6 or more parts, and play cooperative organized games with rules. Speech is fully intelligible. Red flags at age 5 include inability to print any letters, difficulty being understood by strangers, or avoiding social interaction.

#341 · Pediatric Care

What vaccines are given at birth, and what is the rationale?

Answer

The only vaccine routinely given at birth is Hepatitis B (Hep B), dose 1 of a 3-dose series. The rationale is to prevent perinatal and early childhood hepatitis B acquisition, which has a 90% chance of progressing to chronic hepatitis B infection compared to less than 5% risk in adults. Infants of HBsAg-positive mothers must also receive Hepatitis B Immune Globulin (HBIG) within 12 hours of birth.

#342 · Pediatric Care

What vaccines are given at 2 months, and which ones require multiple doses?

Answer

Vaccines given at 2 months include DTaP (diphtheria, tetanus, acellular pertussis), IPV (inactivated poliovirus), Hib (Haemophilus influenzae type b), PCV15 or PCV20 (pneumococcal conjugate), and RV (rotavirus oral vaccine). All require multiple doses: DTaP needs 5 total doses, IPV needs 4, Hib needs 3-4 depending on brand, and PCV needs 3-4 doses. RV needs 2 doses (Rotarix) or 3 doses (RotaTeq).

#343 · Pediatric Care

When should influenza vaccine start in infants, and what is the initial dosing schedule?

Answer

Influenza vaccine is recommended annually for all individuals 6 months and older. Children under 9 years of age receiving influenza vaccine for the first time require 2 doses given at least 4 weeks apart. In subsequent years, only 1 dose is needed. Inactivated influenza vaccine (IIV) is preferred over live attenuated influenza vaccine (LAIV, nasal spray) in children with asthma, immunocompromised children, and children under 2 years of age.

#344 · Pediatric Care

What vaccines are given at 12-15 months, and which require a booster?

Answer

Vaccines at 12-15 months include MMR (measles, mumps, rubella, dose 1), varicella (dose 1), Hib booster (dose 3 or 4 depending on brand), PCV booster, and Hepatitis A (dose 1 of a 2-dose series). MMR and varicella both receive a second booster dose at age 4-6 years. Hepatitis A dose 2 is given 6-18 months after dose 1. Hep A can be given anytime between 12-23 months.

#345 · Pediatric Care

What vaccines are given at 4-6 years, and which ones are boosters?

Answer

The 4-6 year visit includes booster doses of DTaP (dose 5), IPV (dose 4), MMR (dose 2), and varicella (dose 2). These boosters are critical for maintaining immunity as the primary series response wanes. After the 4-6 year DTaP booster, the next tetanus-containing vaccine is Tdap at age 11-12, followed by Td every 10 years.

#346 · Pediatric Care

What vaccines are recommended at the 11-12 year adolescent visit?

Answer

The 11-12 year platform vaccines include Tdap (one dose, replacing a Td booster), HPV vaccine (2-dose series for those starting at ages 9-14, with second dose at 6-12 months; 3-dose series for those starting at ages 15 or older), and meningococcal vaccine MenACWY (with booster at age 16). Influenza vaccine is also given annually. MenB (meningococcal B) may be offered at age 16-23 using shared decision-making.

#347 · Primary Care Medicine

What vaccines are recommended for adults 65 years and older?

Answer

Adults 65 and older should receive high-dose or adjuvanted influenza vaccine annually, as standard-dose flu vaccine is less effective in this age group. Pneumococcal vaccines recommended are PCV20 alone, or PCV15 followed by PPSV23 at least 1 year later. Recombinant zoster vaccine (RZV, Shingrix) is recommended as a 2-dose series given 2-6 months apart for herpes zoster prevention. Tdap should be given once if not previously received, then Td every 10 years.

#348 · Pediatric Care

What are the diagnostic criteria for acute otitis media (AOM), and when can observation be used instead of antibiotics?

Answer

AOM requires acute onset of symptoms, middle ear effusion (MEE), and signs/symptoms of middle ear inflammation (erythema or bulging tympanic membrane, ear pain, or fever). First-line antibiotic is amoxicillin 80-90 mg/kg/day in 2 divided doses. Observation without antibiotics (48-hour watchful waiting) is an option for children 2 years and older with mild unilateral AOM. Children under 2 years old, those with bilateral AOM, or those with severe symptoms (fever above 39 degrees Celsius, severe pain) always require antibiotics.

#349 · Pediatric Care

How is AOM managed in a child with a penicillin allergy?

Answer

For a child with a non-anaphylactic penicillin allergy (rash only, no hives or systemic symptoms), cephalosporins such as cefdinir (14 mg/kg/day) or cefuroxime can be used, as cross-reactivity is approximately 1-2%. For a child with an anaphylactic penicillin allergy (urticaria, angioedema, anaphylaxis), clindamycin or azithromycin are used, though azithromycin has lower efficacy against *Streptococcus pneumoniae*. Trimethoprim-sulfamethoxazole has significant resistance.

#350 · Pediatric Care

What is the management of recurrent or treatment-failure AOM?

Answer

Recurrent AOM is defined as 3 or more episodes in 6 months or 4 or more in 12 months. Treatment failure AOM (no improvement after 48-72 hours on amoxicillin) should be treated with amoxicillin-clavulanate (90 mg/kg/day of amoxicillin component with 6.4 mg/kg/day clavulanate). Referral to ENT for tympanostomy tube placement is appropriate for recurrent AOM, persistent MEE over 3 months with hearing loss, or chronic AOM with perforation.

#351 · Pediatric Care

What is bronchiolitis, what causes it, and what is the evidence-based management?

Answer

Bronchiolitis is a lower respiratory tract infection causing inflammation and mucus in the small airways (bronchioles), most commonly caused by respiratory syncytial virus (RSV) in infants under 2 years old. It presents with rhinorrhea, cough, fever, and wheezing or crackles. Evidence-based management is supportive only: supplemental oxygen (for saturations below 90-92%), nasal suctioning, and adequate hydration. Bronchodilators (albuterol), steroids, and antibiotics are not routinely recommended and have not shown benefit.

#352 · Pediatric Care

What is croup, and how is its severity managed?

Answer

Croup (viral laryngotracheobronchitis) most commonly caused by parainfluenza virus presents with a sudden-onset barking seal-like cough, inspiratory stridor, and hoarseness, often worse at night. The Westley croup score measures severity. Dexamethasone 0.6 mg/kg PO, IM, or IV (max 16 mg) is recommended for all degrees of croup severity. Nebulized racemic epinephrine 2.25% solution (0.5 mL in 3 mL NS) is reserved for moderate-to-severe croup with significant stridor at rest. Cool mist therapy has not been shown to be effective.

#353 · Pediatric Care

What defines a simple versus complex febrile seizure, and how does management differ?

Answer

A simple febrile seizure is single, generalized (not focal), lasts less than 15 minutes, and does not recur within 24 hours. Management is reassurance and treatment of the underlying fever; lumbar puncture is not routinely needed. A complex febrile seizure is focal, lasts more than 15 minutes, or recurs within 24 hours. Complex febrile seizures warrant a full seizure workup including EEG, neuroimaging (MRI), and lumbar puncture if meningitis is suspected. Prophylactic anticonvulsants are not routinely recommended.

#354 · Pediatric Care

What is the first-line antibiotic treatment for group A streptococcal pharyngitis in children?

Answer

First-line treatment for group A strep pharyngitis (GAS) is amoxicillin 50 mg/kg/day (max 1000 mg/day) once daily or in divided doses for 10 days. Penicillin V (25-50 mg/kg/day in divided doses) for 10 days is also appropriate. In penicillin allergy: non-anaphylactic allergy = cephalexin 20 mg/kg/day (max 500 mg) twice daily for 10 days; anaphylactic allergy = clindamycin 7 mg/kg/dose three times daily for 10 days or azithromycin 12 mg/kg/day for 5 days. Full 10-day course is needed to prevent rheumatic fever.

#355 · Pediatric Care

How should iron deficiency be prevented in breastfed infants, and how is iron deficiency anemia (IDA) treated in toddlers?

Answer

Exclusively breastfed infants require iron supplementation at 1 mg/kg/day of elemental iron drops starting at 4 months of age, continuing until iron-rich foods are introduced. Formula-fed infants receive adequate iron from iron-fortified formula and do not need supplementation. Cow's milk should be limited to 16-24 oz per day in toddlers (12-24 months) to prevent displacement of iron-rich foods. IDA treatment requires elemental iron 3-6 mg/kg/day between meals; hemoglobin should be rechecked in 4 weeks to confirm response.

#356 · Pediatric Care

What are the autism spectrum disorder (ASD) red flags at 18 months?

Answer

Critical ASD red flags at 18 months include no pointing to show interest (protodeclarative pointing), no single meaningful words, no imitation of actions or words, and lack of social smile or response to name by 12 months. The M-CHAT-R/F (Modified Checklist for Autism in Toddlers, Revised with Follow-Up) is the recommended ASD screening tool at 18 and 24-month well visits. Any loss of previously acquired language or social skills at any age is an immediate referral red flag.

#357 · Pediatric Care

What are the mandated reporting obligations for suspected child abuse, and what physical findings raise suspicion?

Answer

Healthcare providers are mandated reporters who must report reasonable suspicion (not proof) of child abuse or neglect to Child Protective Services (CPS). No parental consent is required. Physical findings raising suspicion include bruises in non-ambulatory infants, bruises in unusual locations (ears, neck, torso), fractures inconsistent with developmental stage, bilateral retinal hemorrhages, patterned burns, and injuries in multiple stages of healing. Skeletal survey is the initial imaging evaluation for suspected physical abuse in children under 2 years.

#358 · Pediatric Care

What is the acute management of a sickle cell vaso-occlusive crisis (VOC)?

Answer

Management of sickle cell VOC includes intravenous hydration (1.5 times maintenance rate with normal saline or D5W with NS), opioid analgesia (morphine 0.05-0.1 mg/kg IV q3-4h with PCA for severe pain), supplemental oxygen only if the patient is hypoxic (O₂ sat below 95%), and incentive spirometry to prevent acute chest syndrome. NSAIDs can be added for mild-to-moderate pain. If the patient is febrile, empiric broad-spectrum IV antibiotics (ceftriaxone 50-75 mg/kg) are started immediately due to functional asplenia and high sepsis risk.

#359 · Pediatric Care

What is the step-up treatment for moderate persistent asthma in children, and what constitutes good asthma control?

Answer

Moderate persistent asthma (daily symptoms, activity limitation, nighttime awakening more than once per week) in children is treated with medium-dose inhaled corticosteroids (ICS) plus a long-acting beta-2 agonist (LABA) such as salmeterol or formoterol. Good asthma control means fewer than 2 symptomatic days per week, no nighttime awakening, no activity limitation, and rescue inhaler use no more than 2 days per week. Montelukast can be added as adjunct therapy. LABAs should never be used without ICS in children.

#360 · Pediatric Care

What diagnostic workup is required for a first UTI in a male child?

Answer

Any UTI in a male child is unusual and warrants further structural evaluation. After treating the acute UTI, a renal and bladder ultrasound (RBUS) is indicated to detect hydronephrosis, renal scarring, or anatomic abnormalities. Voiding cystourethrogram (VCUG) is performed to evaluate for vesicoureteral reflux (VUR), which is present in approximately 30-40% of children with UTI. Grade III or higher VUR may require antibiotic prophylaxis (trimethoprim-sulfamethoxazole or nitrofurantoin) or surgical correction.

#361 · Pediatric Care

What is Osgood-Schlatter disease and how is it managed?

Answer

Osgood-Schlatter disease is a traction apophysitis of the tibial tubercle, caused by repetitive quadriceps contraction during periods of rapid adolescent growth, typically in active children ages 10-15 years. It presents with unilateral or bilateral anterior knee pain and tenderness over the tibial tubercle, worsened by running, jumping, and kneeling. Management includes activity modification (reducing high-impact activities), NSAIDs (ibuprofen 10 mg/kg every 6-8 hours), stretching/physical therapy, and ice. It is self-limiting, resolving with skeletal maturity.

#362 · Pediatric Care

What is pyloric stenosis and how is it diagnosed and treated?

Answer

Pyloric stenosis is hypertrophy of the pyloric muscle causing gastric outlet obstruction, typically presenting at 2-8 weeks of age (rarely up to 5 months). Classic presentation is projectile non-bilious vomiting after every feeding, with a hungry infant despite vomiting. An olive-shaped mass may be palpated in the right upper quadrant. Laboratory findings show hypochloremic, hypokalemic metabolic alkalosis. Abdominal ultrasound (pyloric length greater than 16 mm, muscle thickness greater than 4 mm) is diagnostic. Treatment is surgical pyloromyotomy after electrolyte correction.

#363 · Pediatric Care

What is precocious puberty and how is it evaluated and treated?

Answer

Precocious puberty is defined as breast development (thelarche) before age 8 in girls or testicular enlargement (greater than 4 mL or greater than 2.5 cm) before age 9 in boys. Evaluation includes bone age X-ray, LH, FSH, estradiol (girls) or testosterone (boys), DHEA-S, and pelvic ultrasound in girls. GnRH stimulation test differentiates central (gonadotropin-dependent) from peripheral (gonadotropin-independent) precocious puberty. Referral to pediatric endocrinology is required. Central precocious puberty is treated with GnRH agonist (leuprolide depot) to halt progression and preserve final adult height.

#364 · Pediatric Care

What are the recommended developmental screening tools at well-child visits?

Answer

The AAP recommends standardized developmental screening at 9, 18, and 30-month well-child visits using validated tools. The Ages and Stages Questionnaire (ASQ-3) screens for developmental delays across 5 domains: communication, gross motor, fine motor, problem-solving, and personal-social. The M-CHAT-R/F is used specifically for autism screening at 18 and 24 months. Positive screens on any validated tool warrant a comprehensive developmental evaluation and referral to early intervention services.

#365 · Pediatric Care

What is the approach to an infant with a first febrile episode under 28 days old?

Answer

Neonates under 28 days with fever (rectal temperature above 38 degrees Celsius) are at high risk for serious bacterial infection (SBI) including sepsis and meningitis. Full sepsis workup is required: CBC with differential, blood culture, urinalysis, urine culture via catheter, CSF analysis (cell count, glucose, protein, culture), and chest X-ray if respiratory symptoms are present. These infants are admitted and started on empiric IV antibiotics (ampicillin plus ceftriaxone or ampicillin plus gentamicin) pending culture results. Empiric acyclovir is added if HSV is suspected.

#366 · Pediatric Care

What is the clinical presentation and management of intussusception in children?

Answer

Intussusception occurs when one segment of intestine telescopes into an adjacent segment, most commonly at the ileocecal junction, in children ages 3 months to 6 years. Classic presentation is intermittent colicky abdominal pain with drawing up of legs, followed by periods of lethargy, and currant-jelly stools (blood and mucus) in later stages. Diagnosis is confirmed with abdominal ultrasound (target sign). Treatment is pneumatic or hydrostatic enema, which is both diagnostic and therapeutic with approximately 90% success rate. Surgical reduction is needed for failed enema or perforation.

#367 · Pediatric Care

What are the clinical features and management of Kawasaki disease?

Answer

Kawasaki disease (mucocutaneous lymph node syndrome) is a vasculitis of unknown cause, most common in children under 5 years old, particularly in Asian descent. Classic diagnosis requires fever for at least 5 days plus at least 4 of 5 criteria: bilateral non-exudative conjunctivitis, mucous membrane changes (strawberry tongue, cracked lips), polymorphous rash, extremity changes (erythema or edema of palms/soles, desquamation), and cervical lymphadenopathy. Treatment is IVIG 2 g/kg single infusion plus high-dose aspirin (80-100 mg/kg/day) to prevent coronary artery aneurysms.

#368 · Pediatric Care

What is the management of a child with suspected ADHD in primary care?

Answer

ADHD evaluation in primary care requires gathering information from multiple settings (home and school) using validated rating scales such as the Vanderbilt Assessment Scale or Conners rating scale, completed by parents and teachers. DSM-5 requires 6 or more inattentive and/or hyperactive-impulsive symptoms in at least 2 settings, present before age 12, and causing functional impairment. For children under 6, behavioral therapy alone is first-line. For children 6 and older, stimulant medications (methylphenidate starting at 5 mg twice daily, or amphetamine salts) combined with behavioral therapy produce the best outcomes.

#369 · Pediatric Care

What is the appropriate management of a child with a closed head injury and what signs suggest significant intracranial injury?

Answer

PECARN (Pediatric Emergency Care Applied Research Network) clinical decision rules guide management of pediatric closed head injuries. Children under 2 years old with GCS less than 15, palpable skull fracture, altered mental status, severe injury mechanism, or worsening headache/vomiting should have CT head. Children 2 and older with GCS less than 15, loss of consciousness, vomiting (2 or more episodes), severe mechanism, or signs of basilar skull fracture need CT. Observation without imaging is appropriate for low-risk injuries. Post-concussive activity restriction includes cognitive and physical rest.

#370 · Pediatric Care

What is the clinical presentation and first-line treatment of cystic fibrosis (CF) exacerbation?

Answer

CF pulmonary exacerbations present with increased cough, change in sputum color or consistency, decreased exercise tolerance, weight loss, and decline in FEV1 by at least 10% from baseline. Most exacerbations are caused by *Pseudomonas aeruginosa*, *Staphylococcus aureus*, or *Burkholderia cepacia*. Treatment typically requires IV antibiotics covering *Pseudomonas* (anti-pseudomonal beta-lactam plus an aminoglycoside or fluoroquinolone), intensive airway clearance therapy, nutritional support, and optimization of CFTR modulator therapy. Hospitalization for 14 days is standard for moderate-to-severe exacerbations.

#371 · Pediatric Care

What are the growth parameters that define failure to thrive (FTT) in infants, and what is the evaluation approach?

Answer

Failure to thrive (FTT), now often called weight faltering, is generally defined as weight below the 3rd or 5th percentile for age, weight-for-length below the 2nd percentile, or weight that crosses 2 major percentile lines downward on the growth chart. The majority of cases (approximately 80%) are non-organic (related to inadequate caloric intake).

Evaluation includes a thorough feeding history, diet recall, observation of feeding, CBC, comprehensive metabolic panel, thyroid function tests, and urinalysis. The most common cause is insufficient caloric intake, and the primary intervention is caloric optimization.

#372 · Pediatric Care

What is the clinical presentation of epiglottitis and how does it differ from croup?

Answer

Epiglottitis (supraglottitis) most commonly caused by *Haemophilus influenzae* type b (before Hib vaccination) or *Streptococcus* species presents with abrupt onset of high fever, severe sore throat, dysphagia (drooling), muffled voice, and inspiratory stridor. The child often sits in the tripod position (leaning forward with neck extended). Unlike croup, epiglottitis does not produce a barking cough. Airway visualization should only be done in a controlled setting (OR) with anesthesia present. Treatment is IV antibiotics (ceftriaxone 50-75 mg/kg) and emergent airway management.

#373 · Pediatric Care

What is nursemaid's elbow (radial head subluxation) and how is it managed?

Answer

Nursemaid's elbow is subluxation of the radial head from the annular ligament, typically occurring in children 1-4 years old following longitudinal traction on the arm (being lifted by the hand or forearm). The child holds the arm in slight flexion and pronation, refusing to use it, with tenderness over the lateral elbow. Radiographs are usually normal and not required if the mechanism is classic. Reduction is accomplished with the hyperpronation technique (forearm pronation while flexing) or the supination-flexion technique. Resolution of symptoms within 10-15 minutes confirms successful reduction.

#374 · Pediatric Care

What are the recommended weight-based acetaminophen and ibuprofen dosing guidelines for pediatric fever?

Answer

Acetaminophen (Tylenol) is dosed at 10-15 mg/kg per dose every 4-6 hours, with a maximum of 5 doses in 24 hours and maximum daily dose of 75 mg/kg/day (not to exceed 4,000 mg/day). Ibuprofen (Advil, Motrin) is dosed at 5-10 mg/kg per dose every 6-8 hours, with a maximum single dose of 400 mg and maximum daily dose of 40 mg/kg/day. Ibuprofen should not be used in infants under 6 months of age. Alternating acetaminophen and ibuprofen (every 3-4 hours) is effective for fever control but requires careful parent education about dosing intervals.

#375 · Pediatric Care

What is the clinical presentation and management of meningococcal disease in children?

Answer

Meningococcal disease caused by *Neisseria meningitidis* presents with sudden-onset fever, severe headache, stiff neck, and a characteristic petechial or purpuric rash that does not blanch with pressure (non-blanching rash), indicating meningococemia. This is a medical emergency with 10-15% mortality. Empiric IV ceftriaxone 100 mg/kg/day should be initiated before lumbar puncture if clinically unstable. Prophylaxis with rifampin 10 mg/kg twice daily for 2 days or ciprofloxacin single dose is given to close contacts. Meningococcal ACWY vaccine at age 11-12 and 16 is critical prevention.

#376 · Pediatric Care

What is the diagnostic criteria and management of type 1 diabetes in children?

Answer

Type 1 diabetes (T1DM) in children presents with polyuria, polydipsia, polyphagia, and weight loss. Diagnostic criteria include fasting glucose above 126 mg/dL, random glucose above 200 mg/dL with symptoms, 2-hour glucose above 200 mg/dL on OGTT, or HbA1c above 6.5%. New-onset T1DM requires referral to pediatric endocrinology. Initial management in DKA includes IV fluid resuscitation (normal saline 10-20 mL/kg), followed by insulin drip starting at 0.05-0.1 units/kg/hour, and potassium replacement. Long-term management requires multiple daily insulin injections or insulin pump plus continuous glucose monitoring.

#377 · Pediatric Care

What is the clinical approach to a child with a murmur, and which features suggest a pathologic murmur?

Answer

Innocent murmurs in children are common (present in up to 80% of children at some point) and are characterized by being systolic, soft (grade I-II/VI), musical or vibratory in quality, loudest at the left lower sternal border or pulmonary area, and without radiation. The Still murmur (musical, vibratory, low-pitched) is the most common innocent murmur. Features suggesting pathologic murmur include diastolic timing, grade III or higher, harsh quality, radiation to the neck or back, associated clicks or S4, and abnormal cardiac symptoms (cyanosis, syncope, chest pain with exertion). Referral to pediatric cardiology is warranted for pathologic features.

#378 · Pediatric Care

What is the recommended approach to childhood obesity screening and management in primary care?

Answer

BMI percentile for age and sex is the recommended screening tool for childhood obesity. Overweight is defined as BMI between the 85th and 94th percentile; obesity is BMI at or above the 95th percentile; severe obesity is BMI at or above the 120% of the 95th percentile. First-line management is intensive behavioral and lifestyle treatment with 26 or more hours of contact over 3-12 months, involving the family. Pharmacotherapy with orlistat (above age 12) may be considered for adolescents with obesity and comorbidities. Metabolic/bariatric surgery referral is appropriate for adolescents with severe obesity (BMI above 40 or above 35 with significant comorbidities).

#379 · Pediatric Care

What is the recommended newborn screening and what conditions are detected?

Answer

Universal newborn screening (NBS) using heel-stick blood collection is performed on all newborns ideally at 24-48 hours of life. Each state determines its specific panel, but the ACMG recommends at least 35 core conditions. Common conditions screened include phenylketonuria (PKU), congenital hypothyroidism (most common treated condition), galactosemia, biotinidase deficiency, congenital adrenal hyperplasia, hemoglobin disorders (sickle cell disease), cystic fibrosis (IRT), and medium-chain acyl-CoA dehydrogenase (MCAD) deficiency. Critical congenital heart disease (CCHD) is screened with pulse oximetry in the newborn nursery.

#380 · Pediatric Care

What is the recommended approach to neonatal jaundice and when is phototherapy indicated?

Answer

Physiologic jaundice appears after 24 hours of life and peaks at day 3-5 in term infants. Total serum bilirubin (TSB) or transcutaneous bilirubin (TcB) should be measured in all neonates before discharge, and phototherapy thresholds are determined based on hour-specific nomograms and neurotoxicity risk factors (gestational age, isoimmune hemolytic disease, G6PD deficiency, sepsis). In a term newborn without risk factors, phototherapy is generally started at TSB levels around 15-18 mg/dL. Pathologic jaundice appearing within the first 24 hours requires immediate evaluation for hemolytic disease.

#381 · Pediatric Care

What is the recommended breastfeeding guidance for newborns and when should it be avoided?

Answer

The AAP recommends exclusive breastfeeding for approximately 6 months, with continued breastfeeding alongside complementary foods through 2 years of age or beyond.

Breastfeeding is contraindicated in HIV-positive mothers in high-income countries, mothers with active untreated tuberculosis, galactosemia in the infant, mothers actively using illicit drugs (cocaine, heroin), and mothers taking certain medications (chemotherapy agents, radioactive iodine). Hepatitis B and C are not absolute contraindications. Neonates should feed 8-12 times per day in the first weeks.

#382 · Pediatric Care

What are the clinical features and treatment of impetigo in children?

Answer

Impetigo is a highly contagious superficial bacterial skin infection most commonly caused by *Staphylococcus aureus* and *Streptococcus pyogenes*. Non-bullous impetigo presents as honey-colored crusted lesions on the face, around the nose and mouth. Bullous impetigo (caused by *S. aureus* toxins) presents with larger fluid-filled blisters. Localized disease is treated with topical mupirocin 2% cream or ointment three times daily for 5-7 days.

Extensive or bullous impetigo requires oral antibiotics: dicloxacillin or cephalexin 25-50 mg/kg/day in 4 divided doses for 7-10 days, or trimethoprim-sulfamethoxazole if MRSA is suspected.

What is the clinical presentation and management of eczema (atopic dermatitis) in infants and children?

Answer

Atopic dermatitis is a chronic inflammatory skin condition presenting with intensely pruritic erythematous patches, papules, and lichenification. In infants, lesions occur on the face and extensor surfaces; in older children and adults, the flexural surfaces (antecubital and popliteal fossae) are typically affected. First-line treatment is regular emollient application (ceramide-containing moisturizers), plus low-potency topical corticosteroids (hydrocortisone 1-2.5%) for mild disease and medium-to-high potency TCS for moderate-severe disease. Tacrolimus 0.03% or pimecrolimus ointment (calcineurin inhibitors) are used for sensitive areas and steroid-sparing. Dupilumab is FDA-approved for children 6 months and older with moderate-to-severe disease.

What is the recommended approach to the 15-month and 2-year well-child visit in terms of developmental assessment?

Answer

At the 15-month visit, developmental surveillance should assess walking, at least 3 words (by 15 months), pointing, and social reciprocity. At the 24-month visit, standardized developmental screening (ASQ-3) is required, plus autism-specific M-CHAT-R/F screening at 18 and 24 months. Speech delay at 2 years (fewer than 50 words, no 2-word combinations) warrants audiologic evaluation and speech therapy referral. Early intervention services (Part C of IDEA) are available for children under 3 years with developmental delays and should be referred without waiting for a formal diagnosis.

#385 · Pediatric Care

What is the clinical presentation and management of hand, foot, and mouth disease (HFMD)?

Answer

HFMD, most commonly caused by coxsackievirus A16 and enterovirus 71, presents in children under 5 years with fever (38-39 degrees Celsius), painful oral ulcers on the buccal mucosa and tongue, and a non-pruritic vesicular rash on the palms, soles, and buttocks. It is self-limiting and management is supportive: adequate hydration, acetaminophen or ibuprofen for pain and fever, and cool liquids or popsicles for oral comfort. Mouth pain may lead to poor fluid intake and dehydration requiring evaluation. Children should be excluded from daycare until afebrile and vesicles have dried.

#386 · Pediatric Care

What is the clinical approach to a child with suspected developmental dysplasia of the hip (DDH)?

Answer

DDH is screened at all newborn and well-child visits with the Barlow maneuver (to dislocate) and Ortolani maneuver (to reduce a dislocated hip). Risk factors include female sex, breech presentation, and family history. Hip ultrasound (not X-ray) is the preferred imaging modality in infants under 4-6 months because the femoral head is not yet ossified. X-ray is appropriate after 4-6 months of age. Treatment for DDH detected in infants under 6 months is the Pavlik harness, which maintains the hip in a position of flexion and abduction to allow normal acetabular development.

#387 · Pediatric Care

What are the signs of increased intracranial pressure (ICP) in an infant versus an older child?

Answer

In infants, elevated ICP presents with a bulging fontanelle (most reliable sign), irritability, high-pitched cry, vomiting, poor feeding, and macrocephaly from head circumference crossing upward percentiles. Sunset sign (eyes deviated downward with sclera visible above the iris) indicates severe ICP. In older children, ICP presents with the classic triad of headache (worse in the morning or with Valsalva), vomiting, and papilledema. Cushing triad (hypertension, bradycardia, irregular respirations) is a late, ominous sign of herniation. Urgent CT head without contrast is the initial imaging study.

#388 · Pediatric Care

What is the clinical presentation and management of periorbital versus orbital cellulitis in children?

Answer

Periorbital (preseptal) cellulitis is anterior to the orbital septum, presenting with eyelid swelling, erythema, and tenderness without proptosis, pain with eye movement, or limited extraocular motility. Orbital (postseptal) cellulitis is posterior to the septum and presents with proptosis, limited and painful eye movement, decreased vision, and fever. CT orbit with contrast is required to differentiate the two and detect subperiosteal abscess. Periorbital cellulitis in mild cases is treated with oral augmentin; orbital cellulitis requires IV antibiotics and urgent ophthalmology and ENT consultation for possible surgical drainage.

#389 · Pediatric Care

What is the treatment for pinworm infection (*Enterobius vermicularis*) in children?

Answer

Pinworm infection presents with perianal and perineal itching, particularly at night, when female worms migrate to lay eggs. Diagnosis is made by the scotch tape test (applied to perianal area in the morning before bathing). Treatment is mebendazole 100 mg as a single dose, or albendazole 400 mg as a single dose, or pyrantel pamoate 11 mg/kg (max 1 g) as a single dose. A repeat dose in 2 weeks is recommended due to frequent reinfection. All household members should be treated simultaneously. Rigorous hand washing and daily laundering of bedding are essential.

#390 · Pediatric Care

What is the recommended screening for lead toxicity in children, and what is the action level?

Answer

Blood lead level (BLL) testing is recommended for children at 12 and 24 months in high-risk communities, and for all children between 6 months and 5 years in areas where at least 25% of housing was built before 1960. The CDC and AAP define the blood lead reference value as 3.5 mcg/dL (lowered from 5 mcg/dL in 2021), which triggers public health action and environmental investigation. Chelation therapy (succimer/DMSA) is reserved for BLL above 45 mcg/dL. There is no safe blood lead level; primary prevention (removing lead sources) is the most important intervention.

What is the clinical approach to a child with anaphylaxis, and what is the first-line treatment?

Answer

Anaphylaxis is a severe, life-threatening allergic reaction defined by involvement of skin/mucosa plus either respiratory compromise or cardiovascular collapse, or two or more organ systems (skin, respiratory, GI, cardiovascular, or neurologic) after allergen exposure. First-line treatment is epinephrine 1 mg/mL (1:1000), dosed at 0.01 mg/kg IM into the anterolateral thigh (max 0.5 mg), which can be repeated every 5-15 minutes.

Antihistamines and steroids are adjuncts only and never replace epinephrine. Children should be monitored for 4-6 hours for biphasic reactions. Auto-injector selection is weight-based: approximately 7.5-15 kg uses 0.1 mg, approximately 15-30 kg uses 0.15 mg (EpiPen Jr), and over 30 kg uses 0.3 mg (adult EpiPen). The appropriate auto-injector should be prescribed upon discharge.

What are the criteria for diagnosing attention deficit hyperactivity disorder (ADHD) per DSM-5?

Answer

DSM-5 ADHD requires 6 or more inattentive symptoms and/or 6 or more hyperactive-impulsive symptoms (5 or more if age 17 or older), present in at least 2 settings, with onset before age 12, lasting at least 6 months, and causing significant functional impairment. Subtypes include predominantly inattentive, predominantly hyperactive-impulsive, and combined presentation. Symptoms must not be better explained by another mental disorder, and conditions such as anxiety, learning disability, sleep disorders, and vision/hearing problems must be considered in the differential.

#393 · Pediatric Care

What is the recommended well-child visit schedule in the first 2 years of life?

Answer

The AAP Bright Futures schedule recommends well-child visits at the following intervals in the first 2 years: 3-5 days (newborn visit), 1 month, 2 months, 4 months, 6 months, 9 months, 12 months, 15 months, 18 months, and 24 months. Each visit includes anthropometric measurements (weight, length, head circumference), developmental surveillance, vision and hearing screening, physical examination, anticipatory guidance, and age-appropriate vaccinations. Developmental screening using standardized tools is performed at 9, 18, and 30 months.

#394 · Pediatric Care

What is the clinical presentation and management of Legg-Calve-Perthes disease in children?

Answer

Legg-Calve-Perthes (LCP) disease is avascular necrosis of the femoral head, most common in boys ages 4-10 years. Presentation includes insidious onset of painless limp (or intermittent hip or knee pain), reduced internal rotation and abduction of the hip, and antalgic gait. X-ray initially may be normal; MRI detects early changes. Frog-leg lateral X-ray eventually shows flattening and fragmentation of the femoral head. Management varies by age of onset and severity: younger children (under 6) have better prognosis and may be observed; older children may require bracing or surgical containment (pelvic or femoral osteotomy).

#395 · Pediatric Care

What is the definition, evaluation, and management of infant and toddler constipation?

Answer

Functional constipation in infants and toddlers is defined by the Rome IV criteria as 2 or more of the following for at least 1 month: 2 or fewer stools per week, excessive stool retention, painful or hard bowel movements, large-diameter stools, or fecal incontinence after toilet training. Initial evaluation includes history and physical exam; digital rectal exam assesses stool burden. First-line treatment in infants 6 months or older is PEG (polyethylene glycol) 0.5-1 g/kg/day. In formula-fed infants, corn syrup or prune juice may help. Fiber and fluid intake should be optimized. Hirschsprung disease should be excluded if presentation is severe or from birth.

#396 · Pediatric Care

What are the key features distinguishing colic from other causes of infant crying?

Answer

Infant colic is defined by the Wessel criteria (rule of threes): crying for more than 3 hours per day, more than 3 days per week, for more than 3 weeks in an otherwise healthy, well-fed infant. It typically begins at 2-3 weeks, peaks at 6 weeks, and resolves by 3-4 months. The infant is otherwise thriving with normal growth. Red flags suggesting organic causes include weight loss, vomiting, fever, blood in stool, and abnormal physical exam. Management is reassurance, parental support, and ruling out identifiable causes. Simethicone has not shown consistent benefit; eliminating cow's milk protein may help in some breastfed infants.

What is the clinical presentation and initial management of a child with suspected appendicitis?

Answer

Classic appendicitis in older children and adolescents presents with periumbilical pain migrating to the right lower quadrant (RLQ), anorexia, nausea, vomiting, and low-grade fever. The Pediatric Appendicitis Score (PAS) or Alvarado score risk-stratifies patients. Physical findings include RLQ tenderness, Rovsing sign (RLQ pain with LLQ palpation), psoas sign, and obturator sign. Ultrasound is first-line imaging in children to avoid radiation; CT abdomen/pelvis with contrast is used if ultrasound is inconclusive. Surgical consultation for laparoscopic appendectomy is the definitive treatment. Perforated appendicitis requires IV antibiotics (piperacillin-tazobactam or ceftriaxone plus metronidazole) before surgery.

What is the recommended evaluation for a child with short stature?

Answer

Short stature is defined as height below the 3rd percentile or more than 2 standard deviations below the mean for age and sex. Evaluation begins with plotting the growth curve over time and calculating the mid-parental height (MPH: average of parents' heights adjusted for sex). Bone age X-ray (left hand and wrist) helps distinguish constitutional growth delay (bone age significantly less than chronologic age) from genetic short stature (bone age equals chronologic age). Lab workup includes CBC, CMP, TSH, free T4, IGF-1, celiac panel (TTG-IgA), and karyotype in girls to exclude Turner syndrome (45,XO). Growth hormone deficiency requires growth hormone stimulation testing.

#399 · Pediatric Care

What are the USPSTF and AAP recommendations for adolescent screening at the annual well visit?

Answer

Annual adolescent well visits (ages 11-21) should include height, weight, BMI screening; blood pressure measurement; depression screening using the PHQ-A (Adolescent version of PHQ-9) starting at age 12; substance use screening with CRAFFT; STI screening (chlamydia and gonorrhea in sexually active adolescents); HIV screening starting at age 15; and dyslipidemia screening with a fasting lipid panel between ages 9-11 and again at 17-21. Confidential time alone with the provider is essential starting at approximately age 11-12. HEADSS assessment covers Home, Education, Activities, Drugs, Sexuality, and Suicide.

#400 · Pediatric Care

What is the management of a child with newly diagnosed type 2 diabetes?

Answer

Pediatric type 2 diabetes (T2DM) is increasingly prevalent, particularly in adolescents with obesity and a family history. Initial treatment for most children without metabolic decompensation includes metformin (500 mg once daily, titrated to 1,000-2,000 mg/day in divided doses) plus intensive lifestyle intervention (diet modification and physical activity). Children with HbA1c above 9% at diagnosis, symptoms of hyperglycemia, or DKA should also start insulin therapy. HbA1c target is below 7% in most pediatric patients. Comorbidities including hypertension, dyslipidemia, and non-alcoholic fatty liver disease should be monitored and managed.

#401 · Professional Practice

What are the USPSTF mammography screening recommendations and the evidence grades?

Answer

The USPSTF recommends biennial mammography for women ages 40-74, with the recommendation grade changing: ages 40-49 received a C recommendation (individualized decision) in 2016 guidelines, updated to a B recommendation in 2024 (recommending screening for all women starting at 40). Ages 50-74 carry a B recommendation. Grade A/B recommendations are covered without cost-sharing under the ACA. Evidence does not support recommending for or against mammography beyond age 75, though clinical judgment applies.

#402 · Professional Practice

What are the USPSTF colorectal cancer screening recommendations?

Answer

The USPSTF recommends colorectal cancer (CRC) screening beginning at age 45 for average-risk adults, updated in 2021 from the previous start age of 50. Screening modalities include colonoscopy every 10 years, annual high-sensitivity fecal immunochemical test (FIT), or annual fecal occult blood test (FOBT). CT colonography every 5 years and flexible sigmoidoscopy every 5 years are also acceptable. Screening is recommended through age 75 (B grade); ages 76-85 use shared decision-making (C grade); not recommended after age 85.

What are the USPSTF cervical cancer screening recommendations?

Answer

Cervical cancer screening begins at age 21 regardless of sexual debut. Women ages 21-29 receive Pap smear alone every 3 years. Women ages 30-65 may receive Pap smear alone every 3 years, Pap plus HPV co-testing every 5 years, or HPV primary testing every 5 years. Screening can stop at age 65 if the woman has had adequate prior negative screening (3 consecutive negative Pap smears or 2 consecutive negative co-tests in the prior 10 years, with the most recent within 5 years). Women with hysterectomy (including cervix) do not require screening.

What are the USPSTF recommendations for osteoporosis screening?

Answer

The USPSTF recommends bone density screening (DEXA scan) for all women age 65 and older (B recommendation). For postmenopausal women under age 65, screening is recommended if the FRAX score (Fracture Risk Assessment Tool) indicates a 10-year risk of major osteoporotic fracture equivalent to or greater than that of a 65-year-old white woman with no additional risk factors (approximately 9.3%). The USPSTF does not recommend for or against screening men. Osteoporosis is defined as T-score of -2.5 or lower; osteopenia is T-score between -1.0 and -2.5.

What are the USPSTF recommendations for abdominal aortic aneurysm (AAA) screening?

Answer

The USPSTF recommends a one-time abdominal ultrasound screening for AAA in men ages 65-75 who have ever smoked (defined as 100 or more cigarettes in their lifetime), which is a B recommendation. AAA is defined as aortic diameter of 3.0 cm or greater. Referral to vascular surgery is indicated for AAA 5.5 cm or larger, or for rapid growth (more than 1 cm per year). The USPSTF found insufficient evidence to recommend screening in women who smoke or in men with no smoking history.

What are the USPSTF recommendations for lung cancer screening?

Answer

The USPSTF recommends annual low-dose CT (LDCT) of the chest for adults ages 50-80 years who have a 20 or more pack-year smoking history and currently smoke or have quit within the past 15 years (B recommendation, updated in 2021). Screening should be discontinued once a person has been tobacco-free for 15 years or develops a health problem substantially limiting life expectancy or the ability to have curative lung surgery. LDCT reduces lung cancer mortality by approximately 20% in high-risk individuals.

#407 · Professional Practice

What are the USPSTF recommendations for depression screening in adults?

Answer

The USPSTF recommends screening for depression in the general adult population (B recommendation), including pregnant and postpartum women, when adequate systems for diagnosis, treatment, and follow-up are in place. The PHQ-2 is an efficient initial screening tool: a positive screen (total score of 3 or more out of 6) warrants follow-up with the PHQ-9 for severity assessment. The USPSTF does not recommend routine screening in children under 12; evidence is insufficient for ages 12-18 per the current recommendations (check for updates as this has evolved).

#408 · Professional Practice

What are the USPSTF recommendations for diabetes screening?

Answer

The USPSTF recommends screening for prediabetes and type 2 diabetes in adults ages 35-70 who are overweight (BMI 25-29.9) or obese (BMI 30 or above), which is a B recommendation. Screening should be offered every 3 years for those with normal results. The ADA recommends screening starting at age 35 for all adults, and at any age for those with risk factors including first-degree relative with diabetes, high-risk race/ethnicity, hypertension, hyperlipidemia, PCOS, or GDM history. Fasting plasma glucose, HbA1c, or 75g OGTT are acceptable screening tests.

#409 · Professional Practice

What are the USPSTF recommendations for HIV screening?

Answer

The USPSTF recommends screening for HIV in adolescents and adults ages 15-65 (B recommendation), as well as in younger adolescents and older adults at increased risk. Screening should also be offered to all pregnant women. At least one lifetime HIV test is recommended for all adults in this age range. Adolescents and adults at increased risk (injection drug use, sex work, multiple partners without consistent condom use) should be screened more frequently, typically annually. Fourth-generation HIV antigen/antibody combination immunoassays are the recommended initial test.

#410 · Professional Practice

What are the USPSTF recommendations for hepatitis C screening?

Answer

The USPSTF recommends screening for hepatitis C virus (HCV) infection in adults ages 18-79 (B recommendation), regardless of risk factors, and in adolescents at increased risk. The anti-HCV antibody test is the initial screening test; a positive result requires confirmatory HCV RNA (viral load) testing to distinguish active infection from resolved prior infection. Adults born between 1945 and 1965 (baby boomers) have the highest prevalence. HCV is curable in more than 95% of cases with direct-acting antiviral (DAA) regimens.

What are the current ACIP influenza vaccine recommendations for adults, including special populations?

Answer

Annual influenza vaccination is recommended for all persons 6 months and older without contraindications. For adults 65 and older, high-dose quadrivalent influenza vaccine (Fluzone High-Dose), adjuvanted influenza vaccine (Fluad), or recombinant influenza vaccine (Flublok) are preferred over standard-dose vaccines due to better immunogenicity. Live attenuated intranasal influenza vaccine (LAIV, FluMist) should not be used in immunocompromised persons or pregnant women. Egg allergy of any severity is not a contraindication or precaution for any licensed influenza vaccine, including LAIV, per current ACIP guidance. Influenza vaccination is especially important in pregnant women (any trimester), elderly adults, and persons with chronic medical conditions.

What are the ACIP recommendations for Tdap and Td vaccines in adults?

Answer

Adults who have not previously received Tdap should receive one dose of Tdap regardless of the interval since their last Td booster. Tdap is recommended during every pregnancy, preferably between 27-36 weeks of gestation, to protect the newborn from pertussis before the infant's own vaccination series begins (cocooning strategy). After the initial Tdap, Td (tetanus-diphtheria) booster is given every 10 years. For tetanus-prone wounds, Td is given if more than 5 years have passed since the last dose.

What are the ACIP pneumococcal vaccine recommendations for adults 50 and older?

Answer

Adults 50 and older (lowered from 65 in October 2024) who have not previously received pneumococcal vaccine should receive either PCV20 alone (as a single-dose strategy), or PCV15 followed by PPSV23 given at least 1 year later (or at least 8 weeks in high-risk immunocompromised patients). PCV20 provides broader coverage against 20 pneumococcal serotypes. Adults who have already received PPSV23 should receive PCV20 or PCV15 at least 1 year after the last PPSV23 dose. Adults with certain high-risk conditions (asplenia, HIV, cochlear implant, CSF leak, immunocompromising conditions) may need additional doses and may be eligible earlier than age 50.

What are the ACIP recommendations for the recombinant zoster vaccine (RZV, Shingrix)?

Answer

Shingrix (RZV) is recommended for adults 50 years and older as a 2-dose series, with the second dose given 2-6 months after the first. RZV is more than 90% effective at preventing herpes zoster (shingles) and post-herpetic neuralgia. It is recommended even for patients who previously had shingles or received the older Zostavax (ZOSTAVAX) live vaccine, as RZV is substantially more effective. RZV is safe and recommended in immunocompromised adults 19 years and older (including those with HIV, hematologic malignancies, or on immunosuppressive therapy).

What are the ACIP HPV vaccine recommendations for adults?

Answer

The 9-valent HPV vaccine (Gardasil 9) is recommended as a 2-dose series (0 and 6-12 months) for individuals starting vaccination at ages 9-14. A 3-dose series (0, 1-2, and 6 months) is recommended for those starting at age 15 or older and for immunocompromised individuals. Routine vaccination is recommended through age 26 (A recommendation). For adults ages 27-45, shared decision-making is used, as benefit decreases due to likelihood of prior HPV exposure (C recommendation). The vaccine does not treat existing HPV infections.

What are the hepatitis B vaccination recommendations for adults?

Answer

The ACIP recommends hepatitis B vaccination for all unvaccinated adults through age 59, and using shared decision-making for adults 60 and older. The standard 3-dose regimen is Engerix-B or Recombivax HB at 0, 1, and 6 months. Heplisav-B is a 2-dose series given at 0 and 1 month, approved for adults 18 and older, and may be preferred for adherence. Adults with diabetes, CKD, HIV, liver disease, injection drug use, or multiple sexual partners are at high risk and should be prioritized. Post-vaccination serology (anti-HBs) is recommended only in high-risk individuals and healthcare workers.

#417 · Professional Practice

What hepatitis A vaccine recommendations apply to adults?

Answer

The ACIP recommends Hepatitis A vaccination for all children at age 1 year, unvaccinated adults at increased risk, and any adult who requests vaccination. Two-dose series (Havrix or Vaqta) at 0 and 6-18 months provides long-lasting immunity. High-risk adults include international travelers, men who have sex with men (MSM), persons with injection or non-injection drug use, those with liver disease (including HCV or HBV infection), and persons experiencing homelessness. Combined Twinrix (Hep A + Hep B) is a 3-dose series at 0, 1, and 6 months for adults requiring both vaccines.

#418 · Professional Practice

How does NP scope of practice vary across practice authority categories in the United States?

Answer

NP practice authority is categorized into three levels. Full practice authority (FPA) states allow NPs to evaluate, diagnose, treat, and prescribe independently without a physician collaboration agreement; approximately 27 states plus DC have FPA. Reduced practice states require NPs to have a collaborative practice agreement with a physician for at least one element of practice. Restricted practice states require physician supervision or team-based management for NP practice. The American Association of Nurse Practitioners (AANP) advocates for full practice authority in all states as evidence supports equivalent patient outcomes.

What are the HIPAA Privacy Rule requirements regarding protected health information (PHI)?

Answer

HIPAA's Privacy Rule protects individually identifiable health information (PHI) in any form, including electronic, written, or spoken. PHI can be disclosed without patient authorization for treatment, payment, and healthcare operations (TPO). Additional permitted disclosures without authorization include public health reporting, court orders, law enforcement, abuse reporting, and research with IRB approval. Patients have the right to access their own PHI, request amendments, and obtain an accounting of disclosures. Violations can result in civil penalties ranging from roughly \$100-\$71,000+ per violation depending on culpability tier (amounts are inflation-adjusted annually under HITECH), with an annual cap around \$2 million for willful, uncorrected violations, and criminal penalties for willful violations.

What is the difference between a living will, a POLST, and a healthcare proxy (durable power of attorney for healthcare)?

Answer

A living will is a written legal document expressing a patient's treatment preferences if they become unable to make decisions (e.g., no CPR, no mechanical ventilation). A POLST (Physician Orders for Life-Sustaining Treatment) is a medical order signed by a clinician, immediately actionable by first responders and healthcare providers, applicable in community settings. A healthcare proxy (durable power of attorney for healthcare) designates a specific person to make medical decisions on the patient's behalf when they lack capacity. The healthcare proxy overrides the living will only when the designated agent is available to make real-time decisions.

#421 · Professional Practice

Under what circumstances can minors consent to their own medical care without parental involvement?

Answer

Most states allow minors to consent independently to sexually transmitted infection (STI) testing and treatment, contraception services, mental health outpatient counseling, and substance abuse treatment. Emancipated minors (married, military service, self-supporting, or court-declared emancipated) can consent to all medical care. Mature minors in some states may consent if deemed capable of understanding and appreciating the nature and consequences of proposed treatment. Emergency medical care can always be provided to minors without parental consent. Providers should be familiar with their specific state laws.

#422 · Professional Practice

What is the PDSA cycle and how is it used in quality improvement?

Answer

The PDSA (Plan-Do-Study-Act) cycle is a structured quality improvement framework used to test changes in clinical practice. Plan: identify a problem, develop a hypothesis, and create an intervention. Do: implement the intervention on a small scale. Study: collect and analyze data to assess whether the change achieved the intended result. Act: if the intervention worked, standardize and expand it; if it did not work, revise the plan and cycle again. Multiple PDSA cycles iteratively refine the intervention. PDSA is the backbone of quality improvement models such as the Model for Improvement.

What is the hierarchy of evidence in clinical research?

Answer

From strongest to weakest evidence: systematic review and meta-analysis of multiple RCTs (highest level), individual randomized controlled trial (RCT), well-designed cohort study (prospective), case-control study (retrospective), cross-sectional study, case series and case reports, expert opinion and consensus statements (lowest level). The Oxford Centre for Evidence-Based Medicine (OCEBM) and GRADE frameworks provide detailed grading. Clinical practice guidelines synthesize evidence across studies and provide actionable recommendations graded by strength of evidence (A = strong, B = moderate, C = weak, D = expert opinion).

What constitutes Medicare/Medicaid billing fraud, and what is upcoding?

Answer

Healthcare billing fraud includes submitting claims for services not rendered, upcoding, unbundling (billing separately for services that should be billed together), and falsifying diagnoses to justify services. Upcoding means billing for a higher level of service (higher CPT code) than was actually provided or documented, resulting in higher reimbursement. The False Claims Act imposes civil monetary penalties of up to \$27,018 per false claim plus treble damages (3 times the amount fraudulently billed). The E/M (evaluation and management) level should reflect documented medical decision-making complexity or total time spent.

#425 · Professional Practice

What are the four principles of biomedical ethics?

Answer

The four principles of biomedical ethics (Beauchamp and Childress, Principles of Biomedical Ethics) are: autonomy (respecting the patient's right to make informed decisions about their own care), beneficence (acting in the best interest of the patient), non-maleficence (avoiding harm to the patient, 'first do no harm'), and justice (fair and equitable distribution of healthcare resources and fair treatment of all patients). Conflicts between these principles commonly arise in clinical practice and require systematic ethical analysis.

#426 · Professional Practice

What are the required elements of informed consent?

Answer

Valid informed consent requires five elements: disclosure of the diagnosis or nature of the condition, the proposed treatment or intervention, the material risks of the proposed treatment, the benefits of the proposed treatment, and the available alternatives (including the option of no treatment). The patient must have decision-making capacity (understanding, appreciation, reasoning, and ability to communicate a choice), consent must be given voluntarily without coercion, and the patient must be competent to consent. Capacity is assessed per clinical judgment and is decision-specific, not global.

#427 · Professional Practice

What conditions require mandatory reporting in most states?

Answer

Mandatory reporting requirements (which vary by state) typically include: suspected child abuse and neglect, elder abuse and neglect, domestic violence (in some states), gunshot wounds and stab wounds, certain communicable diseases (measles, tuberculosis, chlamydia, gonorrhea, HIV, syphilis, hepatitis A/B/C, etc.), known impaired drivers, and occupational diseases. Healthcare providers are mandated reporters for child abuse in all 50 states. Failure to report when legally required can result in civil liability and professional licensing consequences.

#428 · Professional Practice

What is the difference between a failure mode and effects analysis (FMEA) and a root cause analysis (RCA)?

Answer

FMEA (Failure Modes and Effects Analysis) is a proactive quality improvement tool used before an event occurs; it systematically identifies potential failure points in a process and their potential effects, allowing teams to implement preventive measures before harm occurs. RCA (Root Cause Analysis) is reactive, conducted after an adverse event or near-miss; it systematically identifies the underlying root causes (not just surface-level contributing factors) to prevent recurrence. FMEA uses a Risk Priority Number ($RPN = \text{severity} \times \text{occurrence} \times \text{detectability}$) to prioritize which failures to address first.

#429 · Professional Practice

What is the USPSTF recommendation for hypertension screening in adults?

Answer

The USPSTF recommends screening for hypertension in adults 18 and older (A recommendation). Blood pressure should be measured in the clinical setting; adults with elevated office readings (120-129/80 mmHg or above) should have hypertension confirmed with measurements outside the clinical setting (home blood pressure monitoring or ambulatory blood pressure monitoring) before initiating treatment. The 2017 ACC/AHA guideline defines hypertension as blood pressure above 130/80 mmHg; JNC 8 threshold was 140/90 mmHg. Hypertension is the most common condition managed in primary care.

#430 · Professional Practice

What are the USPSTF recommendations for lipid screening and cardiovascular risk assessment?

Answer

The USPSTF recommends using cardiovascular risk assessment (ASCVD 10-year risk calculation) to guide statin prescribing decisions for primary prevention, rather than cholesterol levels alone. The ACC/AHA Pooled Cohort Equations calculate 10-year ASCVD risk using age, sex, race, total cholesterol, HDL, systolic blood pressure, diabetes status, and smoking status. The USPSTF recommends low-to-moderate dose statins for adults 40-75 with 10% or higher 10-year ASCVD risk and at least one cardiovascular risk factor (B recommendation). Adults 76 and older lack sufficient evidence for a recommendation.

#431 · Professional Practice

What is the National Patient Safety Goal related to medication reconciliation?

Answer

The Joint Commission's National Patient Safety Goals require medication reconciliation at all care transitions, meaning a complete and accurate medication list must be compared against any newly ordered medications at admission, transfer between care settings, and discharge. The goal is to prevent medication omissions, duplications, dosing errors, and harmful drug interactions during transitions. At discharge, patients must receive a complete, reconciled medication list in an understandable format. Medication reconciliation is most error-prone at transitions from hospital to home or to post-acute care settings.

#432 · Professional Practice

What is the USPSTF recommendation for obesity screening and counseling in adults?

Answer

The USPSTF recommends screening all adults for obesity using BMI (B recommendation). Adults with BMI 30 or higher should be referred to or offered intensive multicomponent behavioral interventions (at least 12 sessions in the first year) focused on diet, physical activity, and behavioral change strategies. These interventions produce meaningful weight loss and reduce cardiovascular risk. Pharmacotherapy (such as orlistat, phentermine-topiramate, or semaglutide) may be added for adults with BMI 30 or higher, or 27 or higher with weight-related comorbidities, as an adjunct to lifestyle interventions.

What does a sensitivity versus specificity of a diagnostic test mean, and how do positive and negative predictive values differ?

Answer

Sensitivity (true positive rate) is the proportion of people WITH the disease who test positive; a highly sensitive test has few false negatives. Specificity (true negative rate) is the proportion of people WITHOUT the disease who test negative; a highly specific test has few false positives. Positive predictive value (PPV) is the probability that a positive test truly has the disease (depends on disease prevalence). Negative predictive value (NPV) is the probability that a negative test truly does not have the disease. In low-prevalence populations, even highly specific tests will have low PPV; use high-sensitivity tests to rule out disease (SnOut) and high-specificity tests to rule in disease (SpIn).

What are the components of an E/M (evaluation and management) visit that determine the billing level?

Answer

Since the 2021 CMS E/M coding update, outpatient office visit levels (99202-99215) are determined by either medical decision-making (MDM) complexity or total time spent on the day of the encounter (including chart review, documentation, and care coordination). MDM complexity is assessed across three elements: number and complexity of problems addressed, amount and complexity of data reviewed and ordered, and risk of complications/morbidity/mortality. Straightforward MDM = 99202/99212; low complexity = 99203/99213; moderate complexity = 99204/99214; high complexity = 99205/99215.

What is the process for obtaining emergency authorization to treat a patient who lacks decision-making capacity and has no surrogate?

Answer

When a patient lacks decision-making capacity and a healthcare emergency requires immediate intervention, the legal standard of implied consent allows treatment to proceed without a formal surrogate if delay would cause serious harm. For non-emergency decisions, a court-appointed guardian may be sought. In many states, a surrogate hierarchy is legally defined (spouse, adult children, parents, siblings, close friends) if no healthcare proxy document exists. Hospital ethics committees can be consulted when there is surrogate disagreement or clinical uncertainty. Two-physician attestation of incapacity is often required before implementing surrogate decision-making.

What is cultural competence in healthcare, and what are examples of culturally sensitive practices?

Answer

Cultural competence is the ability of healthcare providers to deliver services that effectively meet the social, cultural, and linguistic needs of diverse patient populations. Key practices include using professional medical interpreters (not family members) for patients with limited English proficiency, understanding culturally specific health beliefs and healing practices, addressing implicit bias in clinical decision-making, using a teach-back method to confirm patient understanding, and adapting patient education materials for literacy level. The National CLAS Standards (Culturally and Linguistically Appropriate Services) provide a framework for healthcare organizations.

What are the USPSTF recommendations for fall prevention in older adults?

Answer

The USPSTF recommends exercise interventions (particularly multifactorial exercise programs including balance training, strength training, and gait training) for community-dwelling adults 65 and older at increased risk for falls (B recommendation). Vitamin D supplementation alone is no longer recommended for fall prevention (D recommendation for vitamin D supplementation). The CDC's STEADI (Stopping Elderly Accidents, Deaths, and Injuries) algorithm recommends screening with three questions, and multifactorial assessment including medication review, vision assessment, and home safety evaluation for those at high risk.

What is the role of the NP in pain management and how does the CDC Opioid Prescribing Guideline inform practice?

Answer

The 2022 CDC Clinical Practice Guideline for Prescribing Opioids emphasizes non-opioid and non-pharmacologic therapies (physical therapy, CBT, NSAIDs, SNRIs) as first-line for chronic pain. If opioids are prescribed, the lowest effective dose should be used; MME (morphine milligram equivalents) above 50 MME/day requires reassessment, and above 90 MME/day carries high risk and should generally be avoided. Prescription Drug Monitoring Programs (PDMPs) should be checked before prescribing. Short-acting opioids are preferred over long-acting for acute pain. Opioid treatment agreements and urine drug screens are best practices for chronic opioid therapy.

What are the key components of a safe discharge plan in primary care or hospital settings?

Answer

A comprehensive discharge plan includes a reconciled medication list with explanation of each medication's purpose, follow-up appointment scheduled before discharge (within 1-2 weeks for high-risk patients), clear written instructions for warning signs requiring emergency care, patient education with teach-back confirmation of understanding, referrals to home health or community services as appropriate, and communication with the receiving provider (primary care or specialist). The Centers for Medicare and Medicaid Services (CMS) use 30-day readmission rates as a quality metric; high-risk diagnoses include CHF, COPD, pneumonia, AMI, CABG, and total hip/knee replacement.

What are the USPSTF recommendations for skin cancer prevention?

Answer

The USPSTF recommends counseling children, adolescents, and young adults ages 6 months to 24 years with fair skin about minimizing UV exposure (B recommendation). For adults 25 and older, the USPSTF found insufficient evidence to recommend for or against behavioral counseling for skin cancer prevention (I statement). The USPSTF does not recommend routine skin cancer screening with total body skin examination in asymptomatic adults (I statement for evidence insufficiency). The American Academy of Dermatology recommends annual total body skin exam, particularly for high-risk individuals (previous skin cancer, family history, multiple atypical nevi).

#441 · Professional Practice

What is the role of preconception care in primary care practice?

Answer

Preconception care optimizes maternal and fetal health outcomes by identifying and managing risks before pregnancy. Key components include folic acid 400-800 mcg/day (4 mg/day for high-risk patients such as prior neural tube defect pregnancy), review and adjustment of teratogenic medications (valproate, isotretinoin, ACE inhibitors, warfarin, methotrexate, statins), vaccination update (MMR, varicella if non-immune, influenza), screening for diabetes, hypertension, thyroid disease, HIV, and STIs, and counseling on smoking cessation, alcohol avoidance, and healthy weight. Ideal BMI before conception is 18.5-24.9 kg/m².

#442 · Professional Practice

What is a collaborative practice agreement (CPA) and when is it required?

Answer

A collaborative practice agreement (CPA) is a formal, written agreement between a nurse practitioner and a supervising or collaborating physician that defines the scope of the NP's practice, prescribing authority, and mechanisms for physician consultation and referral. CPAs are required in reduced practice authority states for at least one element of NP practice, and in restricted practice states for all NP activities. In full practice authority states, CPAs are not legally required but may be voluntarily used. CPAs typically specify geographic limitations, on-call coverage requirements, chart review percentages, and prescription formulary restrictions.

#443 · Professional Practice

What are the USPSTF recommendations for intimate partner violence (IPV) screening?

Answer

The USPSTF recommends screening women of reproductive age (approximately 14-46 years) for IPV and referring those who screen positive to ongoing support services (B recommendation). Validated screening tools include the HITS (Hurt, Insult, Threaten, Scream) questionnaire, the WAST (Woman Abuse Screening Tool), and the Danger Assessment. Screening should be done in private, without the partner present. In most states, IPV is not mandated to be reported unless a firearm injury is involved; patient safety and autonomy guide reporting decisions. All providers must be familiar with local IPV resources and safety planning.

#444 · Professional Practice

What is the Triple Aim and how does it guide healthcare system improvement?

Answer

The Triple Aim (Institute for Healthcare Improvement, IHI) provides a framework for optimizing health system performance with three simultaneous goals: improving the patient experience of care (including quality and satisfaction), improving the health of populations, and reducing the per capita cost of healthcare. An additional fourth aim, improving the work life of healthcare providers to prevent burnout, has been added to create the Quadruple Aim. FNP's contribute to Triple/Quadruple Aim goals through preventive care, chronic disease management, care coordination, and reducing unnecessary hospitalizations.

#445 · Professional Practice

What are the components of motivational interviewing (MI) and how is it applied in preventive care?

Answer

Motivational interviewing is a person-centered counseling approach using four core processes: engaging (building rapport), focusing (identifying a change target), evoking (eliciting change talk from the patient), and planning (developing a change plan). Core skills are OARS: Open-ended questions, Affirmations, Reflective listening, and Summaries. The provider expresses empathy, develops discrepancy between current behavior and stated goals, rolls with resistance, and supports self-efficacy. MI is effective for smoking cessation, weight loss counseling, medication adherence, alcohol reduction, and diabetes self-management.

#446 · Professional Practice

What is the USPSTF recommendation for pre-exposure prophylaxis (PrEP) for HIV prevention?

Answer

The USPSTF recommends prescribing PrEP to persons at increased risk for HIV infection (A recommendation). Increased risk includes men who have sex with men (MSM) with a recent STI or inconsistent condom use, heterosexual persons with HIV-positive partners or multiple partners, and persons who inject drugs. Before initiating PrEP, HIV testing must confirm the patient is HIV-negative, along with renal function (eGFR) and hepatitis B status. First-line PrEP is tenofovir disoproxil fumarate/emtricitabine (Truvada, 1 tablet daily) or tenofovir alafenamide/emtricitabine (Descovy, not approved for receptive vaginal sex). Monitoring includes HIV testing, renal function, and STI screening every 3 months.

#447 · Professional Practice

What is the legal standard for negligence in healthcare, and what must a plaintiff prove in a medical malpractice case?

Answer

Medical malpractice is a form of negligence requiring the plaintiff to prove four elements: duty (a provider-patient relationship existed establishing a duty of care), breach of duty (the provider failed to meet the standard of care expected of a reasonably competent provider), causation (the breach directly caused the patient's harm), and damages (actual harm or injury resulted). The standard of care is typically established through expert witness testimony. Nurse practitioners are held to the standard of care of a reasonably competent NP with similar training and in a similar clinical setting, not to the physician standard.

#448 · Professional Practice

What is the difference between Medicare Part A, B, C, and D?

Answer

Medicare Part A covers inpatient hospital care, skilled nursing facility care, hospice, and some home health services. Part B covers outpatient services, preventive care, physician visits, durable medical equipment, and some home health. Part C (Medicare Advantage) is offered by private insurers and covers all Part A and B benefits, often with additional benefits such as dental, vision, and hearing. Part D covers prescription drugs through private plan contracts. Most people qualify for Part A at age 65 without premium payment if they or their spouse worked and paid Medicare taxes for 40 quarters; Part B requires a monthly premium.

What are the USPSTF recommendations for statin use for primary prevention of cardiovascular disease?

Answer

The USPSTF recommends prescribing low-to-moderate intensity statins for primary CVD prevention in adults ages 40-75 who meet all three criteria: one or more CVD risk factors (dyslipidemia, diabetes, hypertension, or smoking), calculated 10-year CVD event risk of 10% or higher, and no prior cardiovascular events (B recommendation). Statins are not recommended when 10-year risk is below 7.5% (I statement, insufficient evidence). The risk discussion should include potential harms (myopathy, elevated liver enzymes, small increased risk of diabetes) and benefits (reduction in MI, stroke, and CVD mortality by approximately 25-35%).

What is the nurse practitioner's role in managing patients with multiple chronic conditions (MCCs)?

Answer

Management of patients with multiple chronic conditions (MCCs) requires a patient-centered, holistic approach that prioritizes the patient's goals and preferences over guideline-based treatment of individual diseases in isolation. Key strategies include comprehensive medication reconciliation to minimize polypharmacy (reviewing all medications for interactions, appropriateness, and burden), advance care planning, care coordination across specialists using integrated health records, functional status and quality-of-life assessment, and shared decision-making. The Chronic Care Model emphasizes self-management support, clinical information systems, delivery system redesign, and community resources as core components.

#451 · Professional Practice

What is the role of the NP in advance care planning, and what tools facilitate these conversations?

Answer

Advance care planning (ACP) is a process of reflection, communication, and documentation of a patient's values, goals, and preferences for future medical care. NPs can initiate ACP conversations using tools such as the POLST (for seriously ill or frail patients), Five Wishes (consumer-friendly advance directive), and Respecting Choices conversation guides. ACP should be documented in the medical record and revisited at each annual visit or when a significant change in health status occurs. ACP is associated with better goal-concordant care, reduced unwanted hospitalizations, and improved family satisfaction at end of life.

#452 · Professional Practice

What is the USPSTF recommendation for cognitive impairment screening in older adults?

Answer

The USPSTF found insufficient evidence to recommend for or against routine cognitive impairment screening in older adults without complaints or concerns (I statement, 2020). However, the Medicare Annual Wellness Visit requires detection of cognitive impairment as a covered benefit, and clinicians should evaluate any patient or family concern about memory or cognition. The Mini-Cog (3-word recall plus clock drawing) is a brief validated screen. The Montreal Cognitive Assessment (MoCA) is the preferred tool for a more sensitive assessment; a score below 26 out of 30 suggests mild cognitive impairment.

What is the difference between incidence and prevalence in epidemiology?

Answer

Incidence is the number of new cases of a disease occurring in a defined population during a specified time period; it measures the rate at which people without the disease develop it. Prevalence is the total number of existing cases (new and old) of a disease in a defined population at a specific point in time; it measures the burden of disease. Prevalence is affected by both incidence and disease duration. For a common chronic disease like hypertension, prevalence is high while incidence may be moderate. $\text{Prevalence} = \text{incidence} \times \text{average disease duration}$ (for stable disease).

What are the key elements of the Annual Wellness Visit (AWV) for Medicare beneficiaries?

Answer

The Medicare Annual Wellness Visit (AWV) is a covered preventive visit (no copay) and includes: health risk assessment, review and update of the medical and family history, vital signs and BMI, cognitive impairment assessment, depression screening (PHQ-2/9), functional ability assessment, fall risk assessment, review of current medications and supplements, creation or update of personalized prevention plan, and advance care planning discussion. It does not include a complete physical exam or management of acute or chronic conditions (a separate E/M visit would be needed and billed separately for those services).

What is the USPSTF recommendation for aspirin use for primary prevention of CVD?

Answer

The USPSTF recommends against initiating low-dose aspirin for primary CVD prevention in adults 60 years and older (D recommendation). For adults ages 40-59 with 10% or higher 10-year CVD risk, the decision to start low-dose aspirin should be individualized (C recommendation). The primary concern is that bleeding risk (GI bleeding, hemorrhagic stroke) increases with age and in many patients outweighs the CVD benefit when aspirin is used for primary prevention. Aspirin remains indicated for secondary prevention (after MI, stroke, or established coronary artery disease).

What is the USPSTF recommendation for vitamin D supplementation?

Answer

The USPSTF found insufficient evidence to recommend for or against vitamin D screening or supplementation for primary disease prevention in asymptomatic adults (I statement). For fall prevention specifically, the USPSTF recommends against supplementing vitamin D (alone or with calcium) in community-dwelling non-institutionalized adults 60 and older to prevent falls (D recommendation), as evidence shows no benefit and possible harm. Vitamin D deficiency treatment (when clinically diagnosed) remains clinically appropriate and separate from population-level prevention recommendations.

What is the NP's role in managing patients at end of life, and what does palliative care versus hospice mean?

Answer

Palliative care focuses on relief of suffering, symptom management, and improving quality of life for patients with serious illness at any stage of disease, and can be provided alongside curative treatment. Hospice is a specific program of palliative care for patients with a terminal prognosis of 6 months or less if the disease runs its natural course, requiring the patient to forgo curative treatment for their terminal diagnosis. Medicare Hospice Benefit covers medications, nursing visits, aide services, chaplaincy, social work, and bereavement counseling. NPs can certify hospice eligibility under the Medicare Conditions of Participation.

What are the key elements of the SOAP note format and how does it support clinical reasoning?

Answer

The SOAP note format organizes clinical documentation into four components. Subjective: patient's reported symptoms, history of present illness, and relevant review of systems. Objective: vital signs, physical exam findings, and relevant diagnostic results. Assessment: differential diagnosis and working diagnosis, with clinical reasoning supporting the diagnosis. Plan: management steps including additional workup, medications with doses and duration, non-pharmacologic interventions, patient education, referrals, and follow-up timeline. The Assessment and Plan (A/P) section reflects clinical decision-making complexity, which drives the E/M billing level and demonstrates the provider's reasoning process.

#459 · Professional Practice

What is the nurse practitioner's obligation when a patient discloses intent to harm an identified third party?

Answer

When a patient discloses a credible, specific threat to harm an identifiable third party, the provider may have a legal duty to warn (Tarasoff v. Regents of University of California) that varies by state law. In Tarasoff duty-to-warn states, the provider must take reasonable steps to protect the potential victim, which may include warning the intended victim, notifying law enforcement, or arranging involuntary psychiatric hospitalization. Providers should consult their facility's legal counsel, risk management, or ethics committee. Documentation of the threat and actions taken is critical.

#460 · Professional Practice

What is the Social Determinants of Health (SDOH) framework and how does it apply to NP practice?

Answer

Social determinants of health (SDOH) are the conditions in the environments where people are born, live, learn, work, play, worship, and age, which affect health outcomes and risks. Five SDOH domains identified by Healthy People 2030 are: economic stability, education access and quality, health care access and quality, neighborhood and built environment, and social and community context. NPs screen for SDOH using tools such as the PRAPARE or AHC (Accountable Health Communities) instrument. Addressing SDOH includes connecting patients to food assistance (SNAP), housing support, transportation, and community health worker programs.

#461 · Pharmacology

Why does warfarin INR rise significantly when fluconazole is added, and how should it be managed?

Answer

Fluconazole is a potent inhibitor of CYP2C9, the primary enzyme responsible for metabolizing the more potent S-warfarin enantiomer. This inhibition significantly reduces warfarin clearance, causing plasma warfarin levels and INR to rise substantially, sometimes doubling within 1-3 days. Management requires reducing the warfarin dose by approximately 25-50% when fluconazole is initiated, checking INR within 3-5 days after starting fluconazole, and monitoring closely until a new stable INR is achieved.

#462 · Pharmacology

Why does rifampin cause warfarin INR to fall, and how much dose adjustment is typically required?

Answer

Rifampin is one of the strongest known inducers of CYP2C9 and CYP3A4, causing a dramatic increase in warfarin metabolism that can reduce warfarin plasma levels by 50-90% within 5-7 days of initiation. INR may fall to subtherapeutic levels, significantly increasing thromboembolic risk. The warfarin dose often must be increased 2-5-fold to maintain therapeutic INR. INR should be monitored weekly during rifampin therapy and for 2-4 weeks after rifampin is stopped, as warfarin requirements will return to baseline.

How does amiodarone interact with warfarin and what dose adjustment is required?

Answer

Amiodarone is a potent inhibitor of CYP2C9 and also inhibits P-glycoprotein, both of which reduce warfarin clearance and raise INR significantly. The interaction is potentiated by amiodarone's extremely long half-life (40-55 days), meaning the interaction may persist for weeks to months after amiodarone is stopped. When amiodarone is initiated in a patient on warfarin, the warfarin dose should be reduced empirically by 30-50%, INR should be checked weekly for the first 1-2 months, and close monitoring should continue. The effect may continue to intensify over weeks as amiodarone accumulates.

Why is combining a statin with clarithromycin dangerous, and which statins are most at risk?

Answer

Clarithromycin is a potent inhibitor of CYP3A4, which is the primary metabolic pathway for simvastatin, lovastatin, and atorvastatin. Inhibiting CYP3A4 dramatically increases plasma statin concentrations, significantly raising the risk of statin-induced myopathy and rhabdomyolysis (characterized by muscle pain, weakness, and myoglobinuria leading to acute renal failure). The safest approach is to temporarily hold the interacting statin during the clarithromycin course, or use azithromycin as an alternative antibiotic. Rosuvastatin and pravastatin are not significantly metabolized by CYP3A4 and are safer alternatives.

#465 · Pharmacology

How does phenytoin affect oral contraceptive efficacy and what counseling is required?

Answer

Phenytoin is a potent inducer of CYP3A4, which metabolizes both estrogen and progestin components of combined oral contraceptives. Enzyme induction significantly lowers contraceptive hormone levels, reducing OCP efficacy and increasing the risk of unintended pregnancy. Patients taking phenytoin should use a reliable barrier method (such as condoms) in addition to hormonal contraception, or consider a higher-dose pill (at least 50 mcg ethinyl estradiol). Long-acting reversible contraception (levonorgestrel IUD or copper IUD) is preferred and not affected by enzyme-inducing drugs.

#466 · Pharmacology

What are the eGFR thresholds for metformin use in CKD, and what is the rationale?

Answer

Metformin is safe and can be continued in patients with eGFR of 45 mL/min/1.73m² or above. Metformin should be used with caution and the patient-provider risk-benefit discussion when eGFR is 30-44 mL/min/1.73m², with careful monitoring and dose reduction. Metformin should be stopped when eGFR falls below 30 mL/min/1.73m² due to the risk of drug accumulation and lactic acidosis. Metformin inhibits mitochondrial complex 1, which normally metabolizes lactate; in patients with reduced renal clearance and tissue hypoperfusion, lactate accumulation can cause life-threatening lactic acidosis.

What dose adjustments are required for gabapentin in patients with CKD?

Answer

Gabapentin is eliminated almost entirely by renal excretion and is not metabolized hepatically. In CKD, gabapentin accumulates proportionally to the degree of renal impairment, causing excessive sedation, confusion, dizziness, and falls, particularly in elderly patients. Dose adjustments are required based on eGFR: eGFR 30-59 mL/min requires 200-700 mg twice daily maximum; eGFR 15-29 mL/min requires 200-700 mg once daily; eGFR below 15 mL/min requires 100-300 mg once daily or every other day. Patients on hemodialysis require supplemental doses after each dialysis session.

Why should nitrofurantoin be avoided in patients with eGFR below 45 mL/min?

Answer

Nitrofurantoin requires adequate renal function for two reasons: first, reduced renal clearance prevents adequate urine concentrations from being achieved, rendering the drug ineffective for treating urinary tract infections; second, drug accumulates systemically in CKD, increasing the risk of serious adverse effects including pulmonary toxicity (acute pulmonary hypersensitivity or chronic interstitial pneumonitis), peripheral neuropathy, and hepatotoxicity. The FDA and clinical guidelines contraindicate nitrofurantoin when eGFR is below 30 mL/min; many experts avoid it below 45 mL/min in clinical practice.

#469 · Pharmacology

Why are bisphosphonates avoided in patients with eGFR below 35 mL/min, and what is the alternative?

Answer

Bisphosphonates (alendronate, risedronate, zoledronic acid) are renally cleared and accumulate to high concentrations in bone and kidneys when eGFR falls below 35 mL/min, potentially worsening renal function and causing adynamic bone disease. Clinical trials excluded patients with severe CKD, so safety data is lacking. Denosumab (Prolia), a RANK ligand inhibitor, is not renally cleared and is the preferred anti-resorptive agent for osteoporosis in patients with CKD stage 4 or 5 and those on dialysis. Calcium and vitamin D supplementation must accompany denosumab to prevent hypocalcemia.

#470 · Pharmacology

What teratogenic effects does isotretinoin cause and what pregnancy prevention program governs its use?

Answer

Isotretinoin (Accutane) is a Pregnancy Category X medication causing severe birth defects including craniofacial abnormalities, cardiac defects, CNS malformations (microcephaly, hydrocephalus), and thymic aplasia in virtually all exposed pregnancies. The iPLEDGE Risk Evaluation and Mitigation Strategy (REMS) program requires all patients of reproductive potential to use two forms of contraception simultaneously and submit to monthly pregnancy testing. Isotretinoin must be stopped at least 1 month before attempting conception. All prescribers, patients, and pharmacies must be enrolled in iPLEDGE.

#471 · Pharmacology

What teratogenic risks does valproate carry, and in which patients is it particularly concerning?

Answer

Valproate (valproic acid, divalproex) is associated with the highest rate of major congenital malformations among antiepileptic drugs, particularly neural tube defects (spina bifida at approximately 1-2%), cardiac defects, craniofacial abnormalities, limb defects, and cognitive impairment or autism spectrum disorder in exposed children. The risk is dose-dependent. Valproate should be avoided in women of childbearing age when alternative treatments (lamotrigine, levetiracetam) are available. If it must be used, high-dose folic acid (4 mg/day) is recommended preconceptionally and in early pregnancy, though this does not eliminate the risk.

#472 · Pharmacology

What are the risks of warfarin use during pregnancy and what anticoagulant is preferred?

Answer

Warfarin crosses the placenta and causes warfarin embryopathy in approximately 5% of pregnancies when used in the first trimester (6-12 weeks), characterized by nasal hypoplasia, bone stippling, and stippled epiphyses. Later in pregnancy it causes fetal hemorrhage (including intracranial hemorrhage) and placental abruption. Low molecular weight heparin (LMWH, enoxaparin) is the preferred anticoagulant throughout pregnancy, as it does not cross the placenta. Unfractionated heparin (UFH) is used near delivery and for mechanical heart valves. Direct oral anticoagulants (apixaban, rivaroxaban) are contraindicated in pregnancy.

#473 · Pharmacology

What cardiac defect is associated with lithium use in pregnancy, and how is the risk managed?

Answer

Lithium use during the first trimester was historically believed to carry a high risk of Ebstein anomaly (downward displacement of a malformed tricuspid valve into the right ventricle). More recent data suggests the absolute risk is small (approximately 0.1-0.2% compared to a background rate of 0.005%), though still higher than in the general population. If lithium must be continued in pregnancy (given high relapse risk with discontinuation), fetal echocardiography is recommended at 18-22 weeks. The risk of bipolar relapse with lithium discontinuation must be carefully weighed against the small fetal cardiac risk.

#474 · Pharmacology

What are the distinguishing clinical features between serotonin syndrome and neuroleptic malignant syndrome?

Answer

Serotonin syndrome is caused by serotonergic drugs and presents with clonus, hyperreflexia, agitation, hyperthermia, tachycardia, and diaphoresis, with rapid onset within hours of drug exposure. NMS is caused by dopamine antagonists (antipsychotics) and presents with lead-pipe rigidity, hyperthermia, diaphoresis, and autonomic instability, with slower onset over 24-72 hours. The key differentiating feature is clonus in serotonin syndrome versus extreme muscle rigidity without clonus in NMS. CK elevation is more pronounced in NMS. Treatment differs: serotonin syndrome uses cyproheptadine; NMS uses dantrolene and bromocriptine.

What is the treatment for serotonin syndrome?

Answer

Treatment begins with immediately stopping all serotonergic agents. Cyproheptadine (Periactin), a serotonin 5-HT_{1A} and 5-HT_{2A} antagonist, is the primary antidote, dosed at 12 mg orally initially then 2 mg every 2 hours as needed (max 32 mg/day). Benzodiazepines (lorazepam 2-4 mg IV) are given for agitation, myoclonus, and neuromuscular hyperactivity. Active cooling is used for hyperthermia (goal temperature below 38.5 degrees Celsius). Severe cases with temperatures above 41 degrees Celsius require ICU admission, sedation, paralysis, and endotracheal intubation. Serotonin syndrome is generally self-limiting and resolves within 24-72 hours with supportive care.

What is andexanet alfa and for which anticoagulants does it provide reversal?

Answer

Andexanet alfa (Andexxa) is a recombinant modified human factor Xa protein that acts as a decoy receptor to bind and neutralize factor Xa inhibitors. It is FDA-approved for reversal of life-threatening or uncontrolled bleeding in patients taking apixaban (Eliquis) or rivaroxaban (Xarelto). It is given as a bolus plus a 2-hour infusion, with dosing varying based on which anticoagulant was taken and time of last dose. A major adverse effect is thromboembolic events (approximately 10% in trials). Cost is extremely high, limiting its availability.

#477 · Pharmacology

What is idarucizumab and what is the dose and indication?

Answer

Idarucizumab (Praxbind) is a humanized monoclonal antibody fragment that binds dabigatran (Pradaxa), a direct thrombin inhibitor, with very high affinity (350 times greater than dabigatran's affinity for thrombin). It is FDA-approved for emergency reversal of dabigatran anticoagulation in patients with uncontrolled bleeding or urgent surgery. The dose is a fixed 5 g IV (given as two separate 2.5 g doses within 15 minutes). Reversal is complete within minutes and lasts at least 24 hours. It does not affect factor Xa inhibitors such as apixaban or rivaroxaban.

#478 · Pharmacology

What is the recommended approach to urgent warfarin reversal for life-threatening bleeding?

Answer

Urgent warfarin reversal for life-threatening bleeding (intracranial hemorrhage, massive GI bleed, trauma) requires 4-factor prothrombin complex concentrate (4-factor PCC, such as Kcentra) plus IV vitamin K 5-10 mg. 4-factor PCC replaces factors II, VII, IX, and X and produces immediate INR reversal within 15-30 minutes. IV vitamin K sustains the reversal effect for 12-24 hours but takes 4-6 hours to produce full effect alone. Fresh frozen plasma (FFP) is a slower alternative requiring larger volumes and risking volume overload. Recombinant factor VIIa is not recommended for warfarin reversal due to thrombosis risk.

What is the dosing for naloxone in opioid overdose reversal, and why may repeated doses be required?

Answer

Naloxone (Narcan) is a competitive opioid receptor antagonist used for reversal of opioid overdose. Standard dosing is 0.4-2 mg IV, IM, or intranasal (IN), with repeat doses every 2-3 minutes if no response, up to a total of 10 mg. Intranasal formulations (4 mg per dose) and auto-injectable devices are available for community use. Repeated doses may be required because naloxone has a relatively short duration of action of 30-90 minutes, which is shorter than the duration of most opioids (including fentanyl analogues and long-acting opioids). Patients should be monitored for at least 4 hours after naloxone administration for recurrence of respiratory depression.

How is protamine sulfate used for heparin reversal, and how effective is it against LMWH?

Answer

Protamine sulfate is a positively charged protein that binds to and neutralizes negatively charged heparin molecules, forming an inactive complex. For unfractionated heparin (UFH), the dose is 1 mg of protamine per 100 units of UFH given in the preceding 2-3 hours (maximum 50 mg IV slowly). Protamine provides complete reversal of UFH anticoagulation. For LMWH (such as enoxaparin), protamine provides only partial reversal of approximately 60-75% of anti-factor Xa activity, and incomplete neutralization is expected because the shorter LMWH chains bind protamine less effectively. Adverse reactions to protamine include hypotension, bradycardia, and anaphylaxis.

#481 · Pharmacology

What are the signs of digoxin toxicity and how is it treated?

Answer

Digoxin toxicity presents with nausea, vomiting, anorexia, visual changes (yellow-green halos, blurred or altered color vision), and cardiac arrhythmias (bradycardia, heart block, atrial tachycardia with block, junctional rhythms, PVCs). The therapeutic digoxin level is 0.5-2.0 ng/mL (target 0.5-0.9 ng/mL for heart failure). Toxicity is more likely with hypokalemia, hypomagnesemia, and hypercalcemia. Treatment includes digoxin-specific antibody fragments (Digibind or DigiFab), with dosing based on known digoxin dose or measured serum level. Atropine is used for symptomatic bradycardia while awaiting Fab fragment preparation.

#482 · Pharmacology

What are the major drug interactions of amiodarone and what dose adjustments are required?

Answer

Amiodarone inhibits CYP2C9, CYP3A4, and P-glycoprotein. This causes clinically significant elevations in warfarin levels (reduce warfarin 30-50%), digoxin levels (reduce digoxin by 50%), and statin levels (simvastatin and lovastatin dose must be capped at 20 mg). Amiodarone also potentiates QT prolongation when combined with other QT-prolonging drugs (antipsychotics, fluoroquinolones). Due to amiodarone's extremely long half-life, these interactions persist for weeks to months after stopping amiodarone. Thyroid function (TSH, free T4) and pulmonary function must be monitored regularly given amiodarone-induced thyroid and pulmonary toxicity.

What is the risk of combining tamsulosin with sildenafil?

Answer

Tamsulosin is an alpha-1 receptor blocker used for benign prostatic hyperplasia (BPH) that causes blood vessel relaxation and blood pressure reduction. Sildenafil (Viagra) is a PDE-5 inhibitor that also causes vasodilation and blood pressure reduction. When used together, their additive vasodilatory effects can cause severe symptomatic hypotension, particularly orthostatic hypotension, leading to dizziness, syncope, and falls. In clinical practice, this combination requires caution: a 4-hour interval between sildenafil and tamsulosin doses is recommended, the lowest effective sildenafil dose (25 mg) should be started, and patients should be counseled about position changes.

What is the tyramine interaction with MAOIs, and which foods must patients avoid?

Answer

MAOIs (monoamine oxidase inhibitors) such as phenelzine and tranylcypromine block breakdown of dietary tyramine in the gut wall and liver. Tyramine is normally metabolized by MAO-A; when inhibited, large amounts of tyramine enter systemic circulation, triggering massive norepinephrine release causing a hypertensive crisis (severe hypertension with headache, palpitations, diaphoresis, and risk of stroke or MI). Foods to avoid include aged cheeses (cheddar, brie, blue cheese), cured and smoked meats (salami, pepperoni), fermented products (sauerkraut, miso), tap beer, red wine, overripe avocados, and broad beans (fava beans).

Why is combining tramadol with an SSRI risky, and what should providers do?

Answer

Tramadol has dual analgesic mechanisms: weak mu-opioid receptor agonism and inhibition of serotonin and norepinephrine reuptake. When combined with SSRIs or SNRIs, the additive serotonergic effect significantly raises the risk of serotonin syndrome. The combination also increases seizure risk, as both SSRIs and tramadol lower the seizure threshold. Providers should avoid this combination when possible; if pain management requires an opioid in a patient on an SSRI, consider using an opioid without serotonergic activity such as oxycodone, hydromorphone, or morphine. If tramadol is unavoidable, use the lowest effective dose and monitor closely for serotonin syndrome symptoms.

What is the interaction between trimethoprim-sulfamethoxazole (TMP-SMX) and methotrexate?

Answer

TMP-SMX creates a dangerous interaction with methotrexate through two mechanisms: it reduces renal tubular secretion of methotrexate, increasing methotrexate plasma levels and exposure, and it independently inhibits dihydrofolate reductase, adding to methotrexate's antifolate effect. The combined antifolate activity can cause severe methotrexate toxicity including myelosuppression (pancytopenia), mucositis, and hepatotoxicity, even at low methotrexate doses used for rheumatoid arthritis or psoriasis. TMP-SMX should be avoided in patients on methotrexate; azithromycin or doxycycline should be used for alternative infections. If the combination is unavoidable, CBC must be monitored closely.

#487 · Pharmacology

What is the potentially fatal interaction between colchicine and clarithromycin?

Answer

Clarithromycin is a potent inhibitor of both CYP3A4 (which metabolizes colchicine) and P-glycoprotein (which transports colchicine out of cells). Concurrent use causes a dramatic increase in colchicine plasma concentrations, leading to severe colchicine toxicity manifesting as multi-organ failure: bone marrow suppression (pancytopenia), rhabdomyolysis, acute kidney injury, hepatotoxicity, neuromuscular toxicity, and cardiovascular collapse. Fatal outcomes have been reported even with brief concurrent use. Colchicine should never be combined with clarithromycin; azithromycin (which does not inhibit CYP3A4 or P-gp significantly) is the preferred macrolide alternative.

#488 · Pharmacology

What is the seizure risk with bupropion and what are the contraindications?

Answer

Bupropion lowers the seizure threshold in a dose-dependent manner, with a seizure incidence of approximately 0.4% at doses up to 450 mg/day. Absolute contraindications to bupropion include bulimia nervosa or anorexia nervosa (risk of electrolyte abnormalities that further lower seizure threshold), history of seizure disorder, and situations predisposing to seizures such as abrupt alcohol withdrawal or abrupt benzodiazepine withdrawal. The maximum daily dose is 450 mg (150 mg per individual dose), and doses should be spaced at least 6 hours apart. Prior history of head trauma or CNS tumor increases seizure risk.

Which classes of drugs are associated with QT prolongation, and what conditions increase the risk?

Answer

QT-prolonging drug classes include fluoroquinolone antibiotics (ciprofloxacin, levofloxacin, moxifloxacin), antipsychotics (haloperidol, quetiapine, ziprasidone), azithromycin (the risk is modest but real), methadone, antiarrhythmics (amiodarone, sotalol, procainamide), and antiemetics (ondansetron, droperidol). Risk of torsades de pointes is significantly increased in patients with baseline QTc above 500 ms, structural heart disease, hypokalemia, hypomagnesemia, bradycardia, and female sex. Avoid combining multiple QT-prolonging agents; obtain baseline ECG when initiating high-risk drugs and correct electrolytes before and during treatment.

What is the mechanism and clinical use of SGLT2 inhibitors, and what are their key adverse effects?

Answer

SGLT2 (sodium-glucose cotransporter 2) inhibitors (empagliflozin, canagliflozin, dapagliflozin) block glucose reabsorption in the proximal renal tubule, causing glycosuria and lowering blood glucose. Beyond glycemic control, they provide cardiovascular benefit (reduce heart failure hospitalization and cardiovascular death) and renoprotective effects. Key adverse effects include genital mycotic infections (most common), urinary tract infections, euglycemic DKA (glucose may be normal), Fournier gangrene (necrotizing fasciitis of the genitalia), hypotension (from volume depletion), and lower limb amputation risk (canagliflozin). SGLT2 inhibitors are contraindicated when eGFR is below 20 mL/min for diabetes indication.

#491 · Pharmacology

What is the mechanism, dosing, and monitoring of methotrexate for rheumatoid arthritis?

Answer

Methotrexate inhibits dihydrofolate reductase, reducing purine and pyrimidine synthesis and suppressing rapidly dividing immune cells. For RA, it is dosed at 7.5-25 mg orally once weekly (never daily in non-oncology use). Folic acid 1-5 mg daily (given on non-methotrexate days) reduces side effects (mucositis, nausea, hair loss) without reducing efficacy. Monitoring includes CBC, comprehensive metabolic panel (liver enzymes, creatinine), and albumin at baseline and every 4-8 weeks during titration, then every 8-12 weeks when stable. Methotrexate is contraindicated in pregnancy, significant hepatic or renal disease, and active infection.

#492 · Pharmacology

What are the key pharmacologic differences between first-generation and second-generation antihistamines?

Answer

First-generation antihistamines (diphenhydramine, chlorpheniramine, hydroxyzine) readily cross the blood-brain barrier, causing significant CNS effects including sedation, cognitive impairment, and anticholinergic effects (dry mouth, urinary retention, constipation, blurred vision). They are listed on the Beers Criteria as potentially inappropriate for adults 65 and older. Second-generation antihistamines (loratadine, cetirizine, fexofenadine) are minimally sedating at standard doses (cetirizine can cause some drowsiness), do not cause significant anticholinergic effects, and are preferred for allergic rhinitis and urticaria. Fexofenadine is the least sedating.

What is the mechanism and clinical use of GLP-1 receptor agonists, and what are their key adverse effects?

Answer

GLP-1 (glucagon-like peptide-1) receptor agonists (semaglutide, liraglutide, dulaglutide, exenatide) mimic endogenous GLP-1, stimulating glucose-dependent insulin secretion, suppressing glucagon, slowing gastric emptying, and reducing appetite. They produce significant weight loss (5-15% with semaglutide at diabetes doses; up to 15-20% with higher-dose semaglutide in obesity). Cardiovascular benefits include reduction in MACE events. Key adverse effects are nausea, vomiting, diarrhea (most common), and rare pancreatitis. Contraindicated in patients with personal or family history of medullary thyroid cancer or MEN type 2 (risk seen in rodent studies). Injectable formulations; oral semaglutide (Rybelsus) is also available.

What are the key drug interactions and adverse effects of fluoroquinolone antibiotics?

Answer

Fluoroquinolones (ciprofloxacin, levofloxacin, moxifloxacin) should not be taken within 2 hours of antacids, calcium, iron, zinc, or sucralfate, as divalent cations chelate fluoroquinolones and reduce absorption by up to 90%. Ciprofloxacin inhibits CYP1A2, raising levels of theophylline and caffeine; it also increases warfarin INR. All fluoroquinolones prolong the QT interval and carry an FDA black box warning for tendinopathy (Achilles tendon rupture), peripheral neuropathy, CNS effects, and aortic aneurysm risk, particularly in elderly patients, those on corticosteroids, and those with renal impairment. They should be reserved for infections without safe alternatives.

What are the classes of drugs used for heart failure with reduced ejection fraction (HFrEF) and their respective mortality benefits?

Answer

Four drug classes form the cornerstone of HFrEF therapy, collectively called the four pillars: ACE inhibitors (or ARBs, or ARNI sacubitril/valsartan), beta-blockers (carvedilol, metoprolol succinate, bisoprolol), mineralocorticoid receptor antagonists (spironolactone, eplerenone), and SGLT2 inhibitors (empagliflozin, dapagliflozin). Each class independently reduces all-cause mortality by approximately 15-25%. ARNI (sacubitril/valsartan, Entresto) reduces cardiovascular mortality more than ACE inhibitors alone in symptomatic HFrEF and is now preferred over ACEi/ARB alone when tolerated. Loop diuretics (furosemide) are used for symptom management of fluid overload but do not have mortality benefit.

What is the mechanism and clinical use of N-acetylcysteine (NAC) in acetaminophen overdose?

Answer

Acetaminophen overdose depletes hepatic glutathione stores, allowing accumulation of the toxic metabolite NAPQI

(N-acetyl-p-benzoquinone imine), which causes hepatocellular necrosis.

N-acetylcysteine (NAC) replenishes glutathione precursors, detoxifies NAPQI, and reduces hepatic injury. It is most effective when given within 8-10 hours of ingestion. The Rumack-Matthew nomogram (serum acetaminophen level plotted against time post-ingestion) determines treatment need. IV NAC protocol: 150 mg/kg over 1 hour (loading), then 12.5 mg/kg/hour for 4 hours, then 6.25 mg/kg/hour for 16 hours (Prescott protocol). Oral NAC (Mucomyst) is an equivalent alternative at 140 mg/kg loading dose then 70 mg/kg every 4 hours for 17 additional doses.

What is the mechanism and clinical use of direct oral anticoagulants (DOACs), and how do they differ from warfarin?

Answer

DOACs include factor Xa inhibitors (apixaban, rivaroxaban, edoxaban, betrixaban) and the direct thrombin inhibitor dabigatran. Unlike warfarin, DOACs have predictable pharmacokinetics, rapid onset (within 1-3 hours), fixed dosing without routine INR monitoring, fewer food and drug interactions, and shorter half-lives. They are FDA-approved for stroke prevention in non-valvular atrial fibrillation, DVT/PE treatment and prevention, and VTE prophylaxis after joint surgery. DOACs are contraindicated with mechanical heart valves and significantly impaired renal function (dabigatran requires eGFR above 30, apixaban is preferred in CKD). Each agent has a specific reversal agent: idarucizumab for dabigatran, andexanet alfa for factor Xa inhibitors.

What is the pharmacologic management of chronic obstructive pulmonary disease (COPD) by GOLD group?

Answer

GOLD A (low symptoms, low risk): short-acting bronchodilator (SABA or SAMA) as needed. GOLD B (high symptoms, low risk): long-acting bronchodilator (LABA or LAMA) daily. GOLD E (high risk, regardless of symptoms): LAMA plus LABA combination; inhaled corticosteroids (ICS) are added for patients with frequent exacerbations or eosinophil counts above 300 cells/mcL. Triple therapy (ICS/LABA/LAMA, such as fluticasone/vilanterol/umeclidinium) is used for persistent symptoms or exacerbations despite dual therapy. Roflumilast (PDE-4 inhibitor) can be added for patients with chronic bronchitis phenotype and frequent exacerbations. Systemic corticosteroids are used for acute exacerbations (prednisone 40 mg daily for 5 days).

What are the key differences between ACE inhibitors and ARBs, and when is one preferred over the other?

Answer

ACE inhibitors (lisinopril, enalapril, ramipril) block conversion of angiotensin I to angiotensin II and also prevent bradykinin breakdown, causing bradykinin accumulation responsible for the class-defining adverse effect of dry cough (in 10-20% of patients) and angioedema (rare but potentially life-threatening). ARBs (losartan, valsartan, irbesartan) block the angiotensin II type 1 receptor directly and do not affect bradykinin, so they do not cause cough and have a lower risk of angioedema. ARBs are preferred in patients who develop ACE inhibitor cough. Both classes are contraindicated in pregnancy (fetal renal toxicity). ARNI (sacubitril/valsartan) is preferred over either alone for HFrEF.

What medications are first-line for the treatment of gout, and how is urate-lowering therapy monitored?

Answer

Acute gout flares are treated with NSAIDs (indomethacin 50 mg three times daily or naproxen 500 mg twice daily), colchicine (1.2 mg then 0.6 mg one hour later, then 0.6 mg twice daily until flare resolution), or prednisone 30-40 mg daily for 3-5 days. Urate-lowering therapy (ULT) is indicated for patients with 2 or more flares per year, tophi, or uric acid nephrolithiasis. First-line ULT is allopurinol, starting at 100 mg daily (50 mg in CKD) and titrated every 2-5 weeks to achieve serum uric acid below 6 mg/dL (below 5 mg/dL for tophaceous gout). Febuxostat is an alternative for patients intolerant of allopurinol. Allopurinol hypersensitivity syndrome (AHS) is more common in Han Chinese, Thai, and Korean patients with HLA-B*5801 allele.